

Paul Rosen

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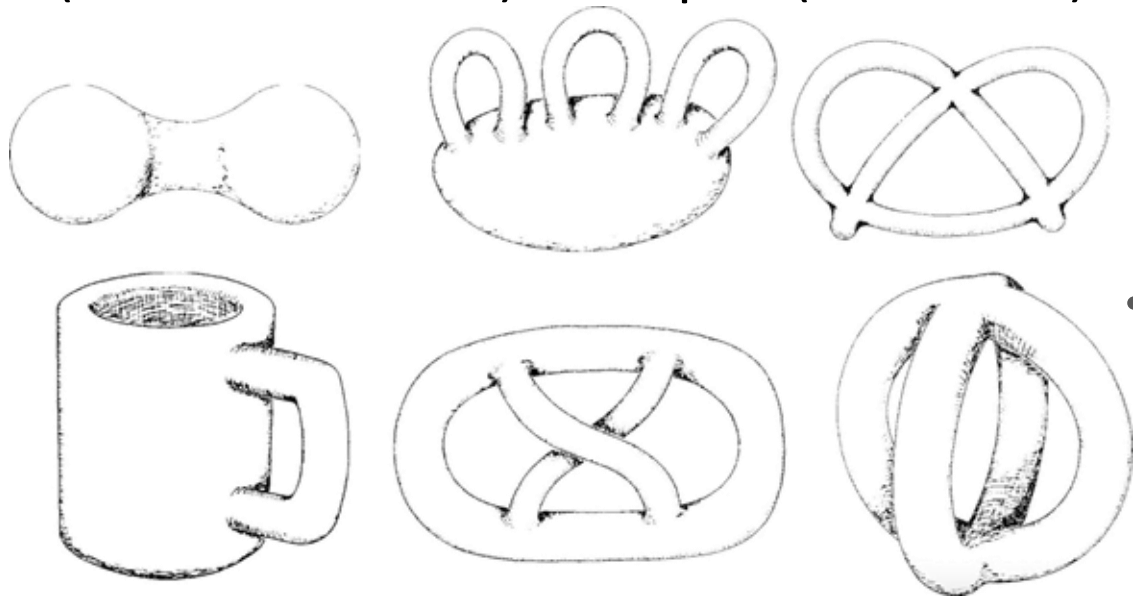
About Paul and the SCI Institute

Paul Rosen

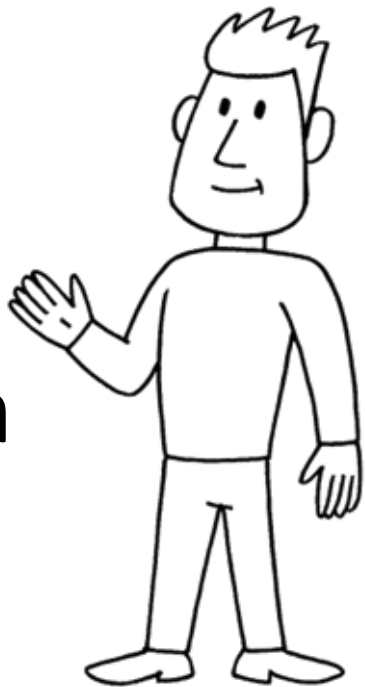
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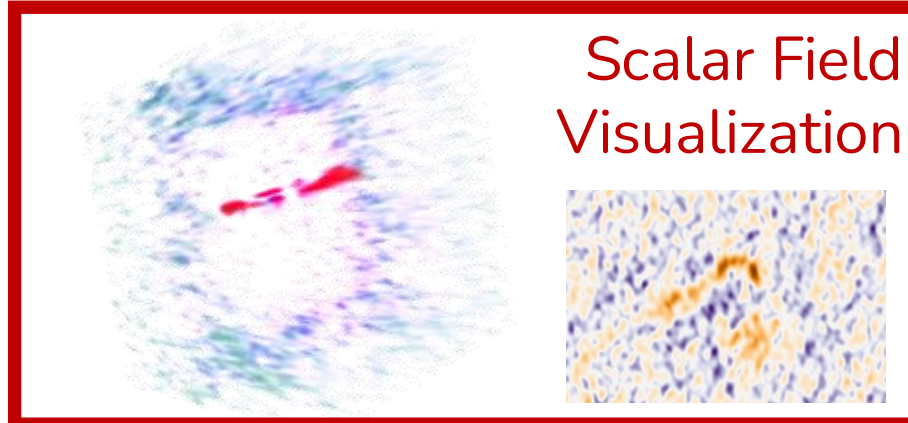
Geometric (rigid) & Topologic
(deformable) Shape (of data)



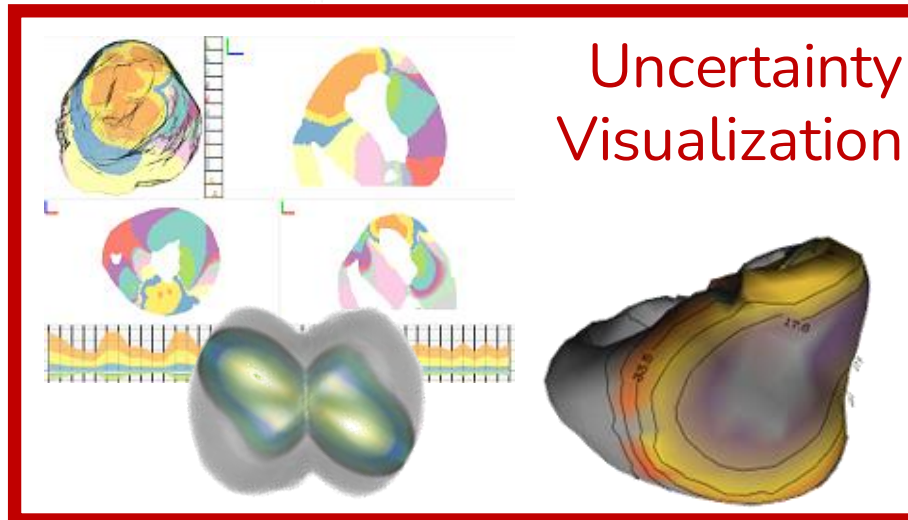
Human-
Centered
Approach



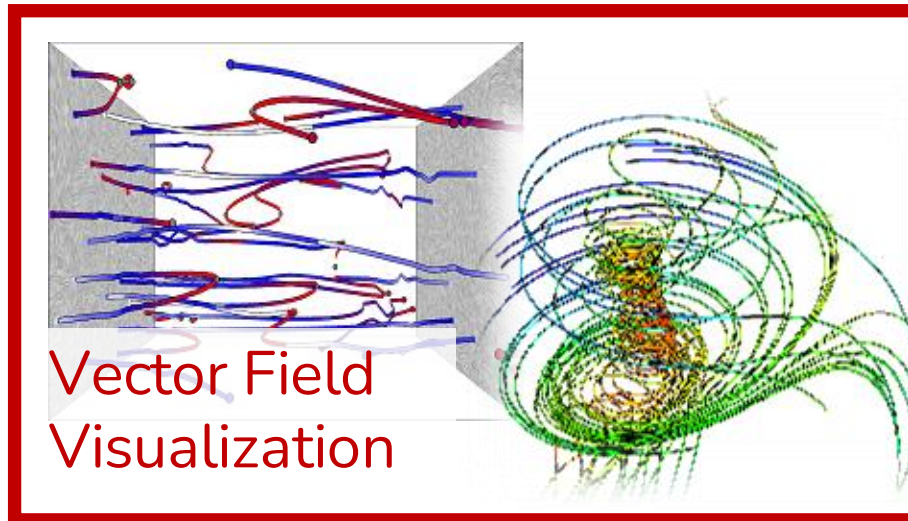
Scalar Field
Visualization



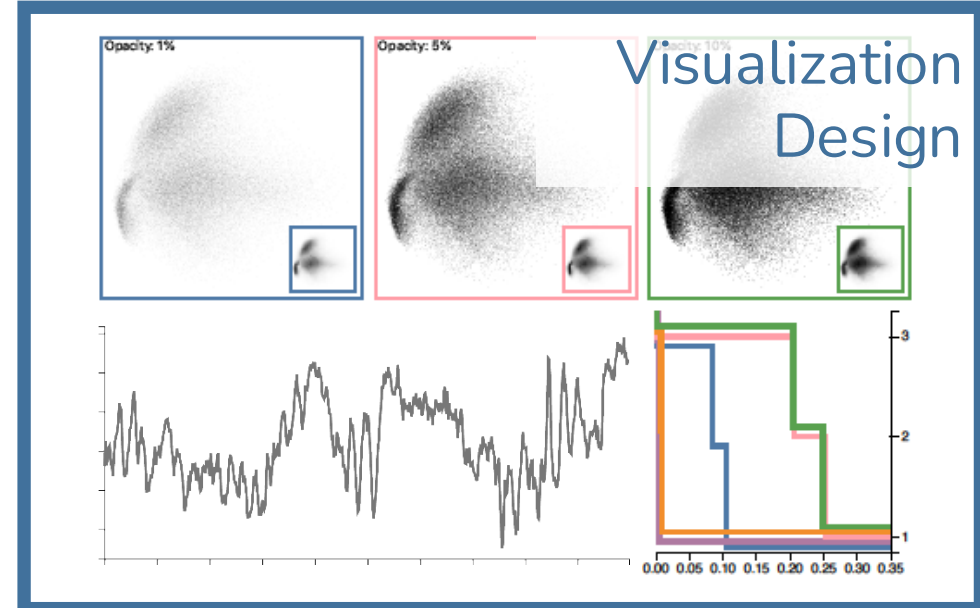
Uncertainty
Visualization



Vector Field
Visualization



Visualization
Design

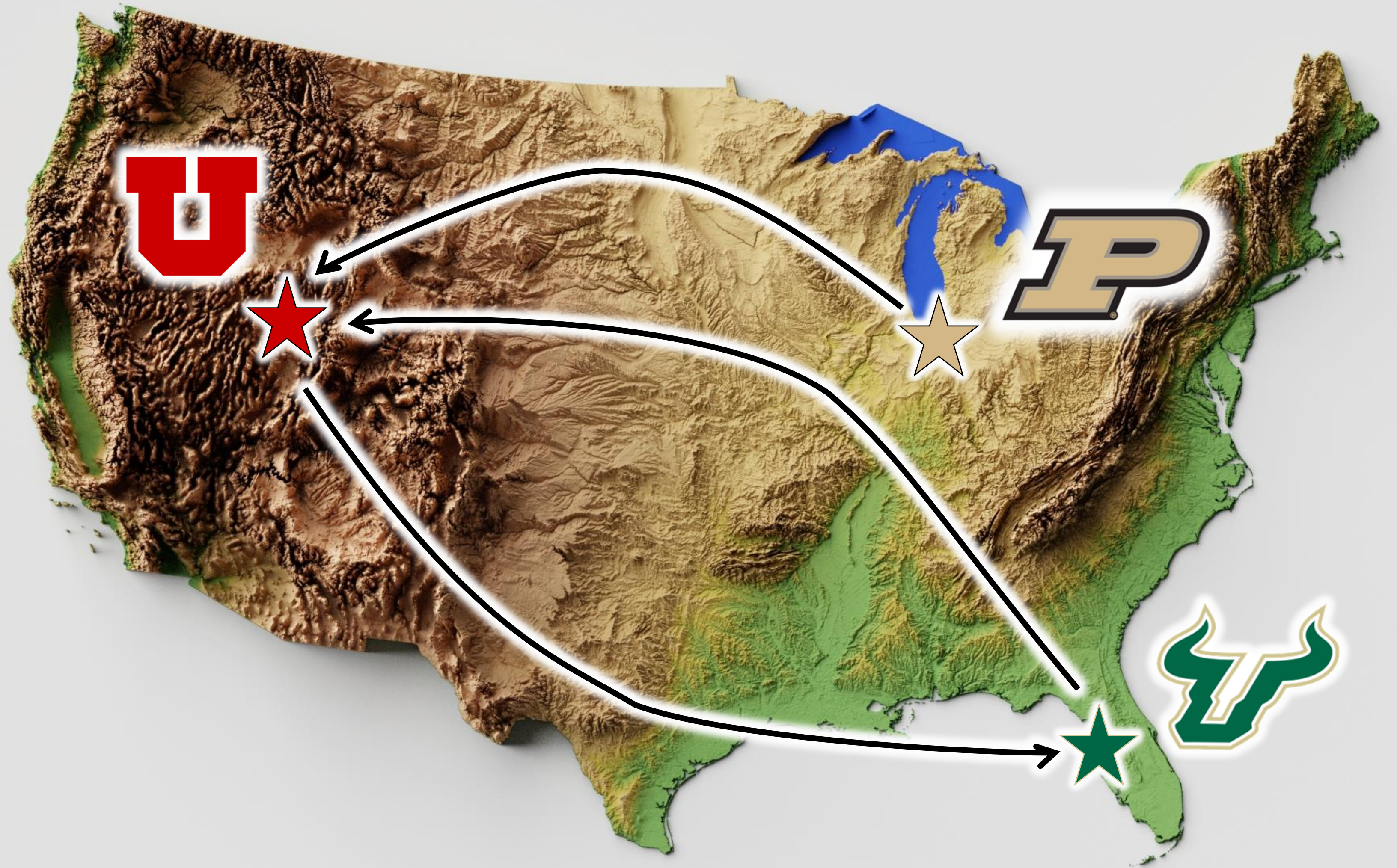


Network
Visualization



Software
Visualization







Scientific Computing and Imaging Institute

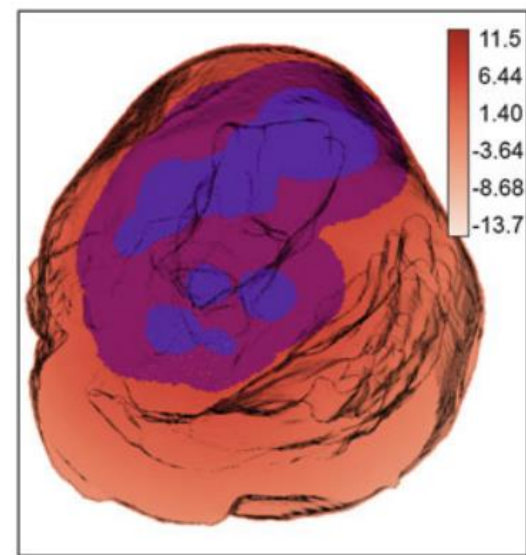
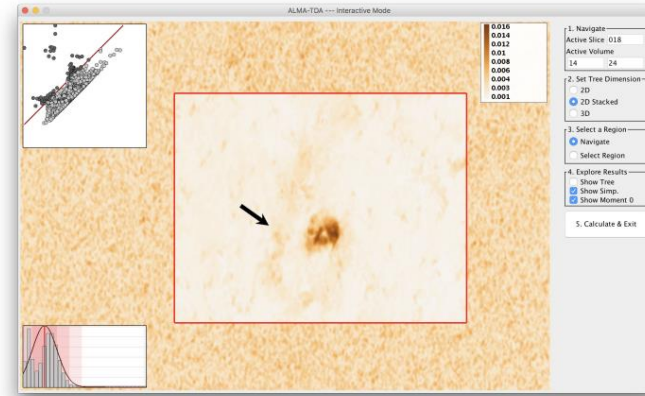
30 years of impact

**Empowering our students and scholars to impact society through
multidisciplinary transformative computing research**

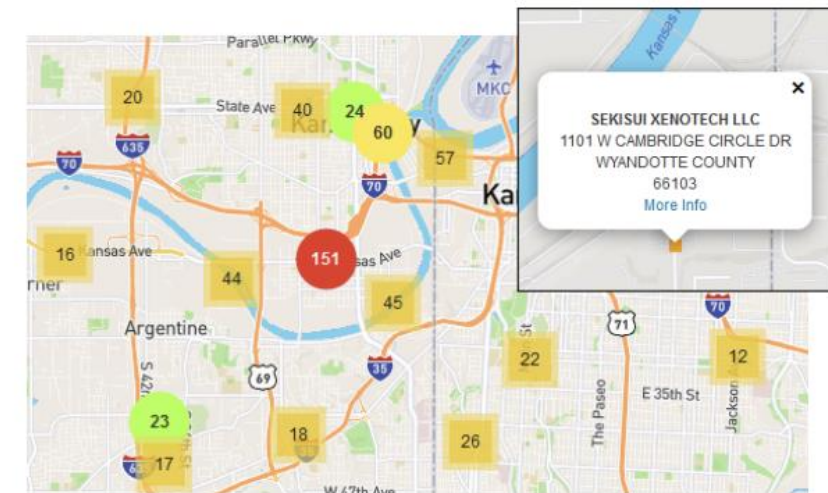
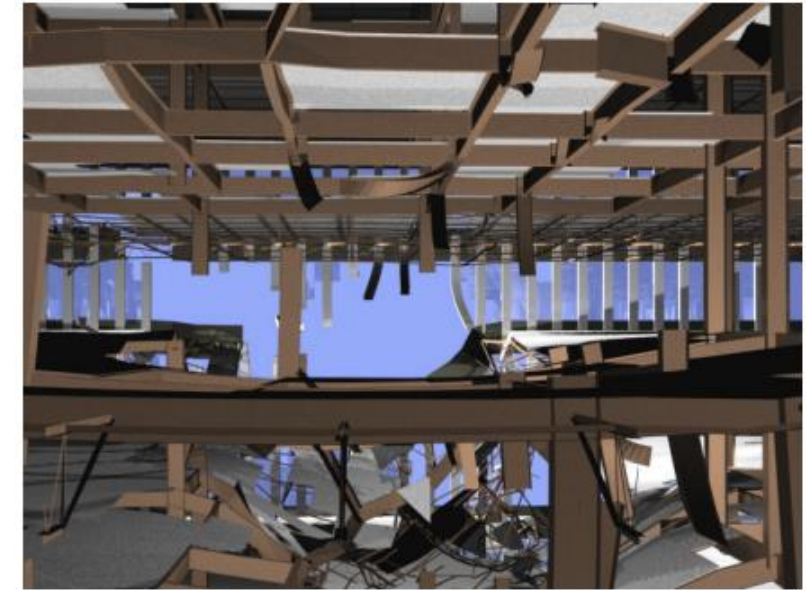


Examples of some collaborations

- Astronomy
 - LSST ('09)
 - NRAO ('16-'17)
 - LSST DSFP ('23, '26)
 - CosmicAI ('26)
- Biomedical Engineering
 - Simulation of TBI
 - Uncertainty in myocardial ischemia and DBS

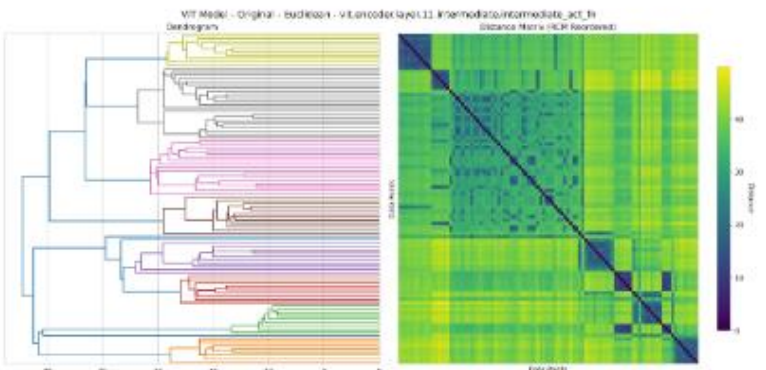
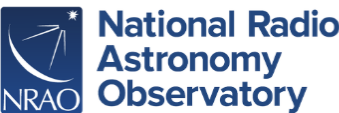


- Civil/Structural Engineering
 - Simulation of building damage
- Public Health
 - Asthma management in adolescents
 - Air quality monitoring
 - Wastewater monitoring

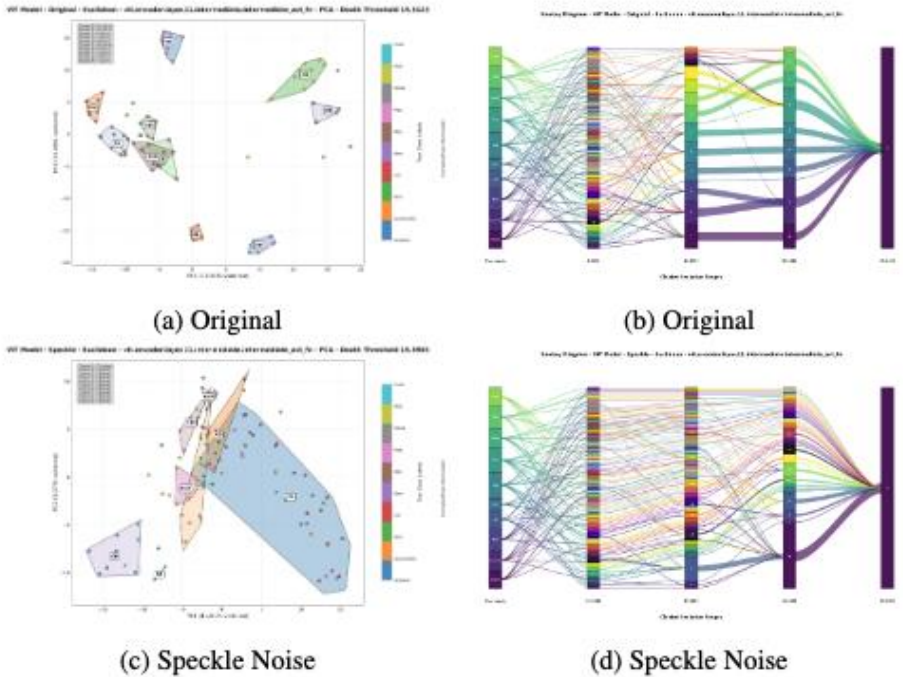
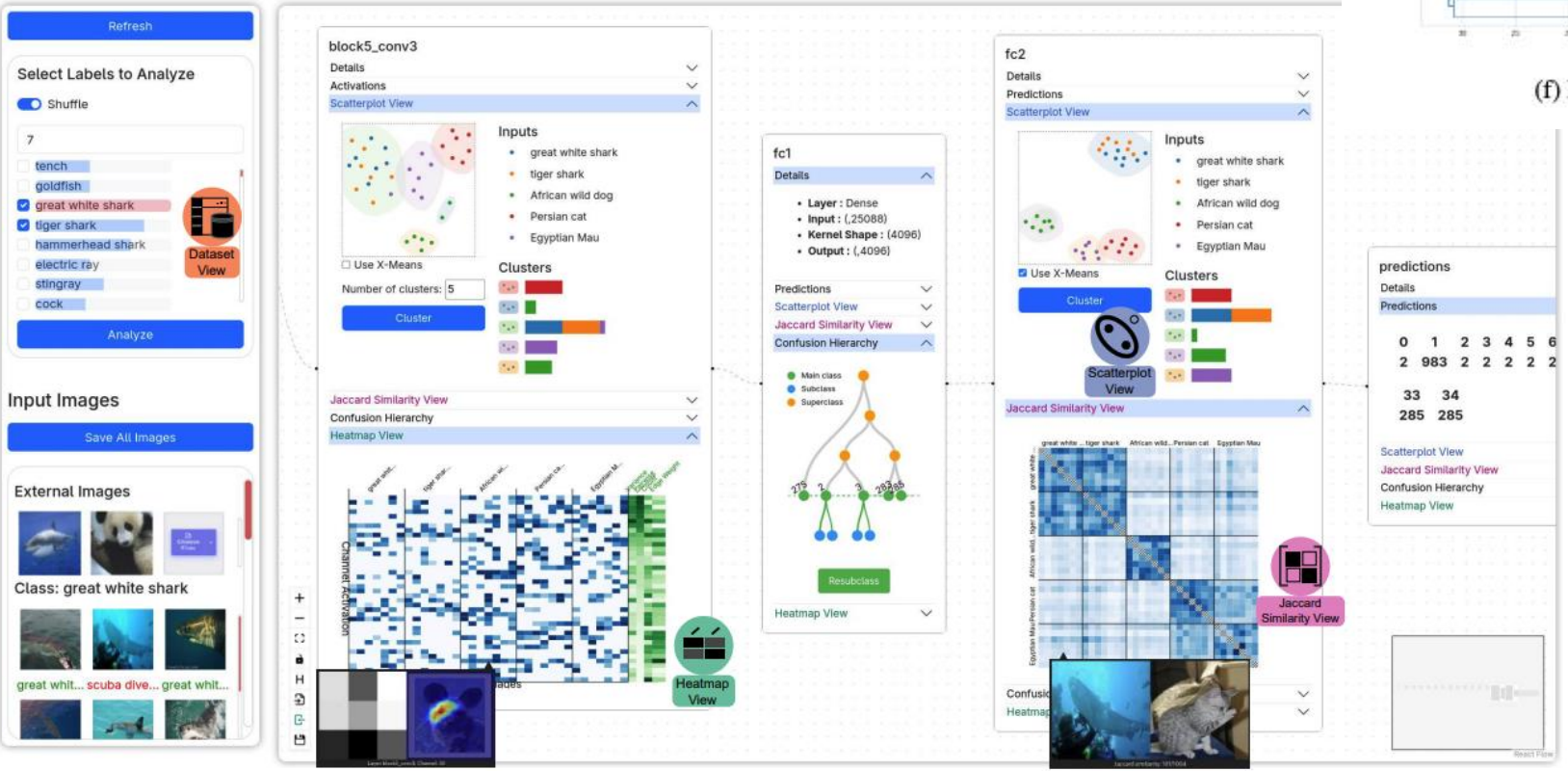


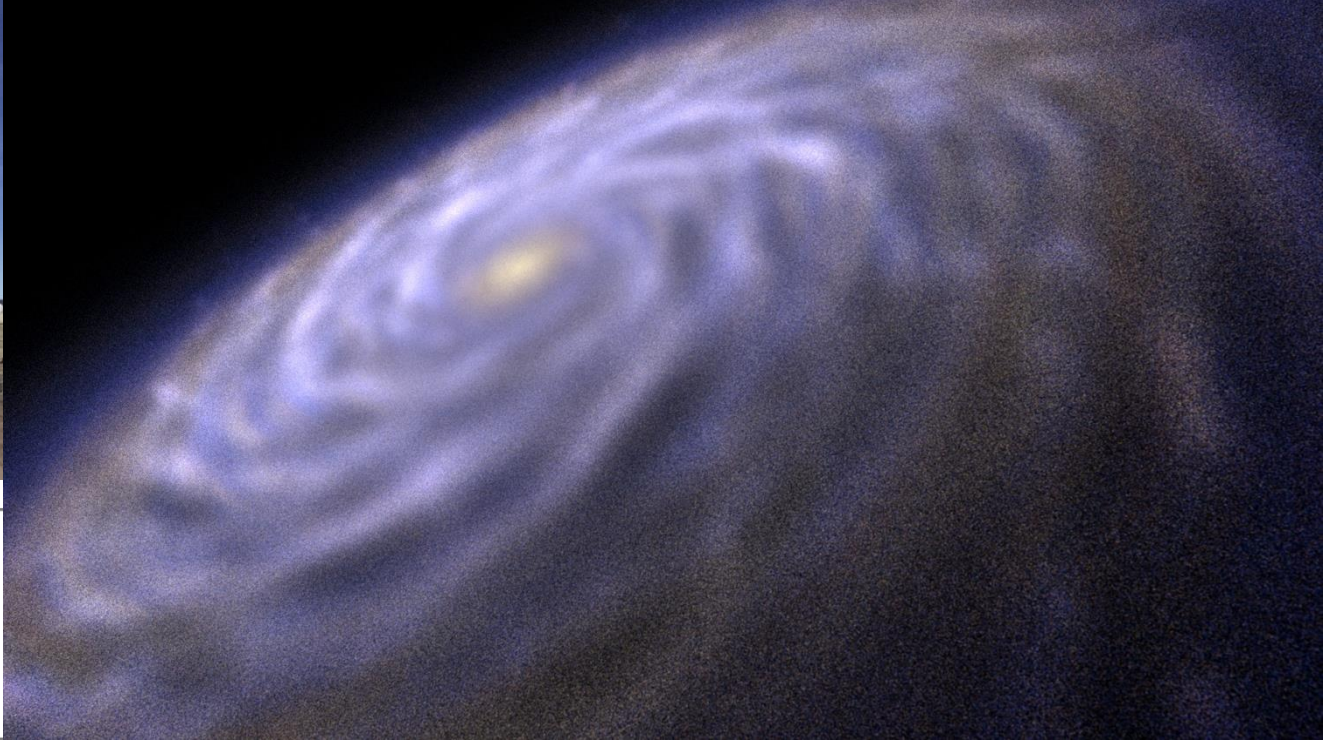
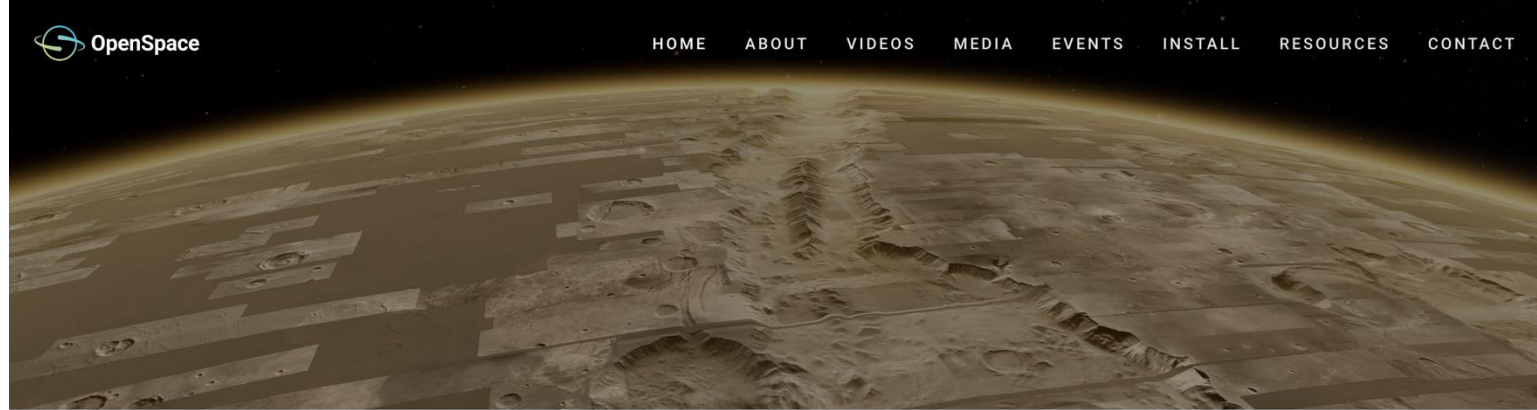
CosmicAI

- Explainability and interpretability of deep neural networks used in astronomy applications.

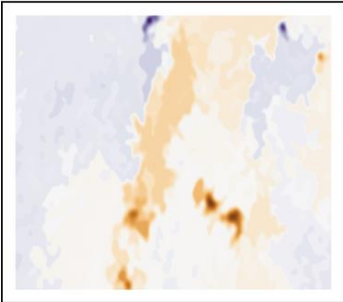
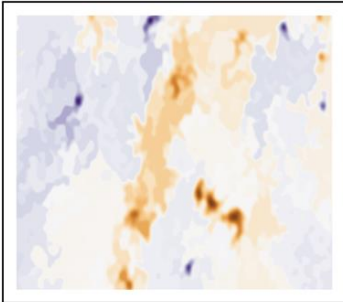
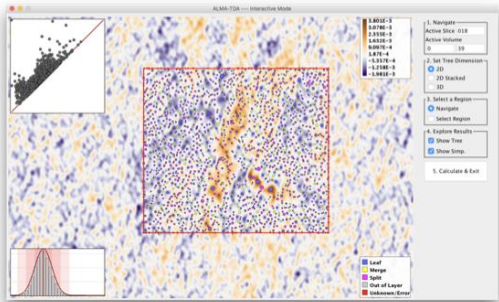
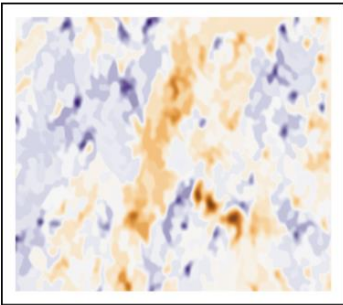
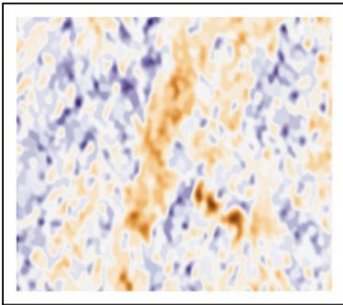
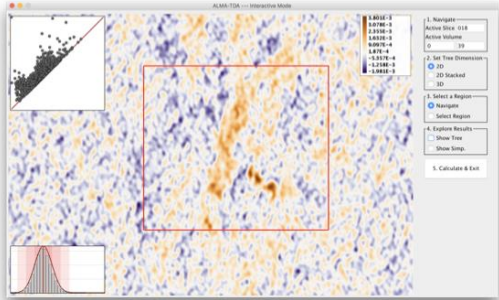


(f) Heatmap Dendrogram layer 11 (Euclidean)





OpenSpace is open-source interactive data visualization software designed to visualize the entire known universe and portray our ongoing efforts to investigate the cosmos.



Task-Driven Categories



Data wrangling



Data exploration



Feature Identification



Object reconstruction



Education & outreach

Technique-Driven Categories



2D/3D plots



2D images



3D rendering



Interactive vis.



Dim. reduction



Uncertainty vis.



New display

Topic-Driven Categories



Extragalactic astronomy



Galactic astronomy



Planetary astronomy



Solar astronomy & astrophysics

Simulation




- Rosen, P., Seth, A., Mills, E., Ginsburg, A., Kamenetzky, J., Kern, J., Johnson, C.R. and Wang, B., 2021. Using contour trees in the analysis and visualization of radio astronomy data cubes. In *Topological Methods in Data Analysis and Visualization VI: Theory, Applications, and Software* (pp. 87-108). Springer International Publishing.
- Lan, F., Young, M., Anderson, L., Ynnerman, A., Bock, A., Borkin, M.A., Forbes, A.G., Kollmeier, J.A. and Wang, B., 2021, June. Visualization in astrophysics: Developing new methods, discovering our Universe, and educating the Earth. In *Computer graphics forum* (Vol. 40, No. 3, pp. 635-663).
- Bock, A., Axelsson, E., Costa, J., Payne, G., Acinapura, M., Trakinski, V., Emmart, C., Silva, C., Hansen, C. and Ynnerman, A., 2019. OpenSpace: A system for astrographics. *IEEE transactions on visualization and computer graphics*, 26(1), pp.633-642.

Bei Wang Phillips

Associate Professor

Paul Rosen

Associate Professor

Chuck Hansen

Distinguished
Emeritus Professor





Paul Rosen

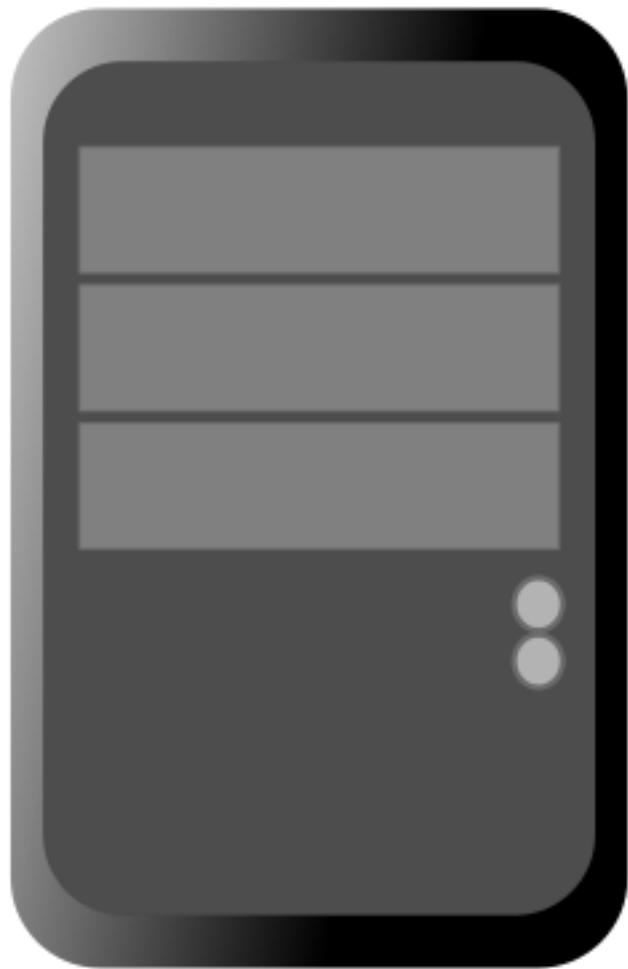
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Perception & Visualization

data

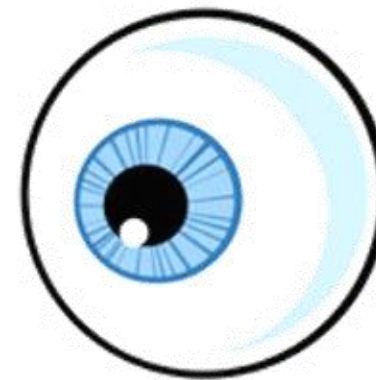
knowledge



(analysis—stats, ML, AI, etc.)



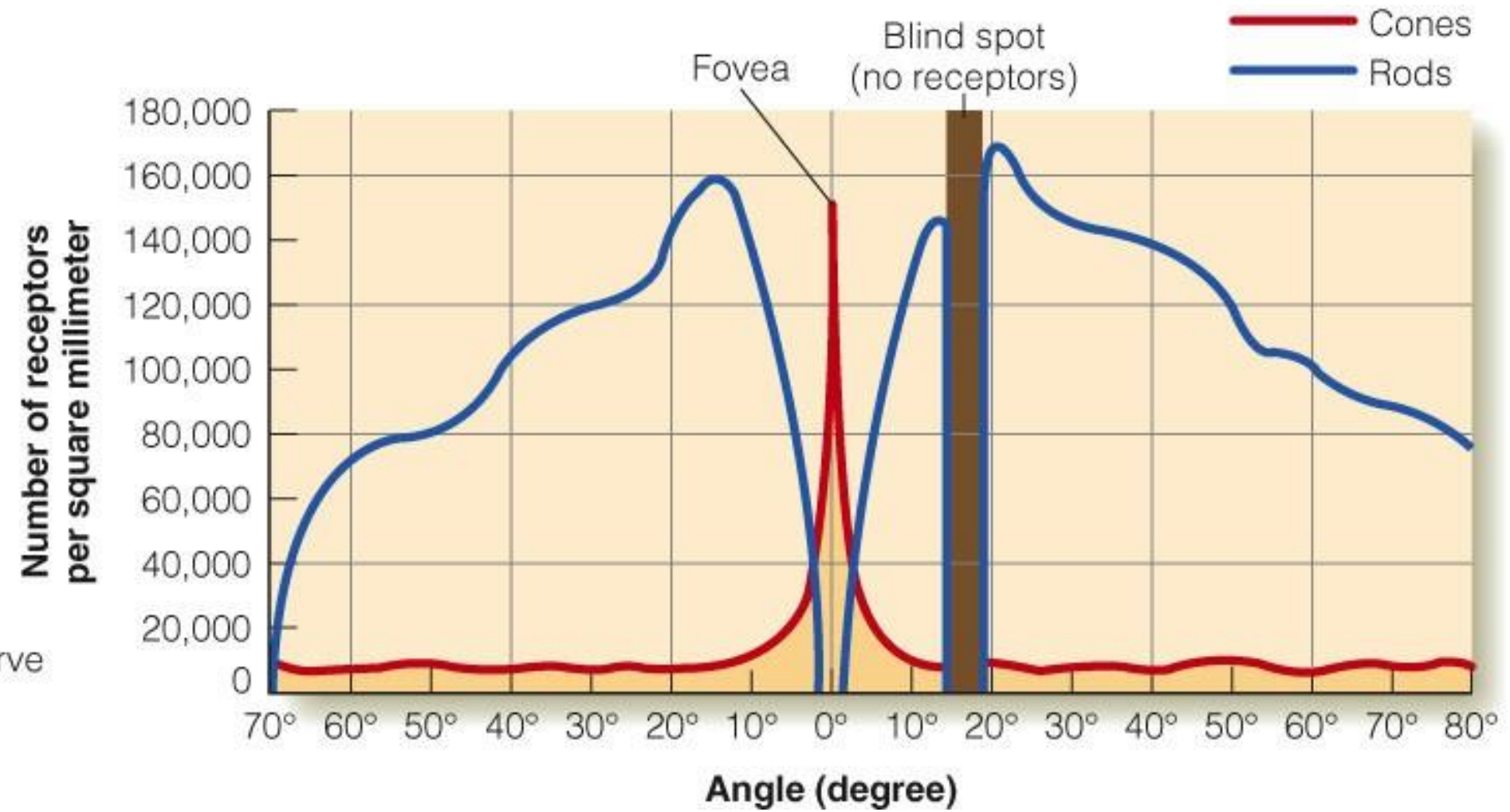
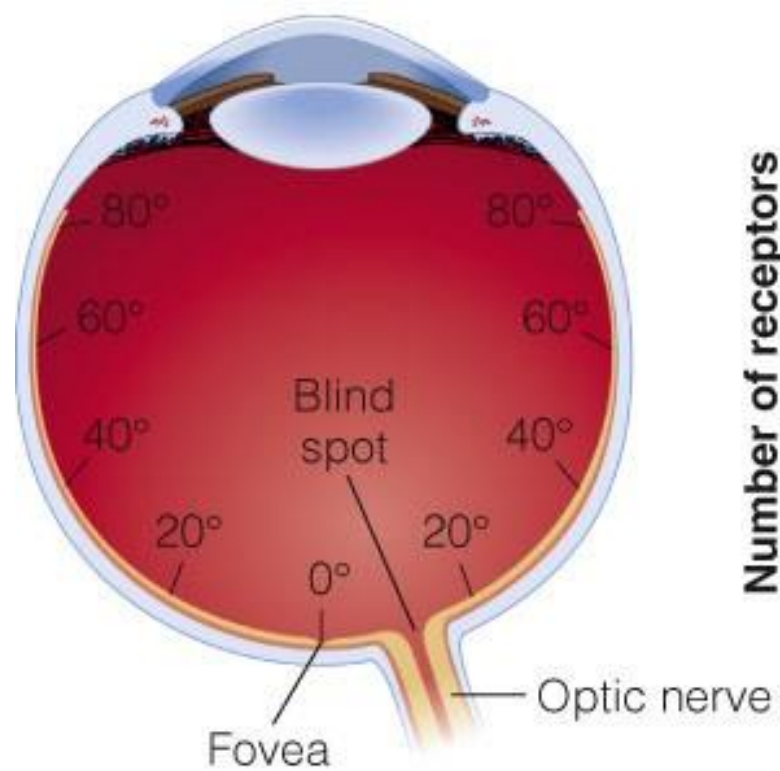
(visualization)



(perception)



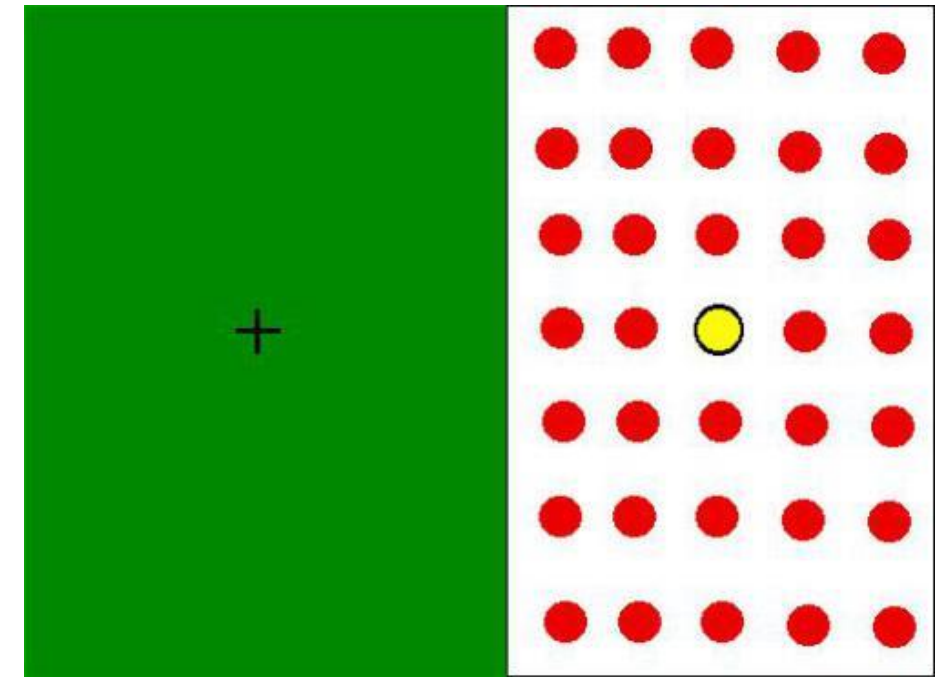
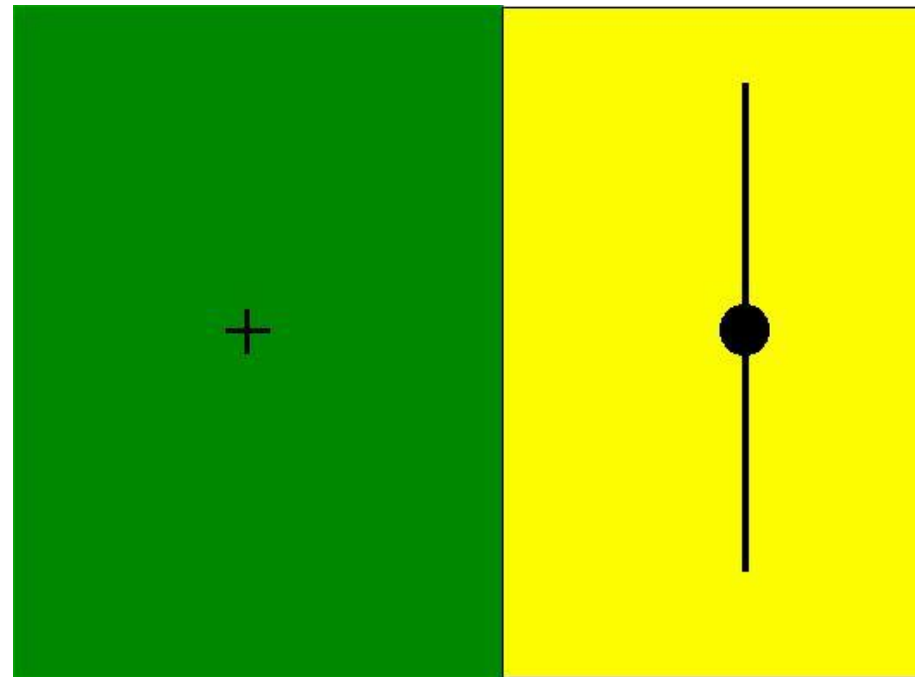
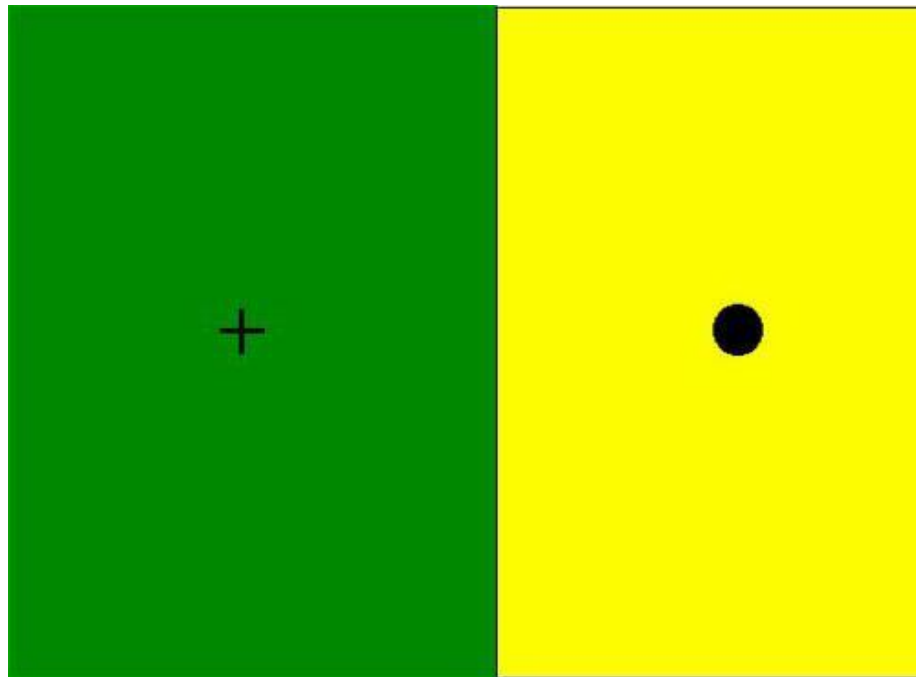
(cognition)

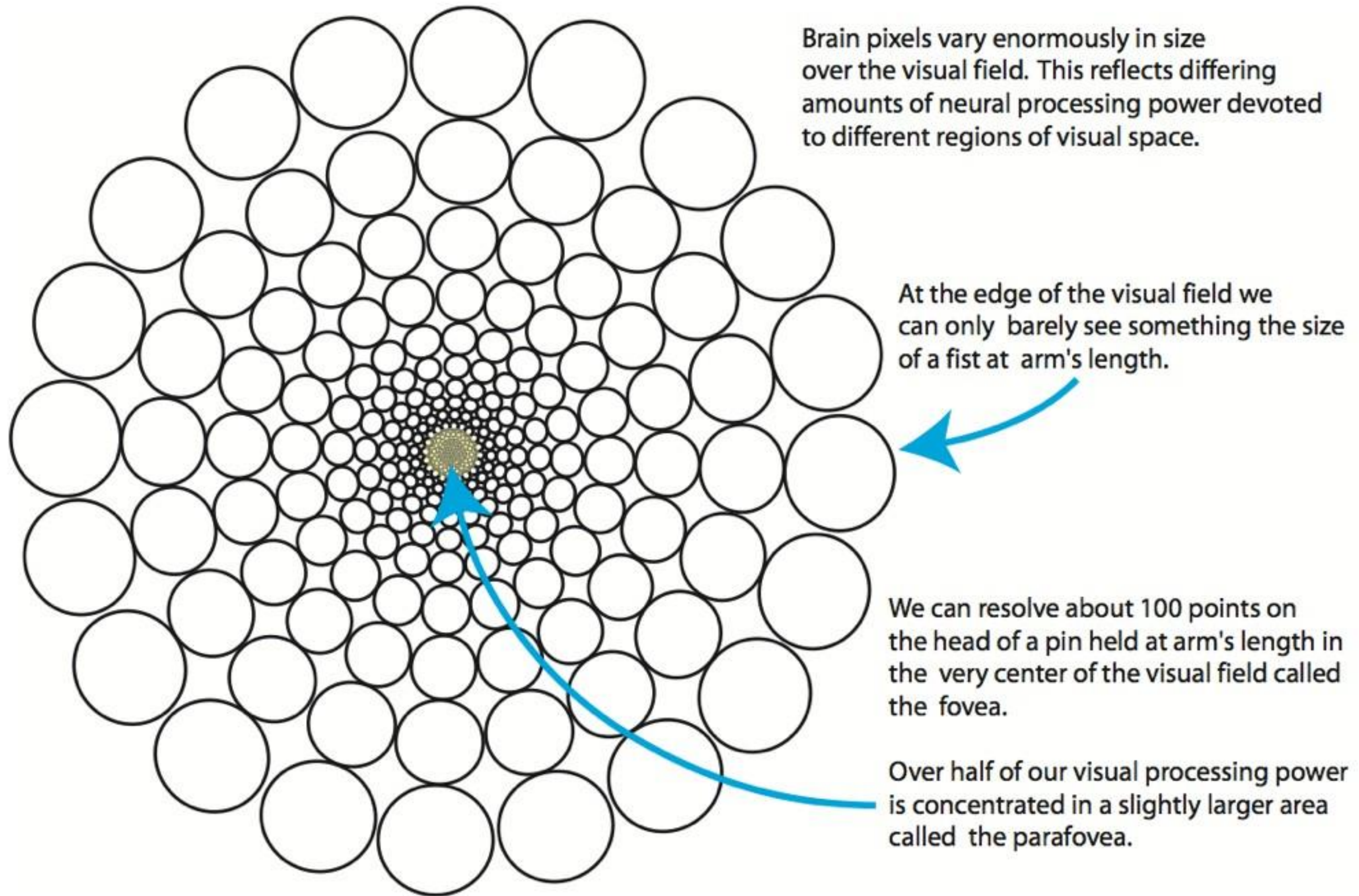


E. B. Goldstein "Sensation and Perception" (Adapted from Lindsay & Norman, 1977)

Blind Spot

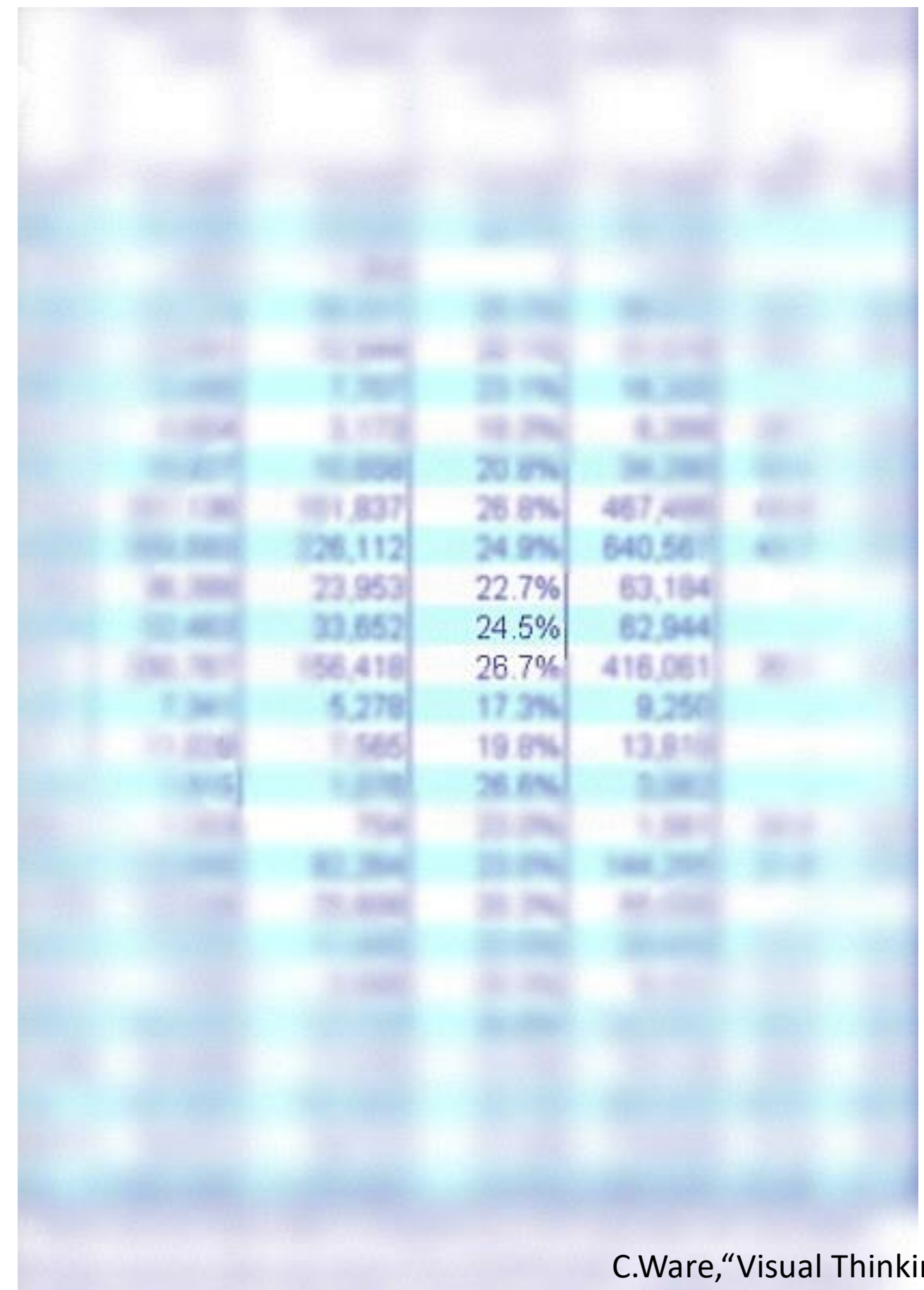
- Close one eye, stare at +, move around until the image changes





Foviation

Foveation is relatively easy to see. The key to recognizing the phenomenon is to stair at a single word **on** the printed page. Then, without moving your gaze, note the blurriness of the surrounding text.

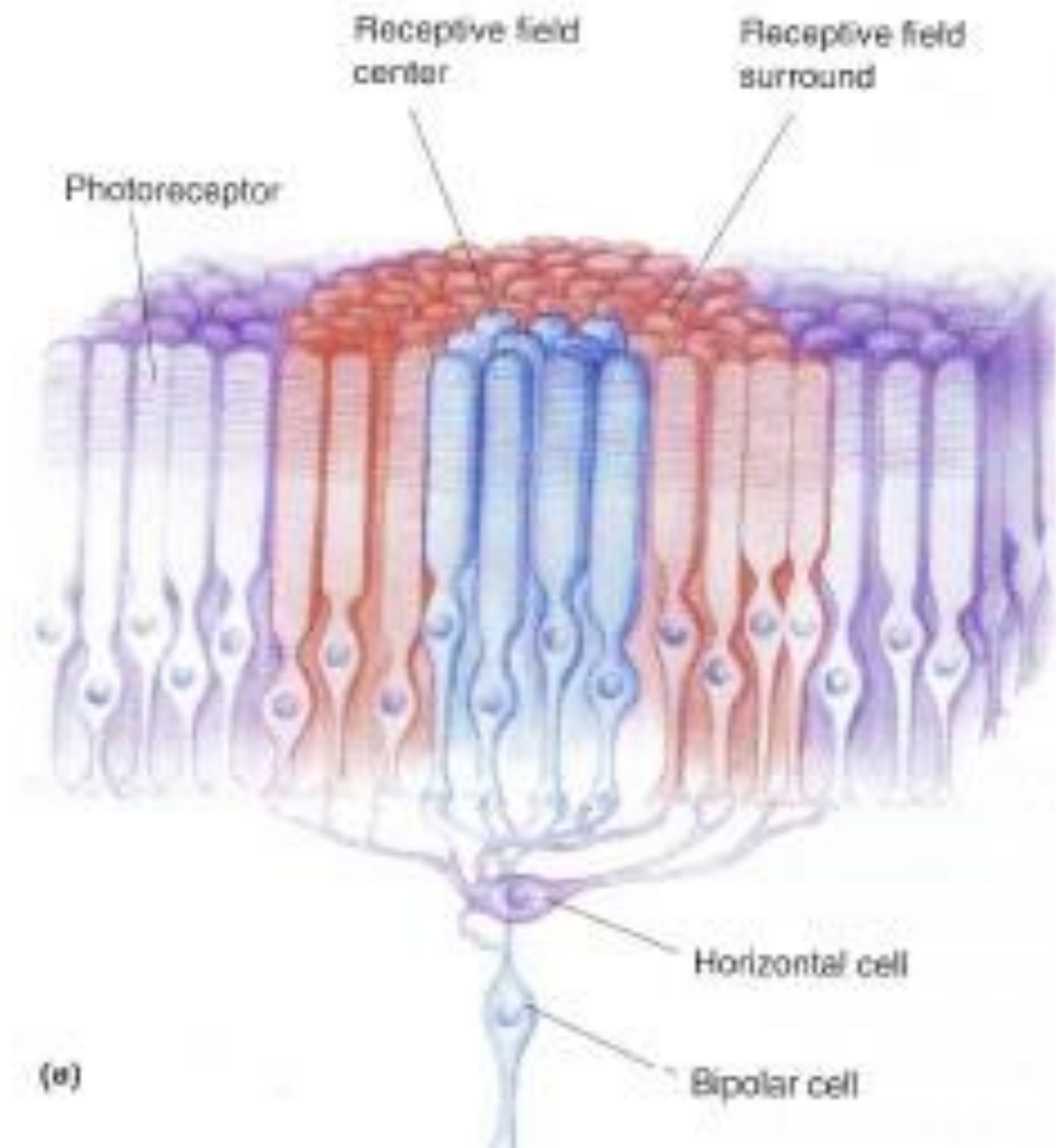


C.Ware, "Visual Thinking for Design"

Takeaway

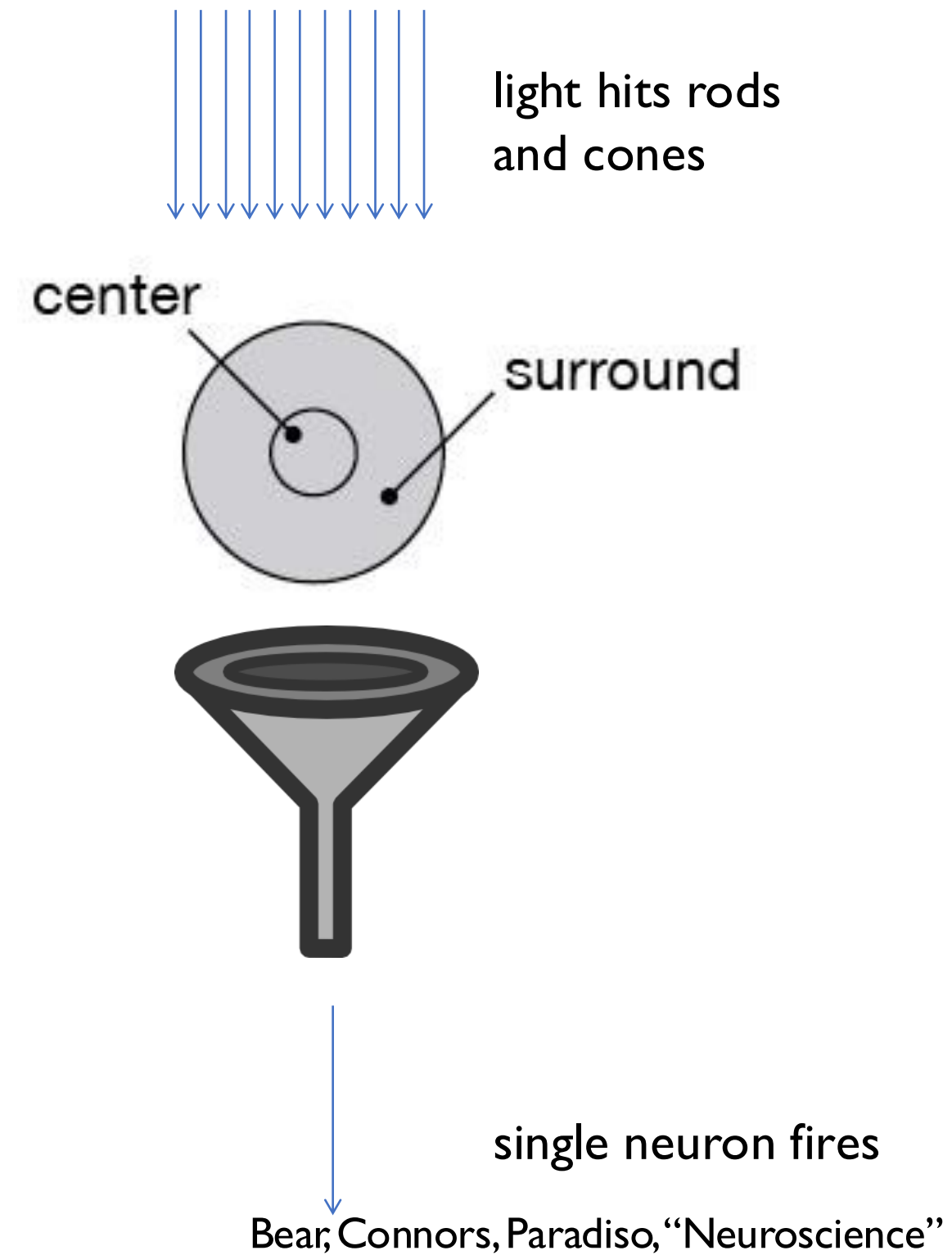
- Our vision at any given moment is relatively limited. Our brain “fills in the missing pieces” using a variety of evolved tools.
- **Be careful placing too much data on the screen. Crisp and clear visualizations will result in the best interpretation.**

receptive field



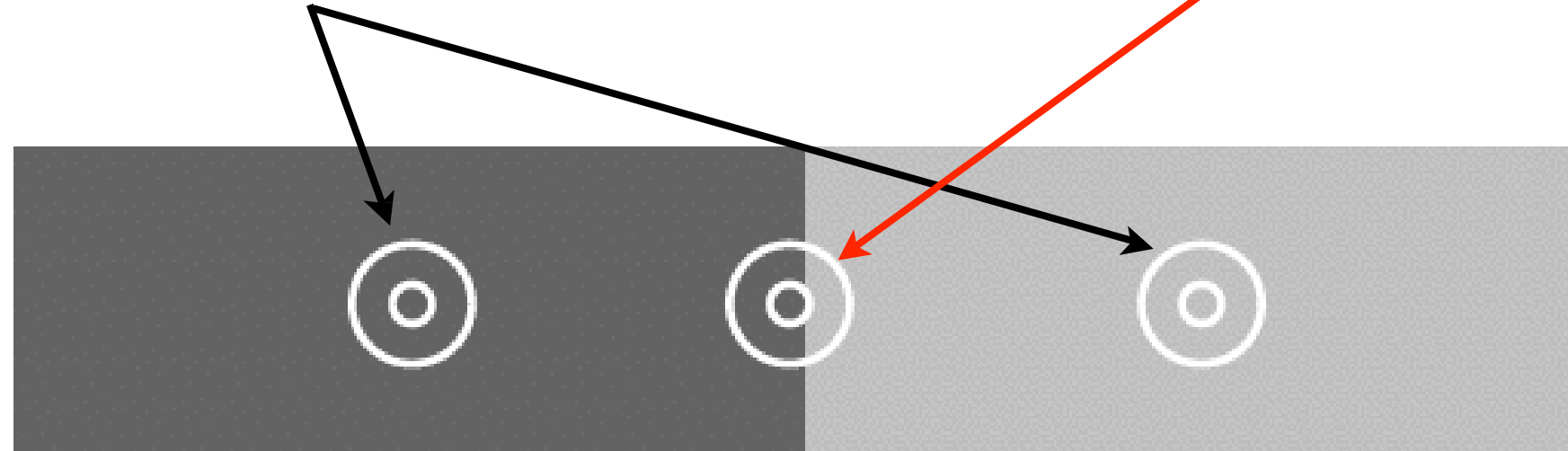
100M rods and cones

1M ganglion cells



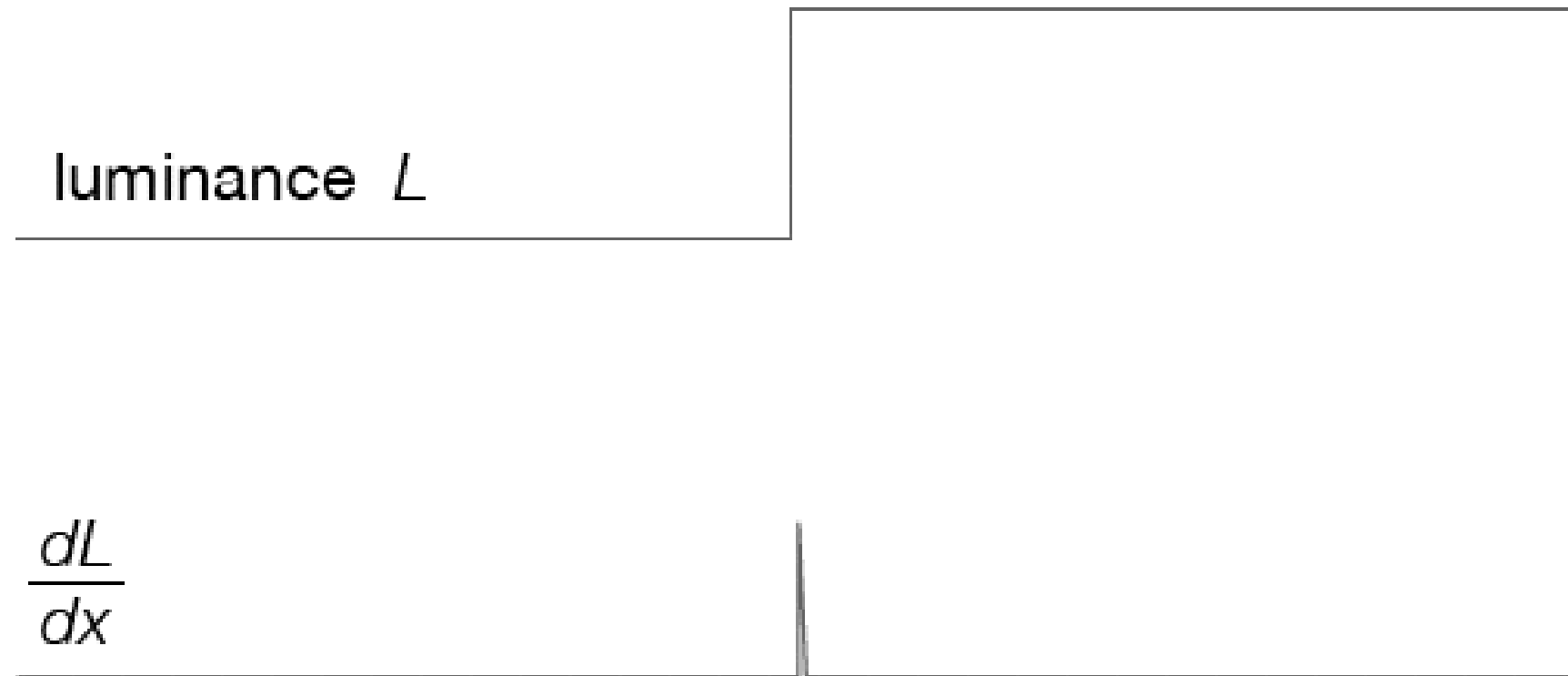
low activity
center and surrounds cancel

**activity increased
or decreased at edges**

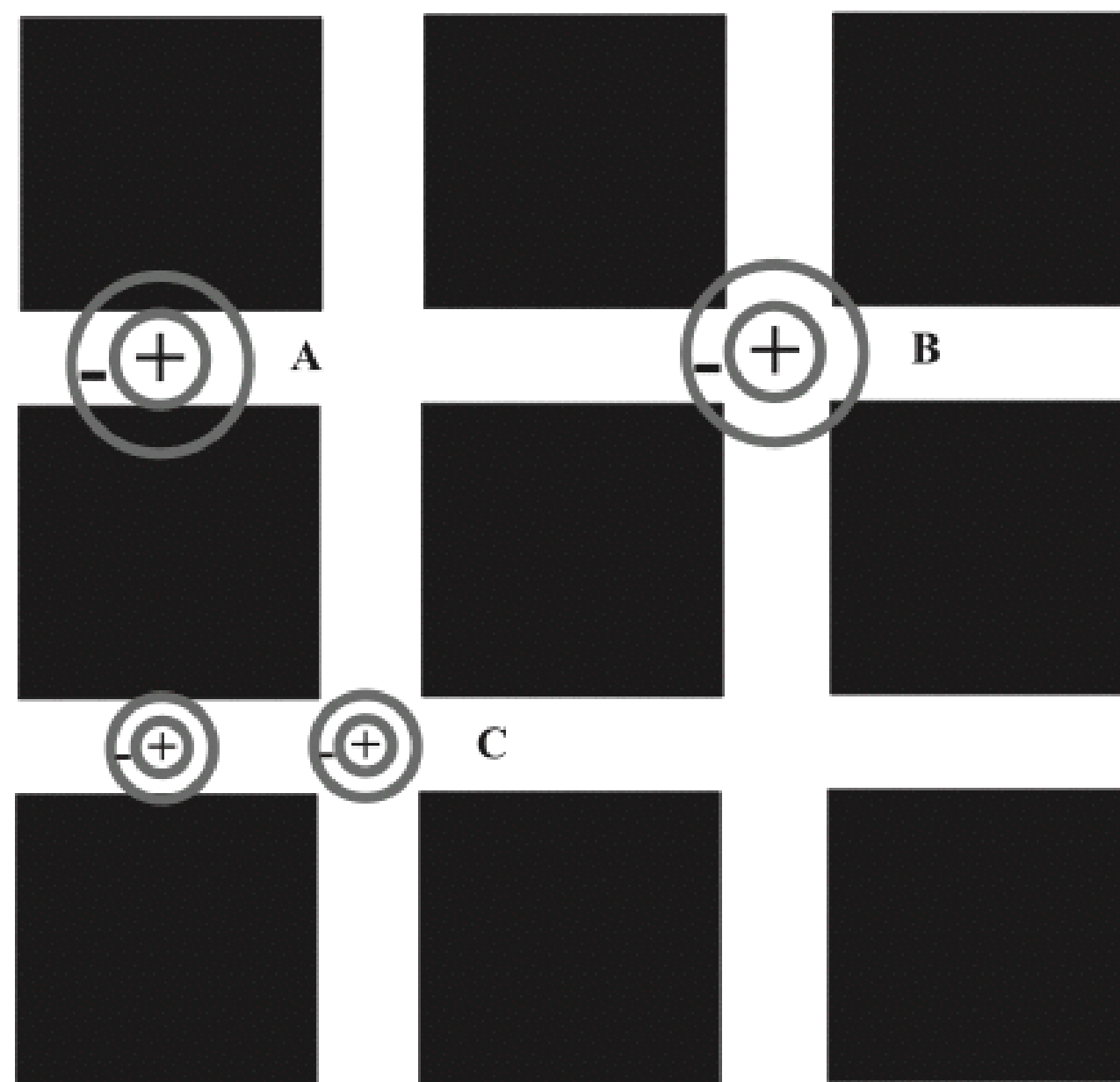


luminance L

$\frac{dL}{dx}$

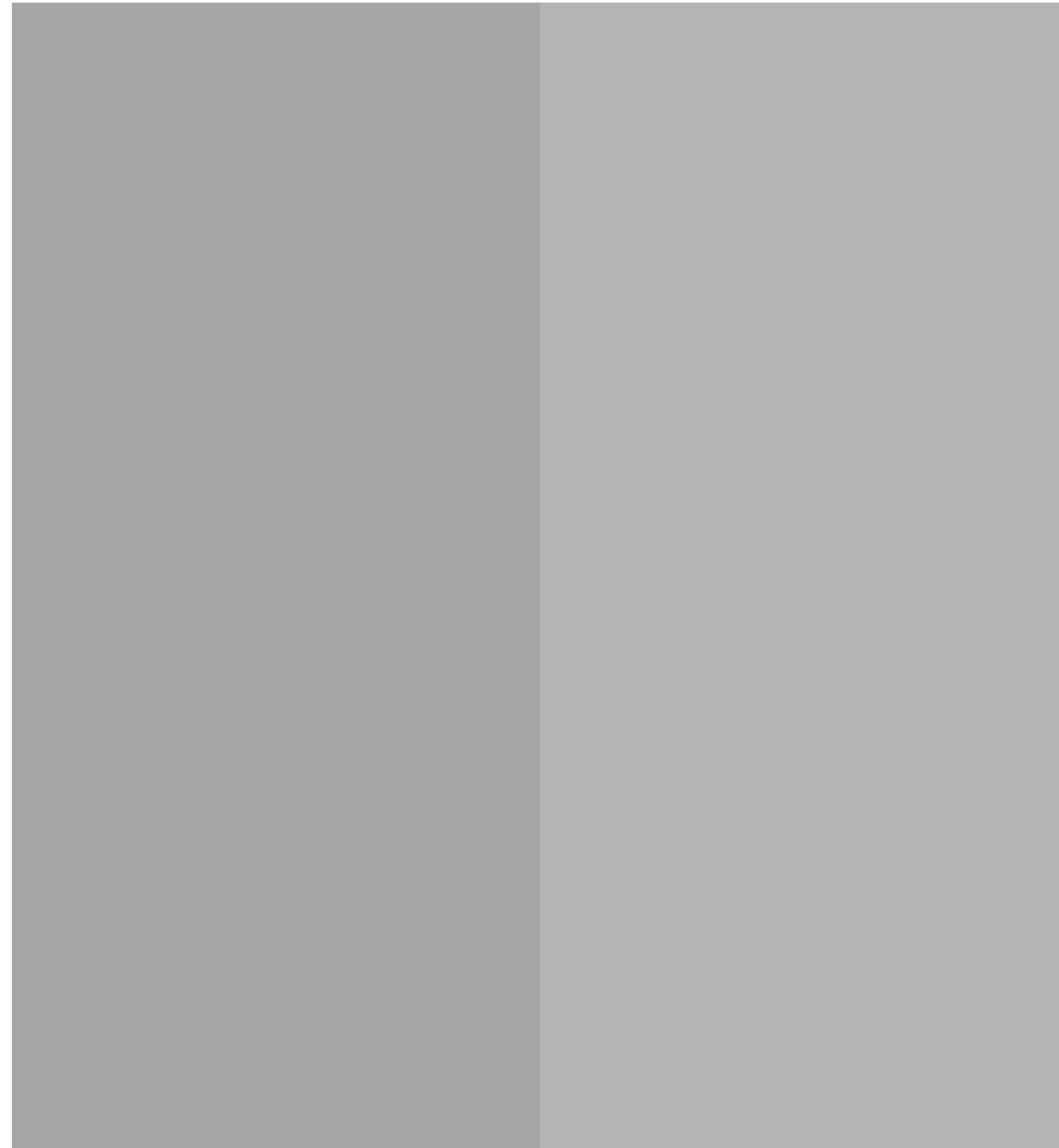


Hermann grid effect

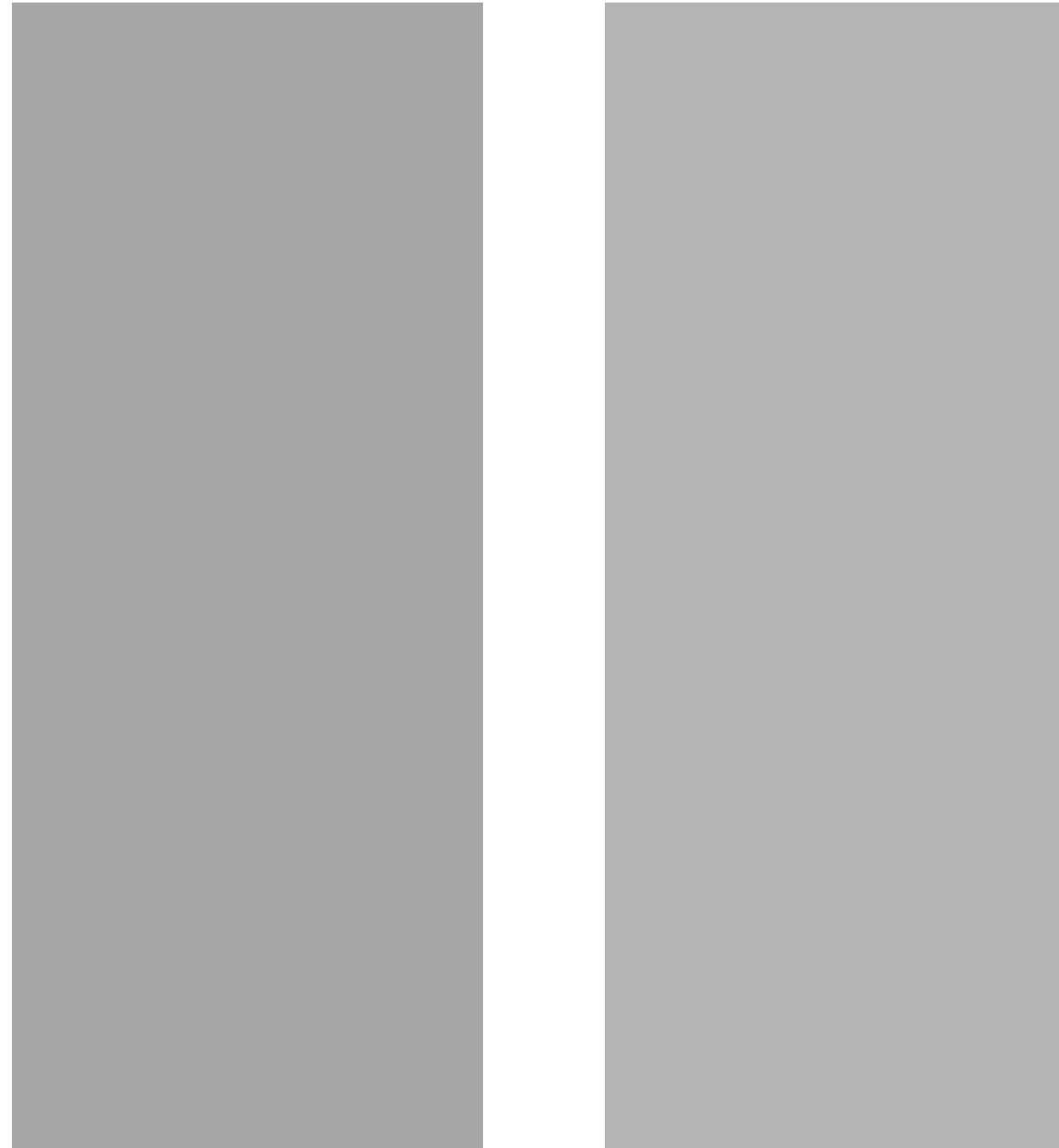


Annie Prud'homme-Généreux

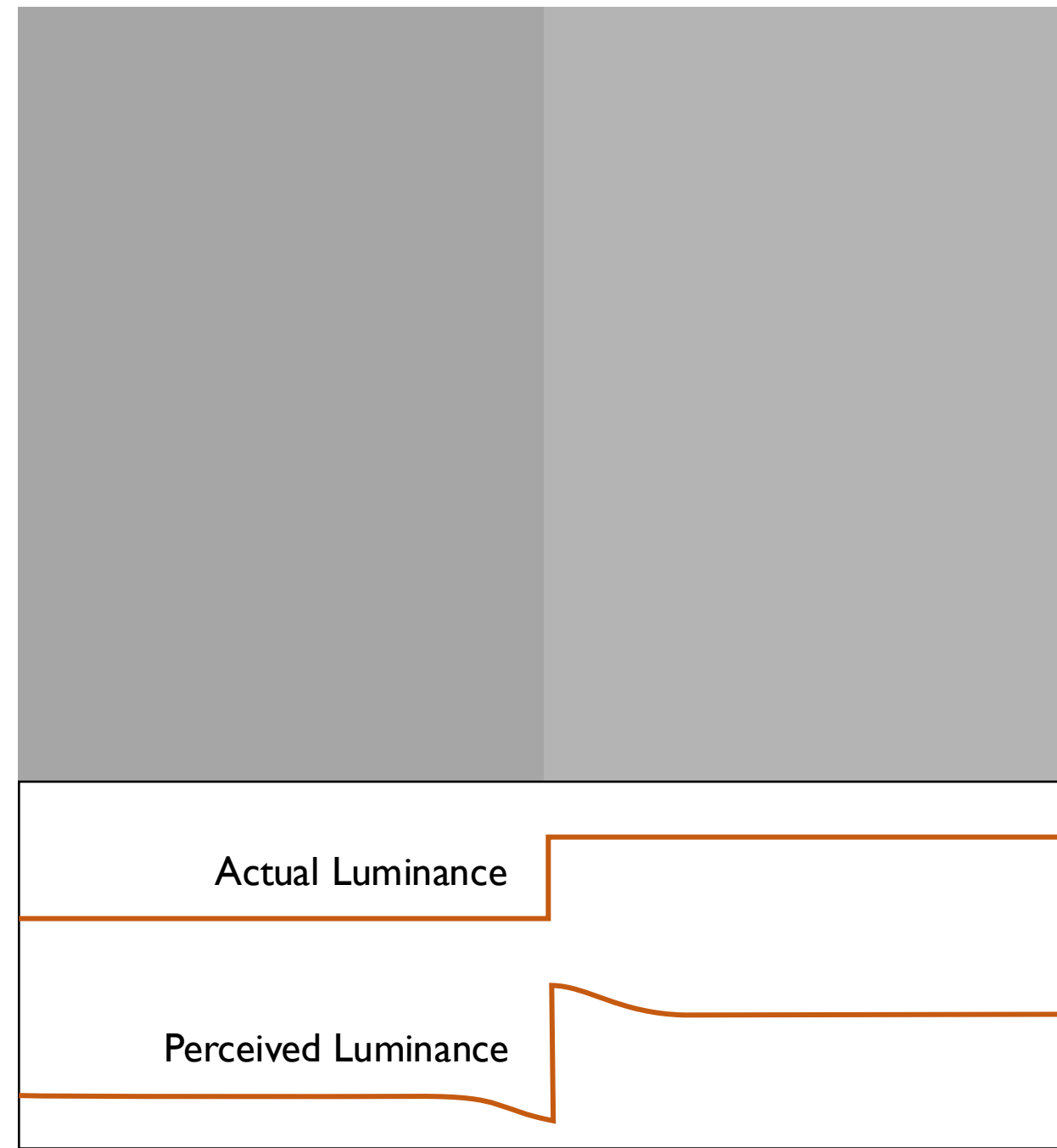
Cornsweet Illusion



Cornsweet Illusion



Cornsweet Illusion



Takeaway

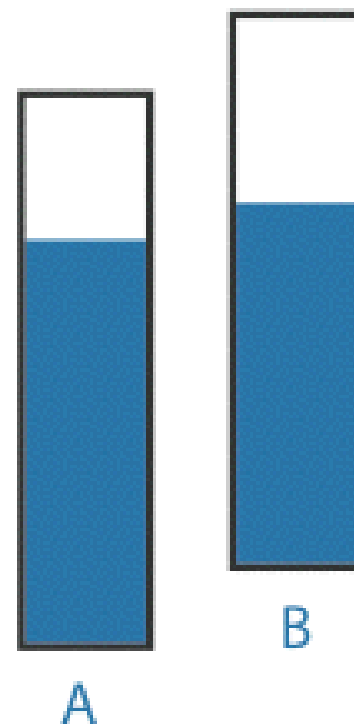
- Our visual system is attracted to edges and is sensitive to differences, not absolute values.
- **Maximize the contrast with the background if the outlines of shapes are important.**

WEBER'S LAW

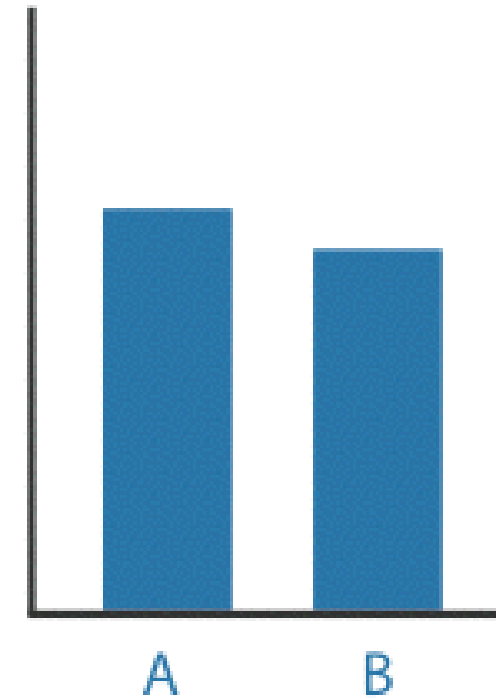
- we judge based on relative, not absolute, differences



Unframed
Unaligned



Framed
Unaligned

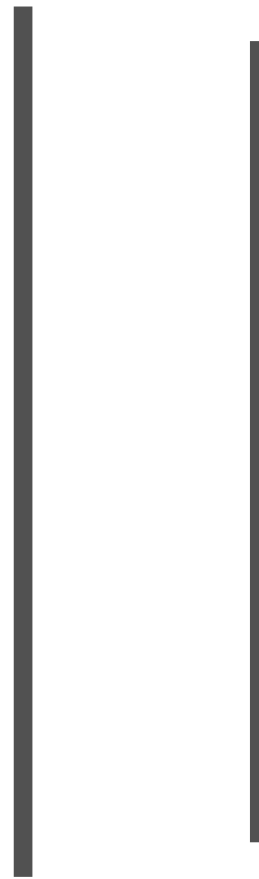


Unframed
Aligned

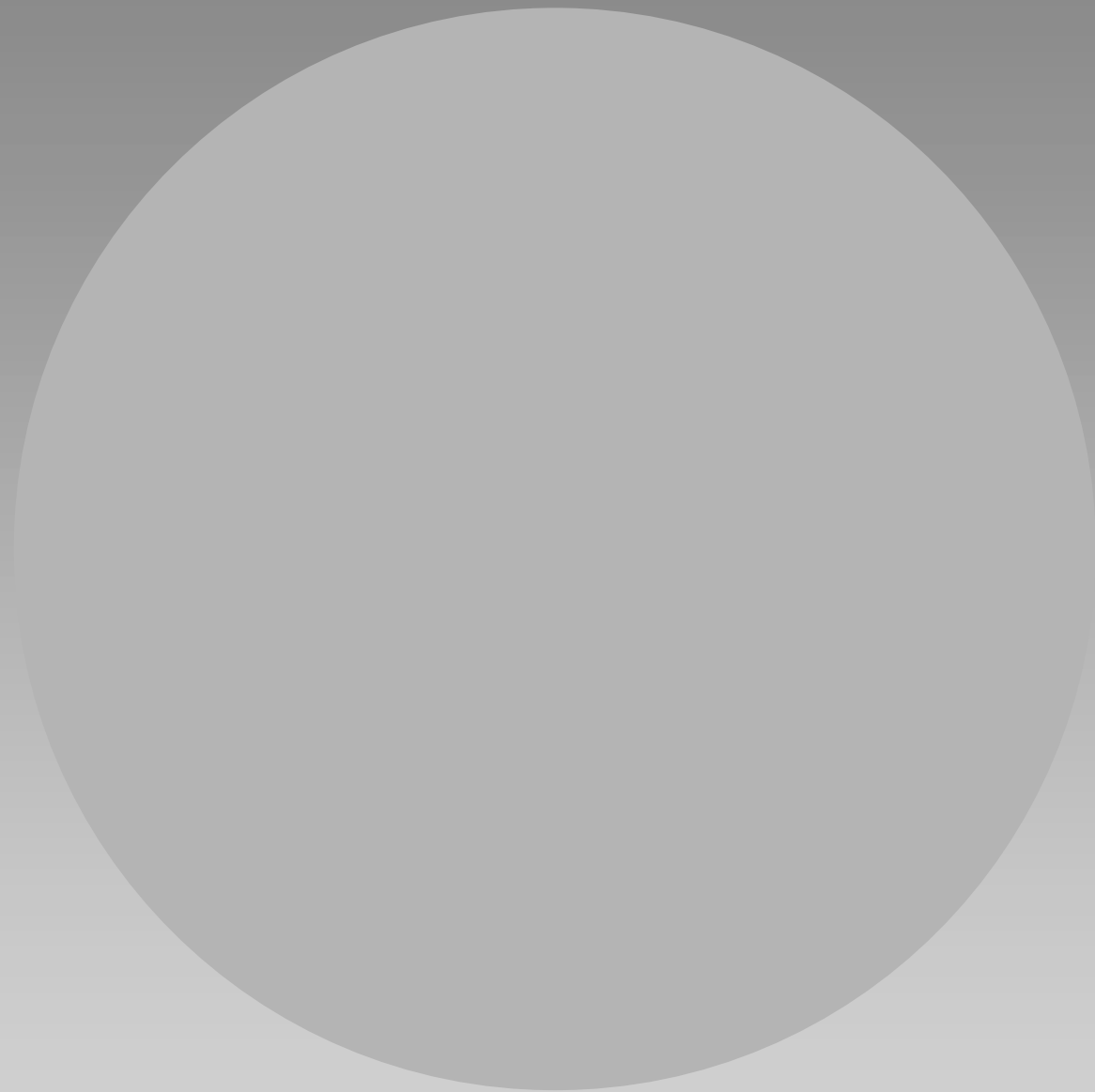
AXIS OF ALIGNMENT



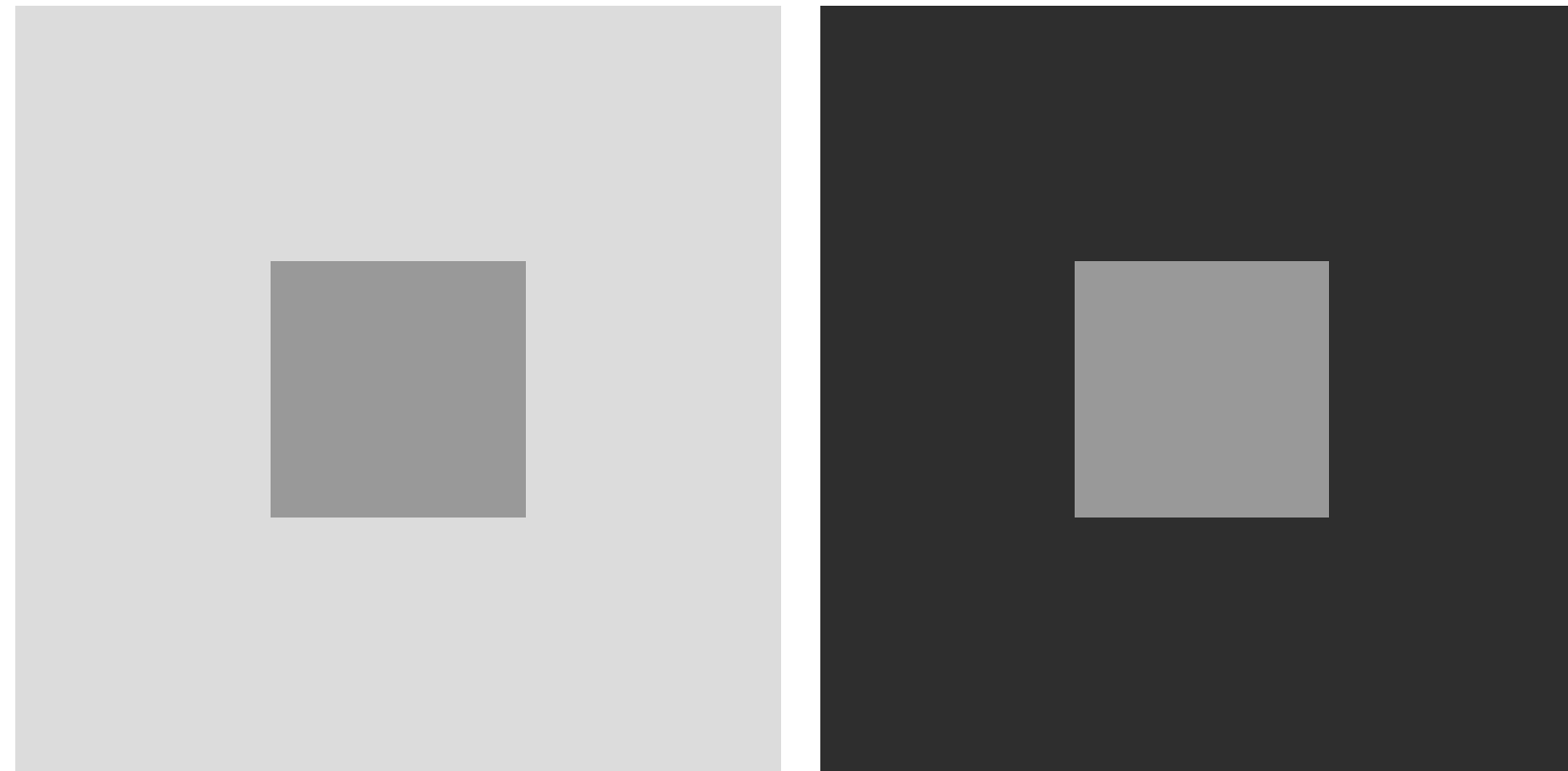
AXIS OF ALIGNMENT



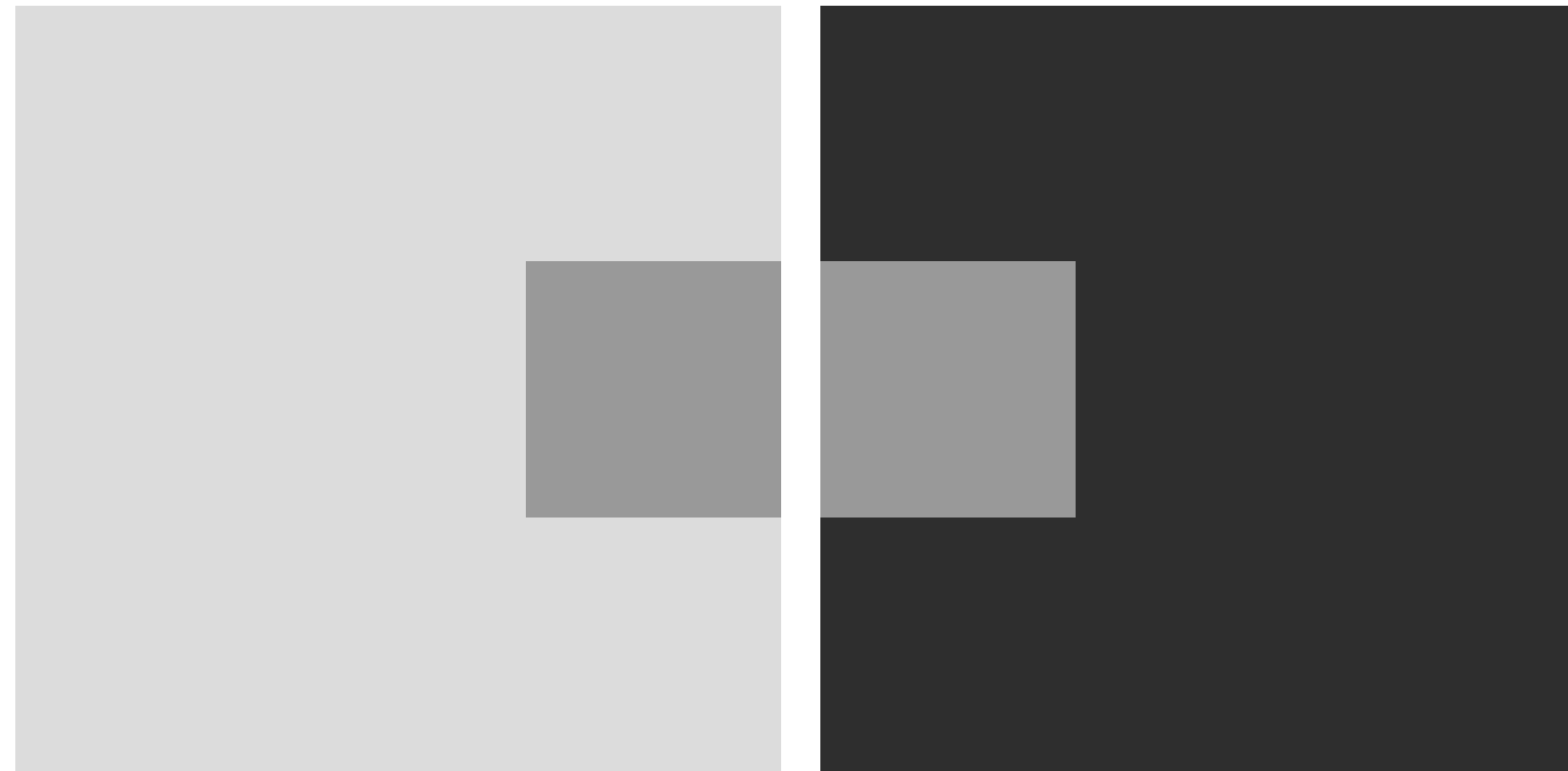
simultaneous contrast



SIMULTANEOUS CONTRAST



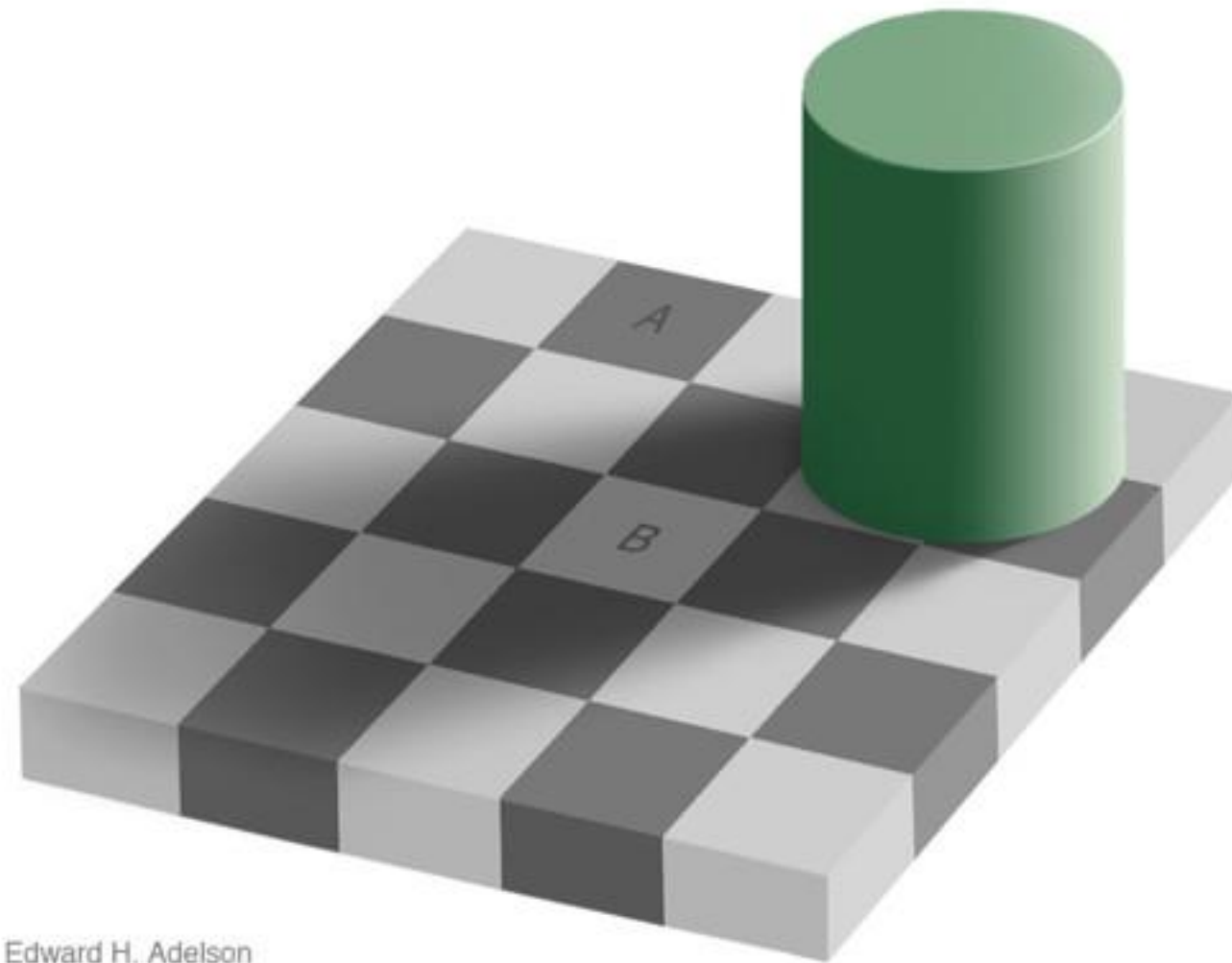
SIMULTANEOUS CONTRAST



SIMULTANEOUS CONTRAST



SIMULTANEOUS CONTRAST



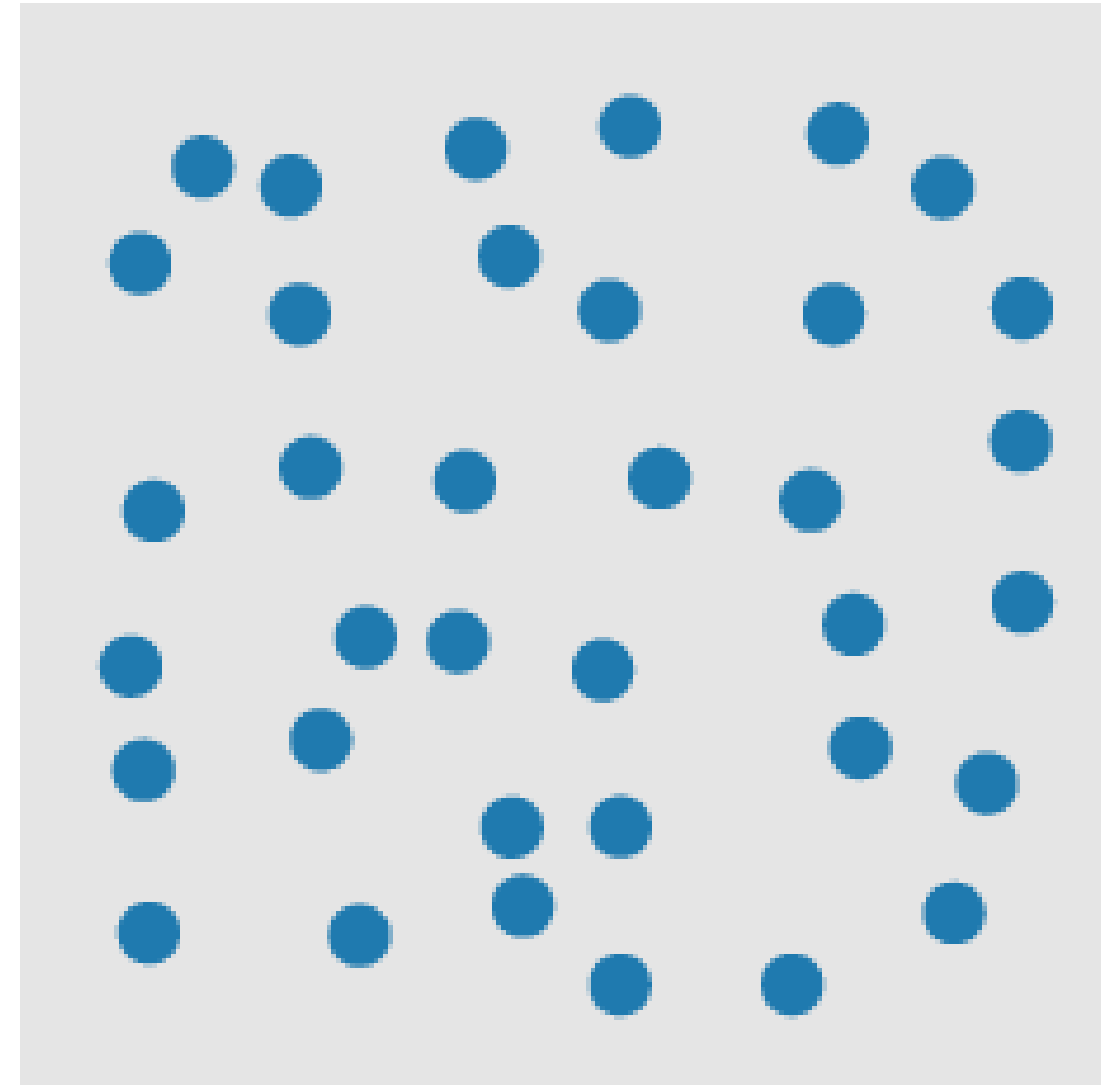
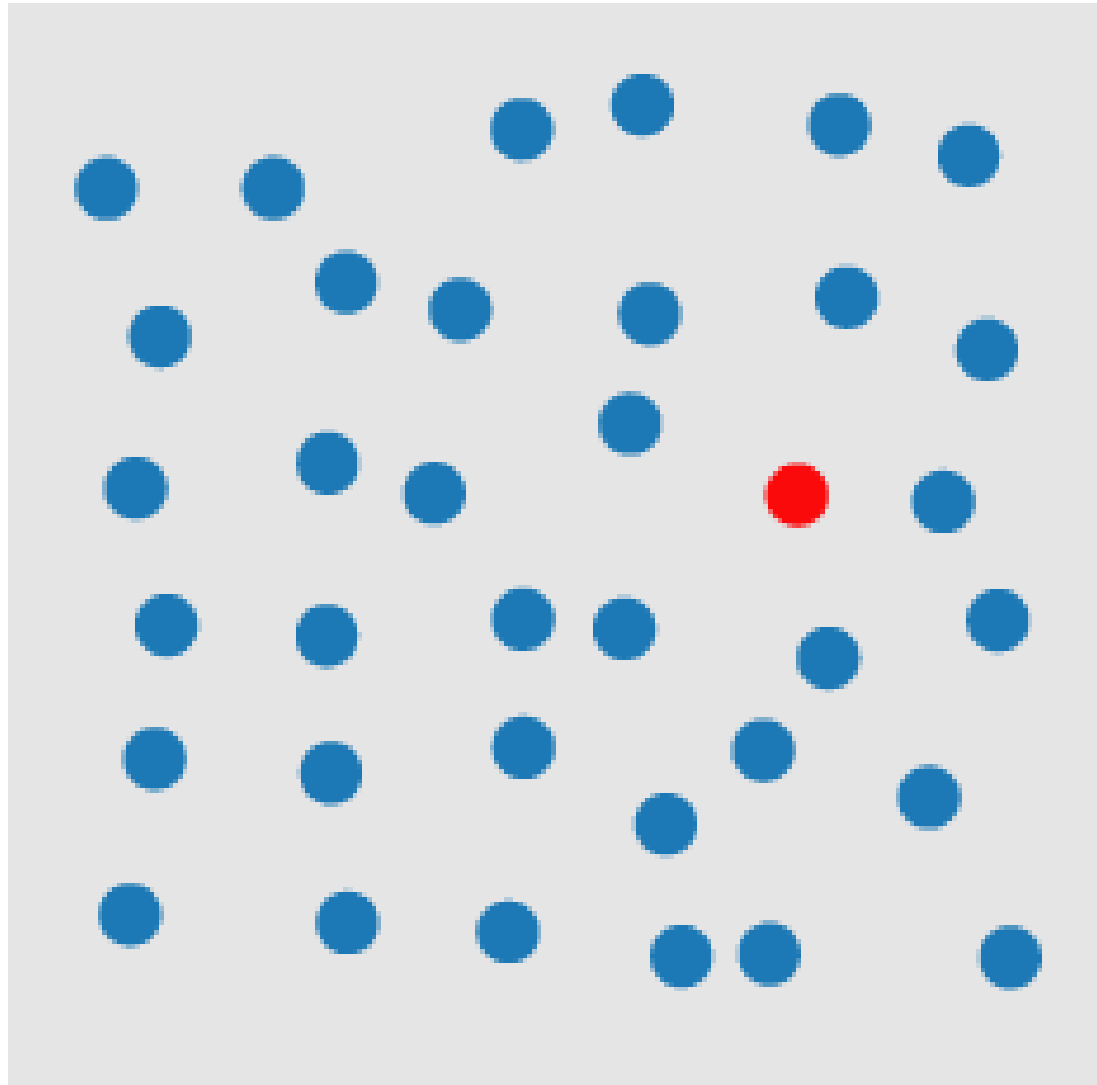
Edward H. Adelson

http://persci.mit.edu/media/gallery/checkershadow_double_full.jpg

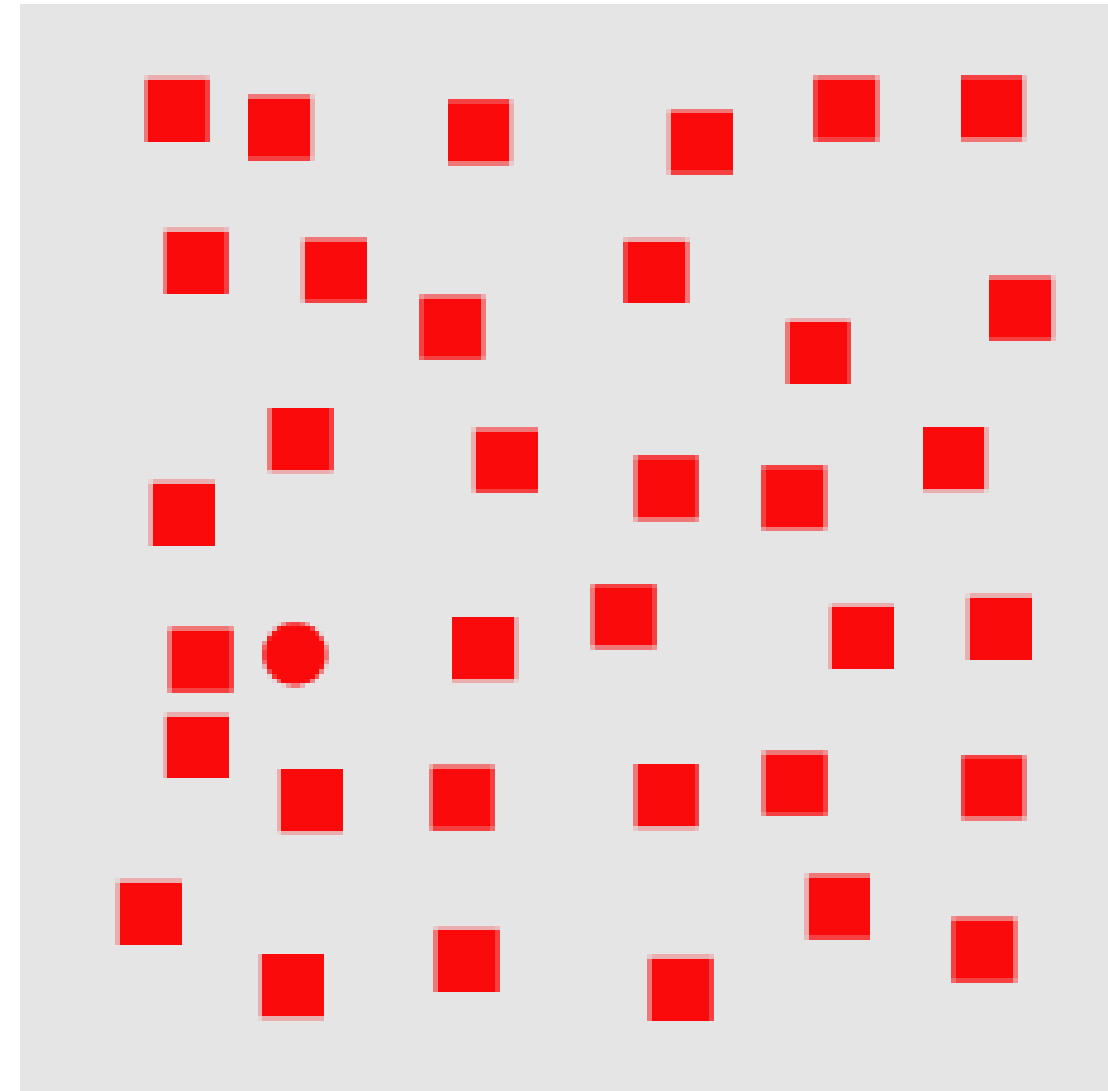
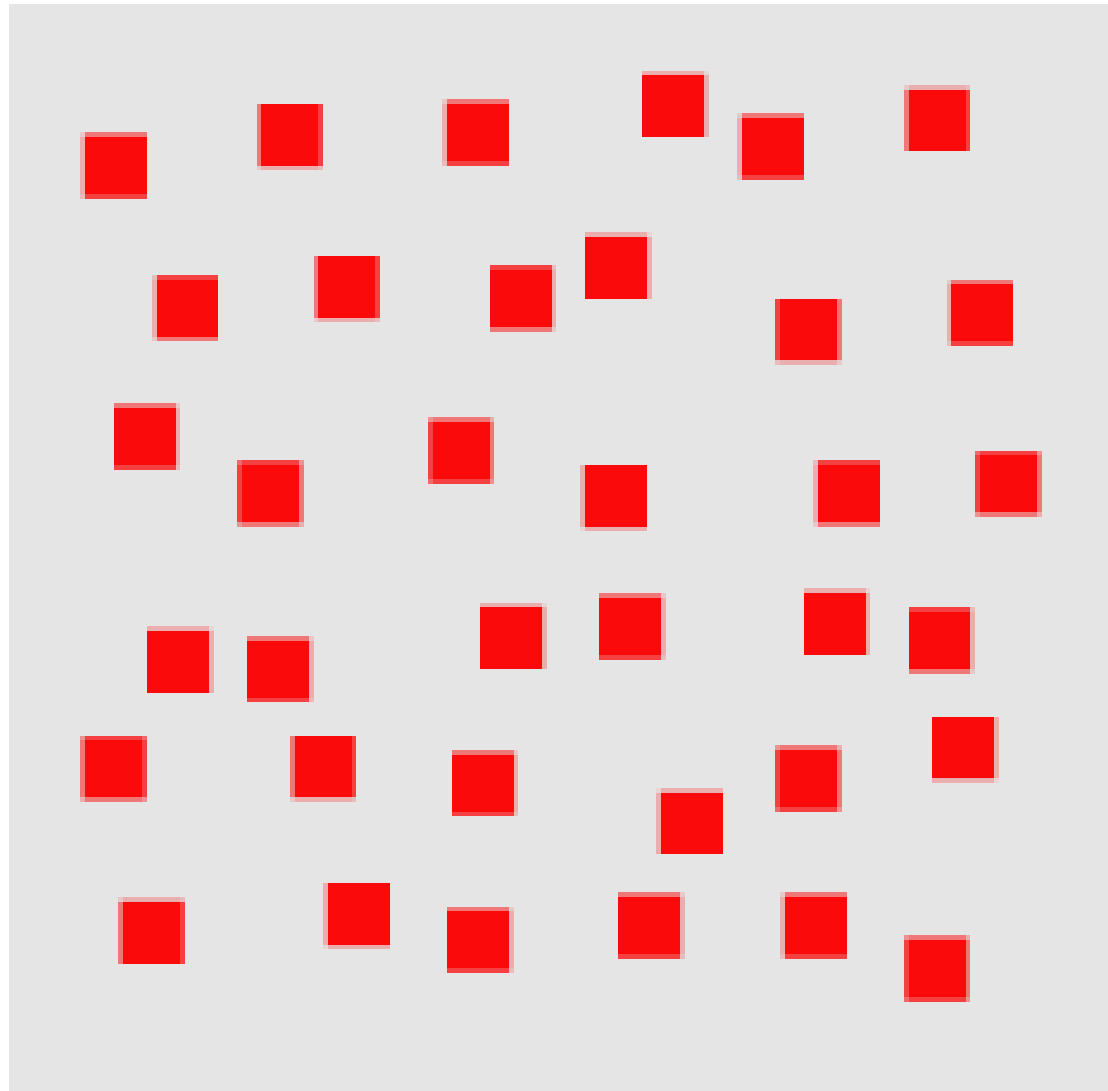
Takeaway

- We have a strong propensity to assume our judgments are absolute, when in fact they are generally relative to the local context.
- **Do your best to not place data in difficult contexts. Choose position and orientation of objects carefully.**

Pick the outlier



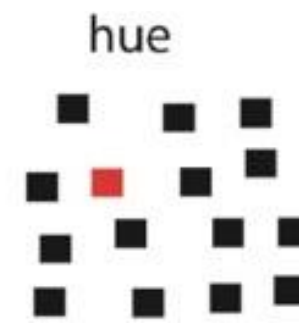
Pick the outlier



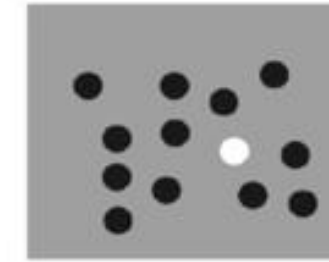
Healy 2007

BASIC POPOUT CHANNELS

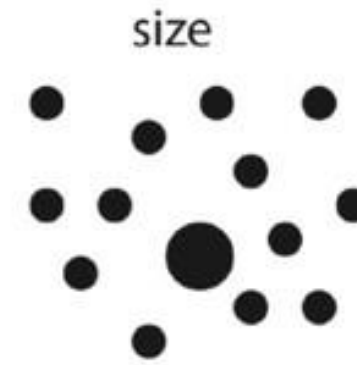
Color



lightness



Elementary
shape



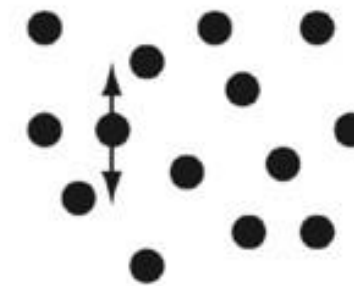
elongation



orientation



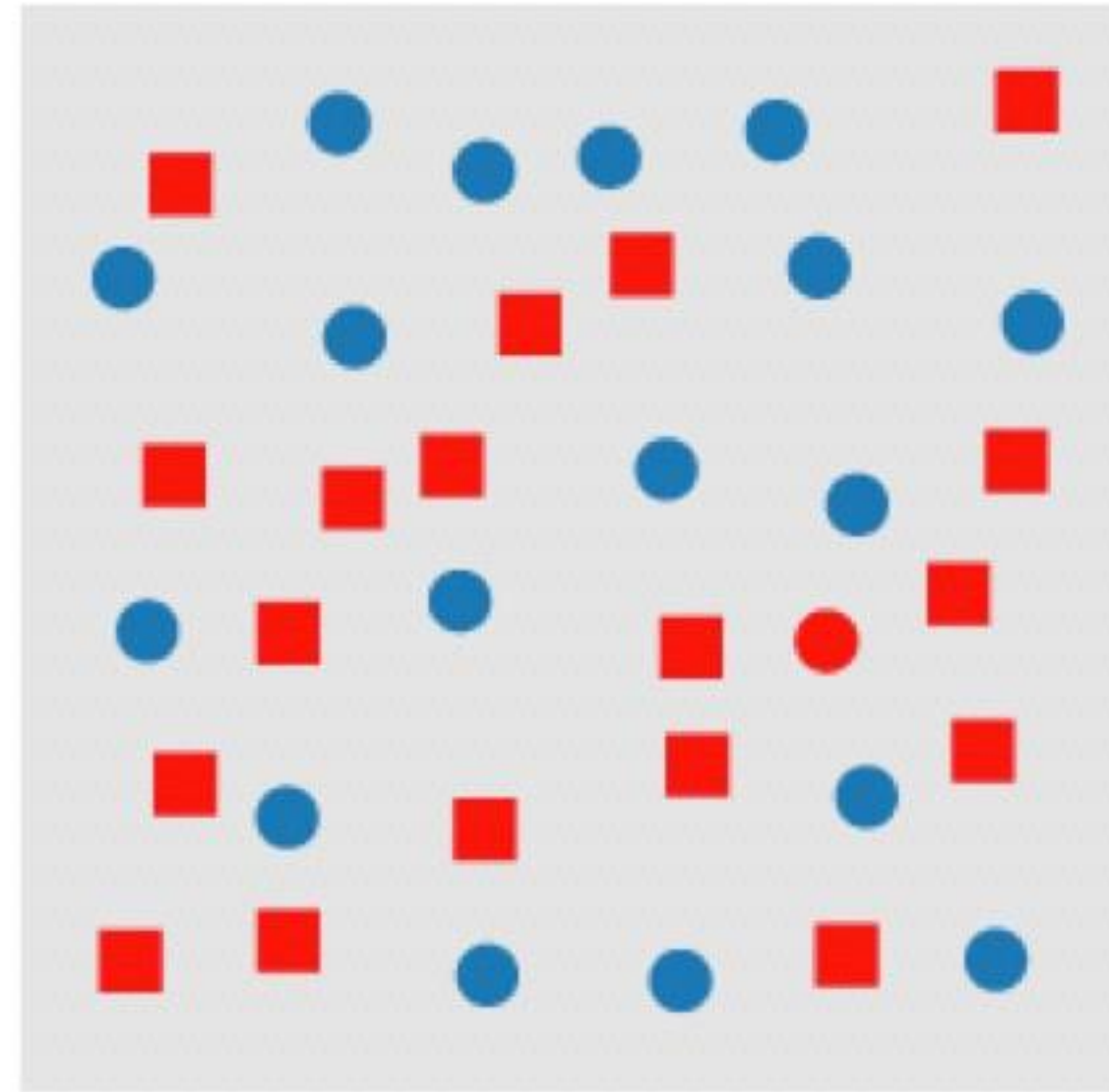
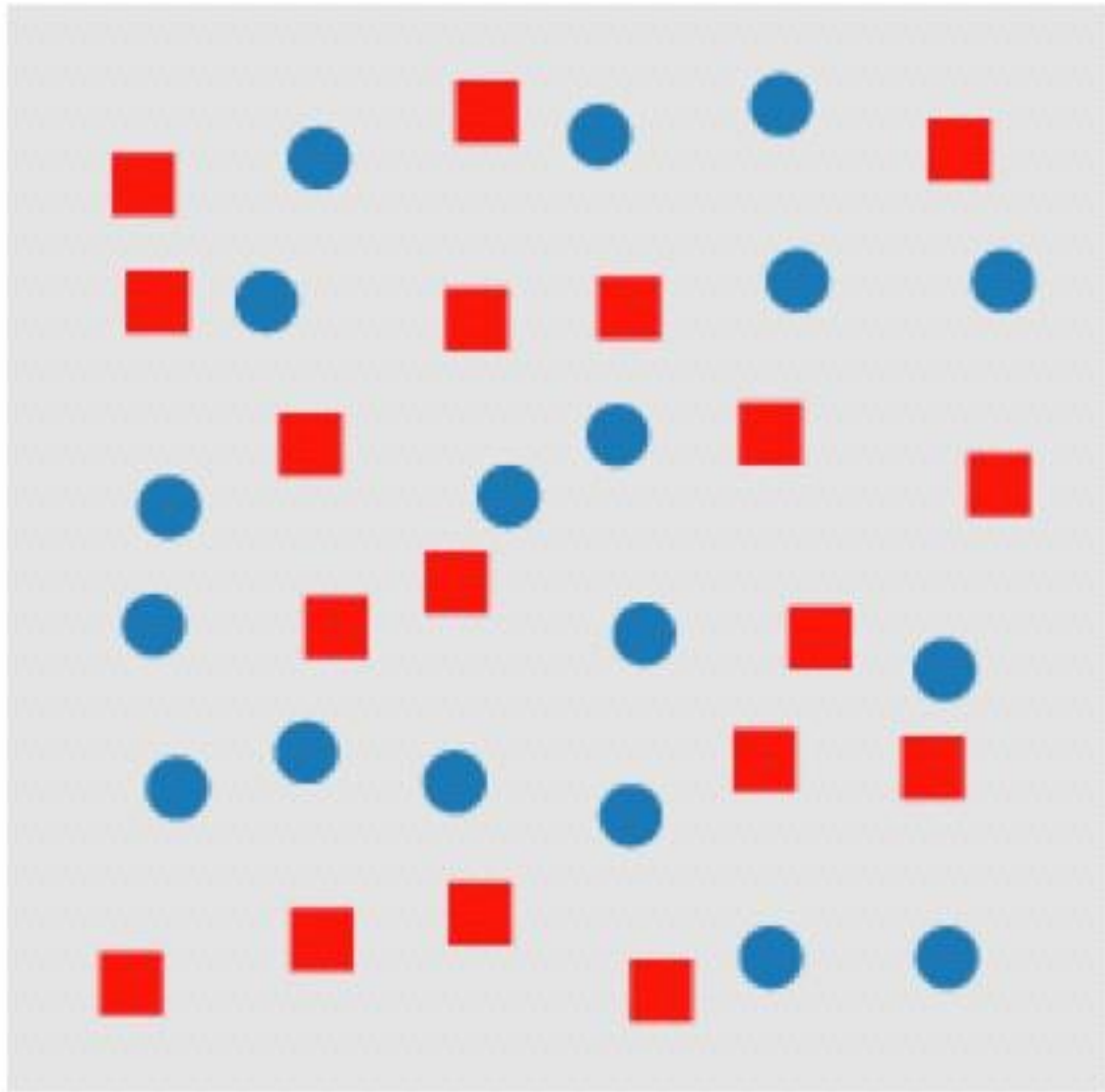
Motion



Spatial
grouping

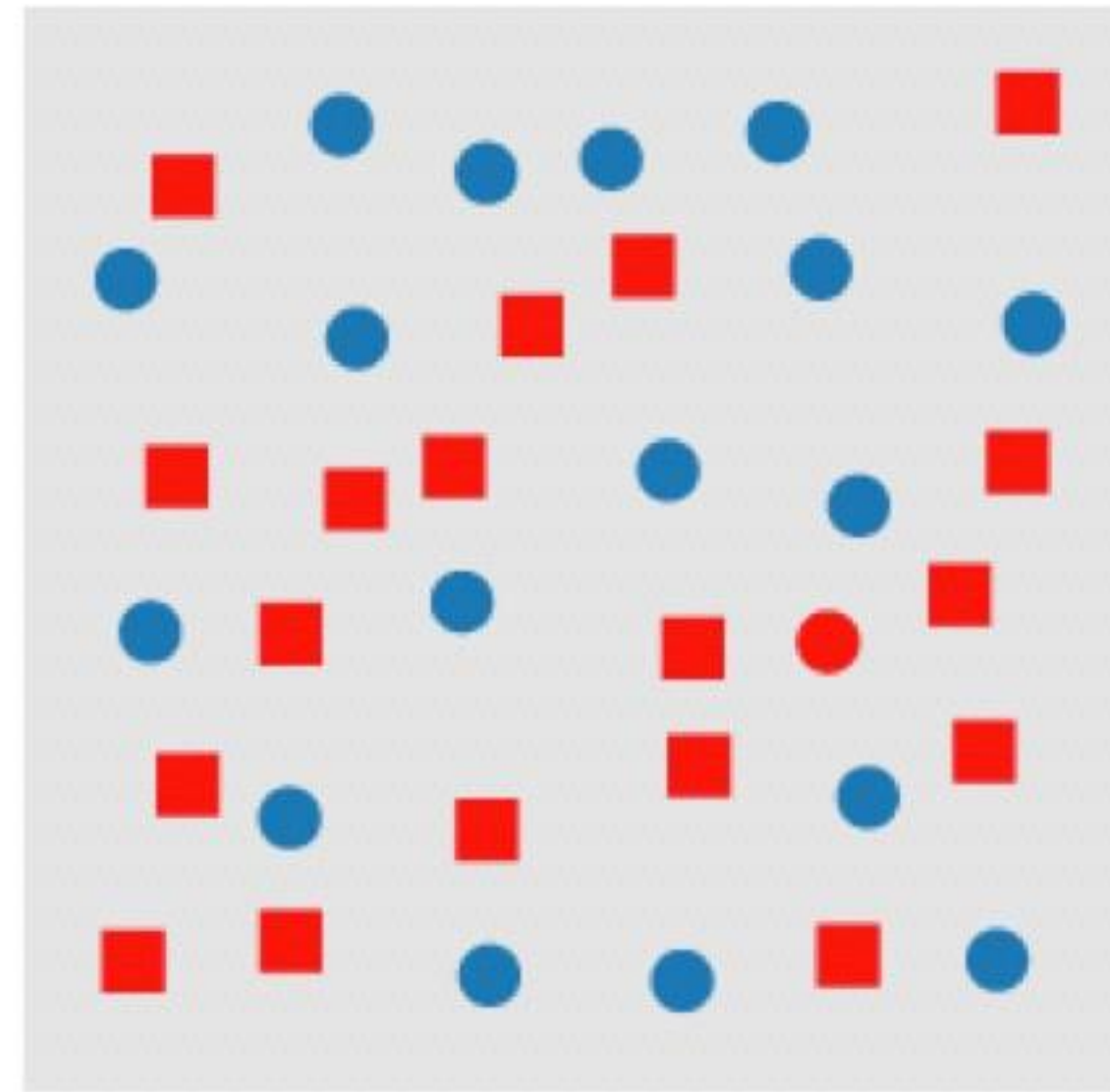
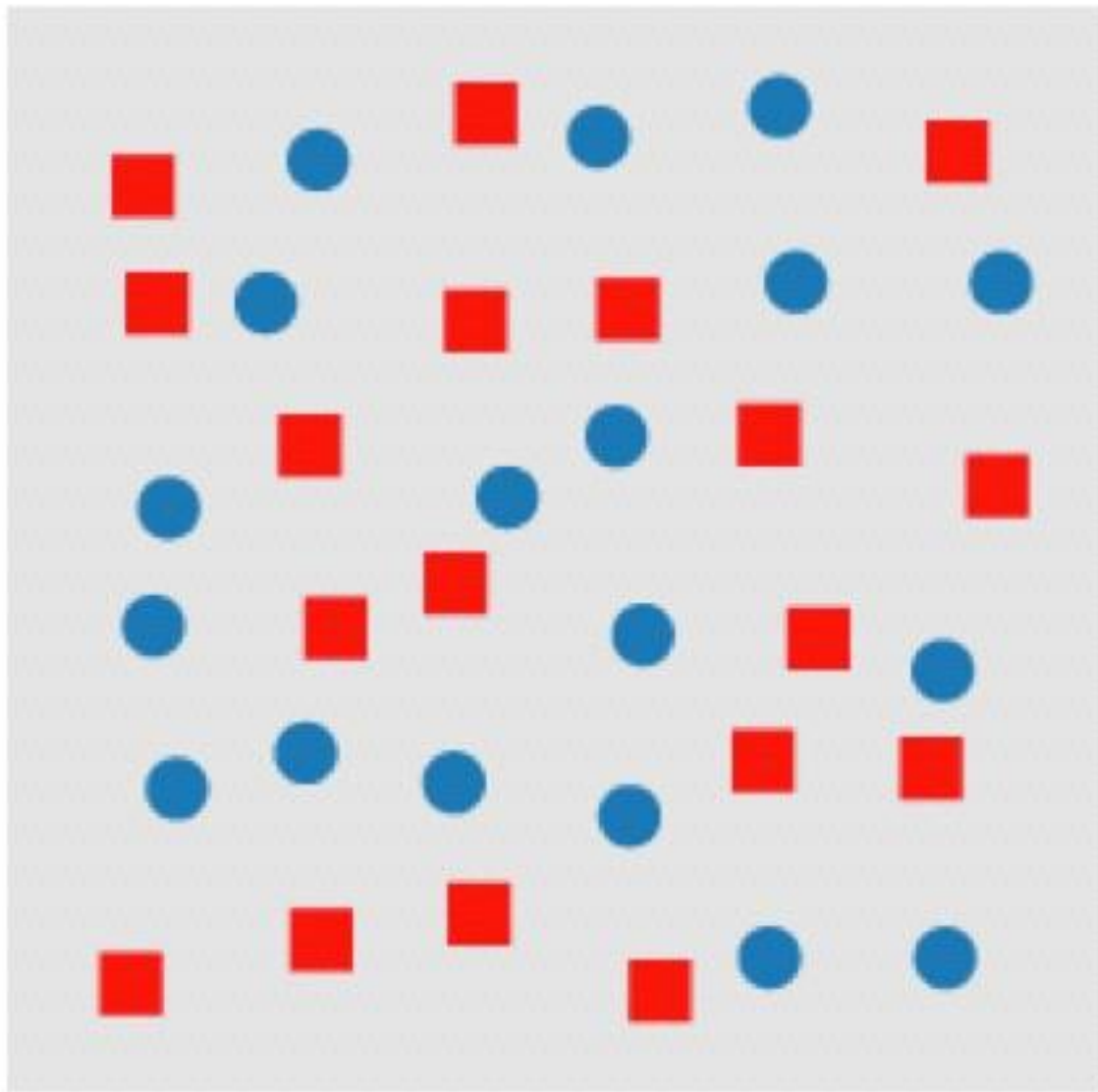


Pick the outlier



Healy 2007

CONJUNCTION (or, why to use a single channel at a time)



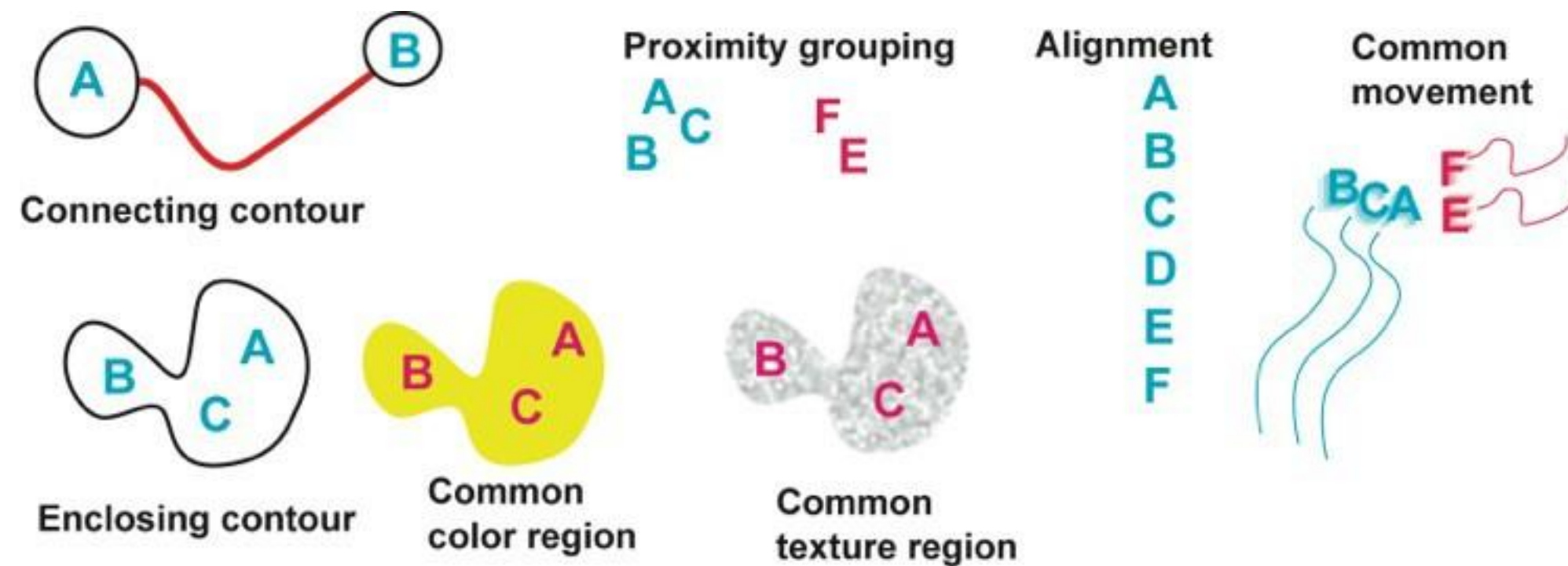
Healy 2007

Takeaway

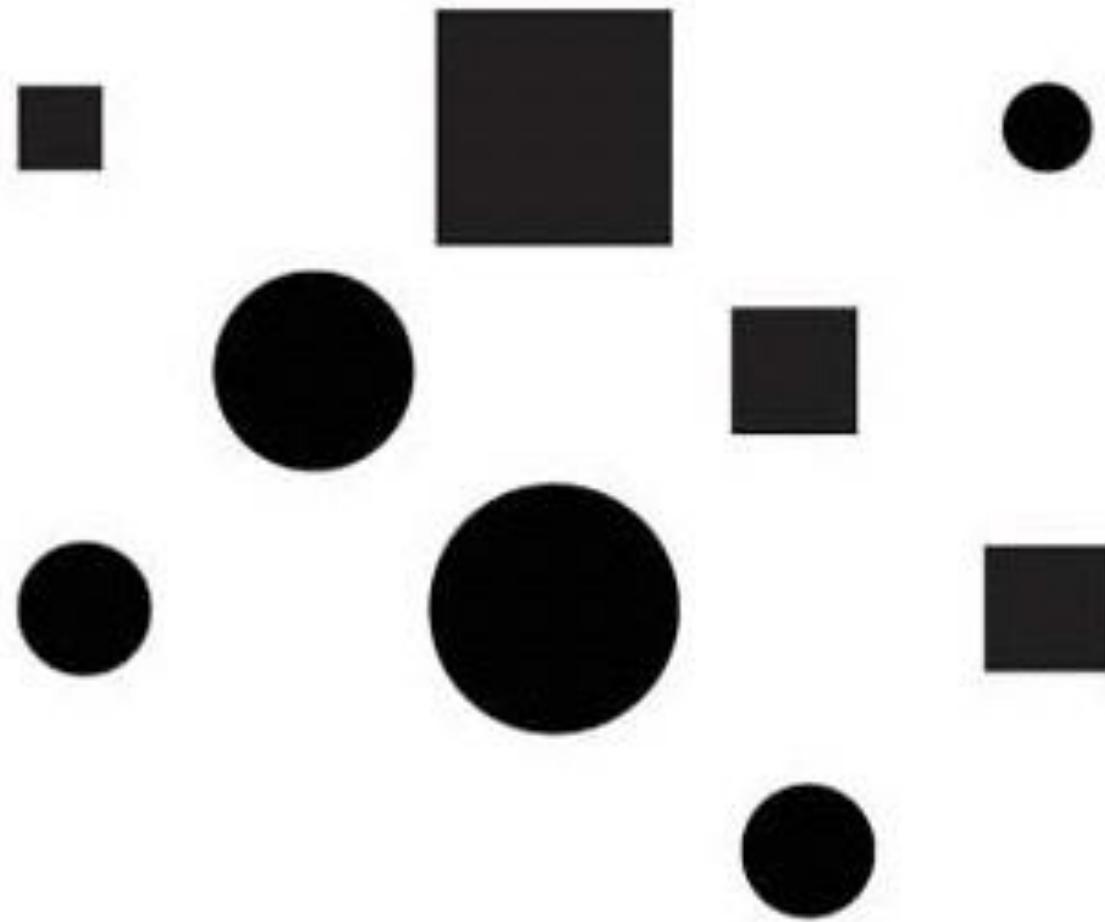
- We can easily see objects that are different in color and shape, or that are in motion.
- **Use color and shape sparingly to make the important information pop out.**

Gestalt principles

- German: “Gestalt” = form
- patterns transcend the visual stimuli that produced them



similarity



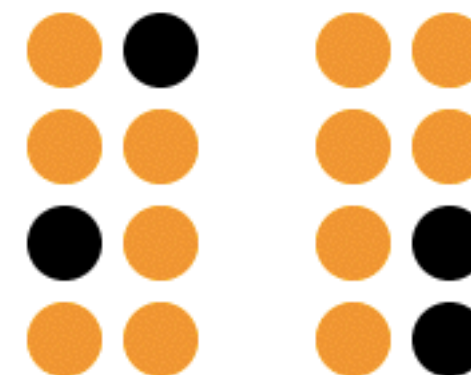
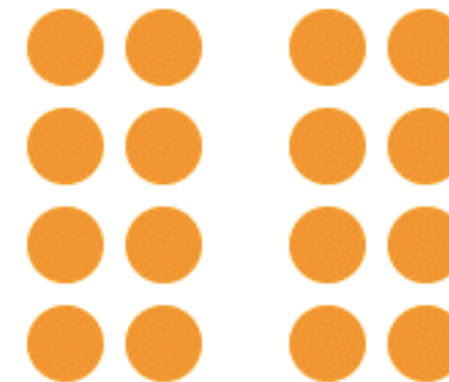
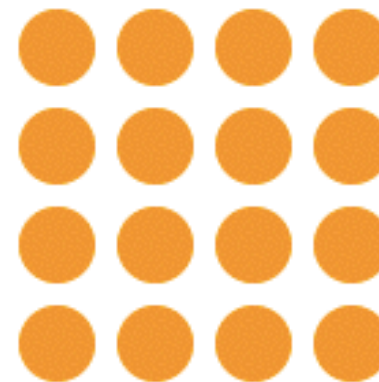
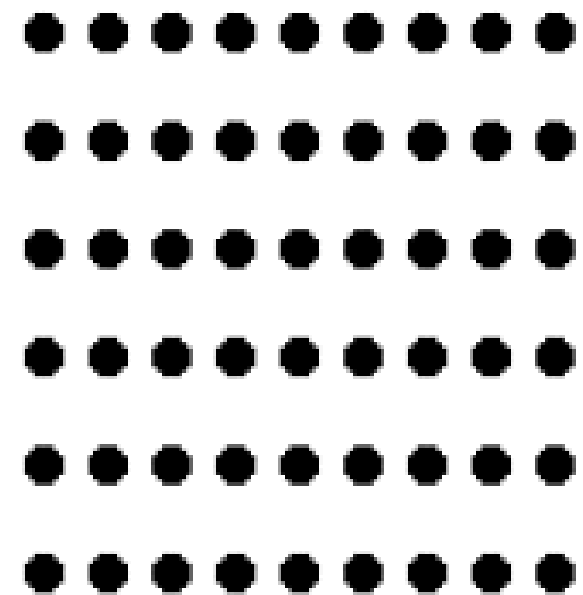
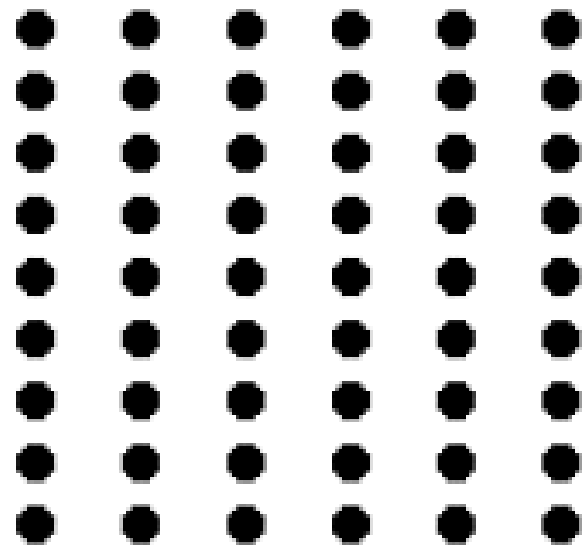
Bang Wong, "Gestalt Principles, I", Nature Methods

similarity



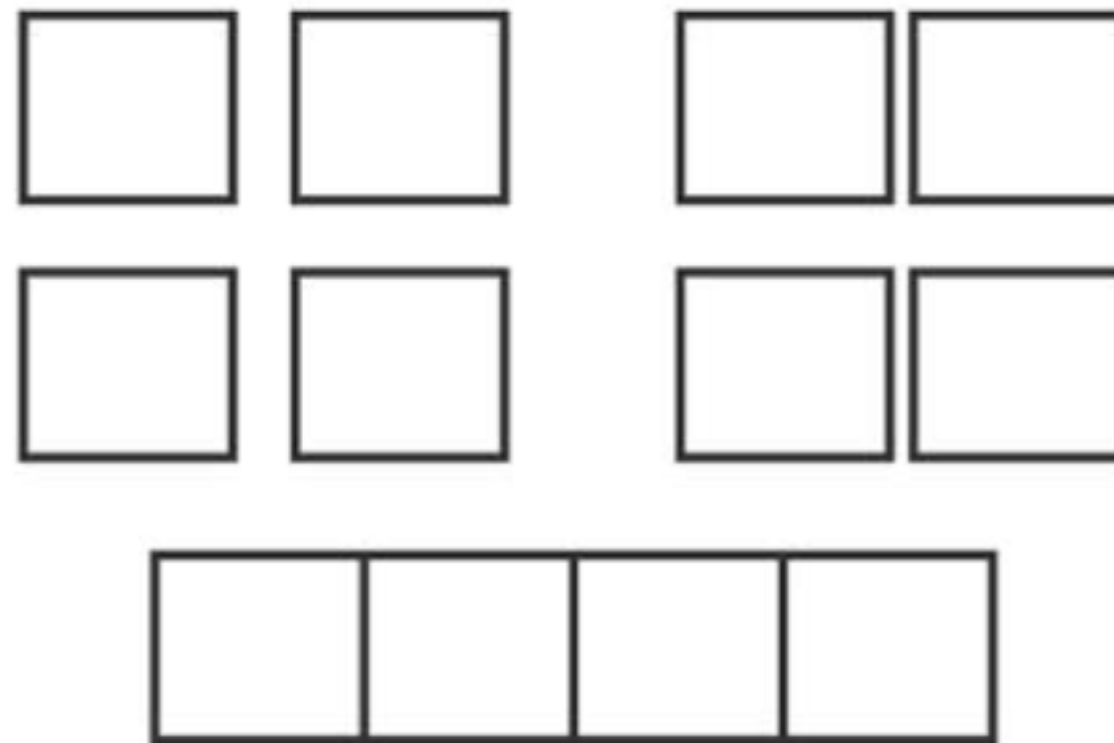
Andy Rutledge, “Gestalt Principles of Perception”

proximity



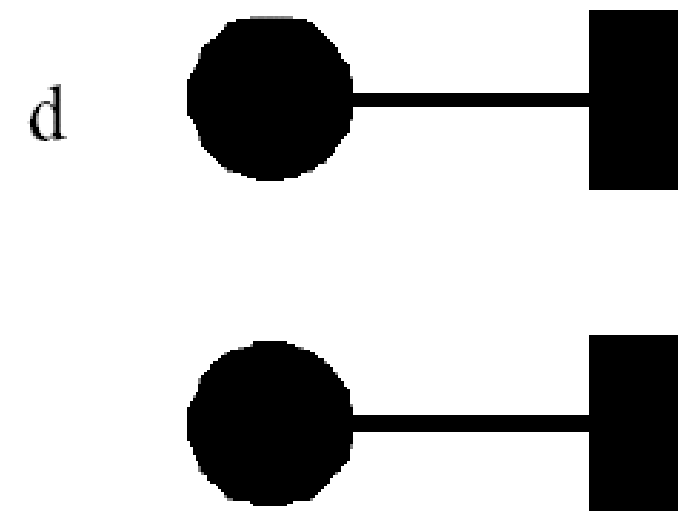
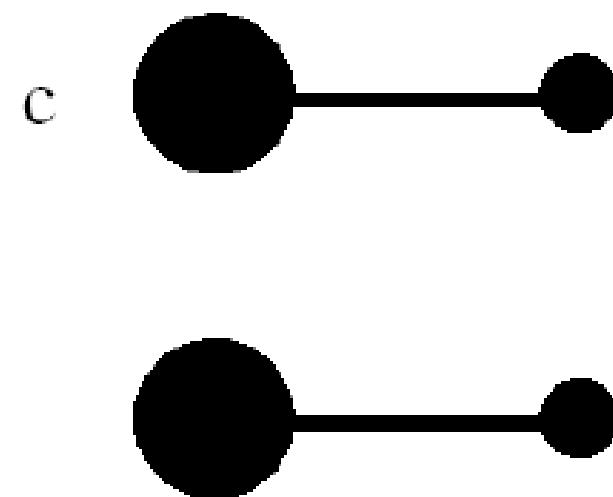
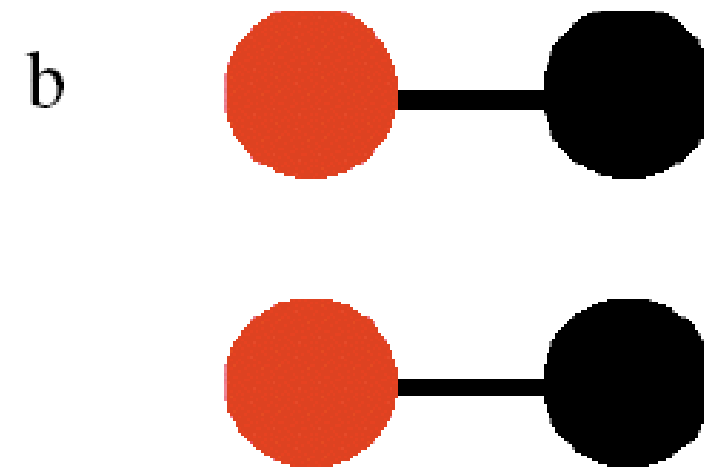
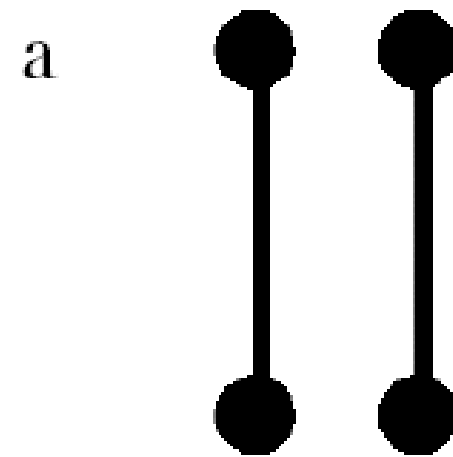
Andy Rutledge, "Gestalt Principles of Perception"

proximity



Bang Wong, "Gestalt Principles, I", Nature Methods

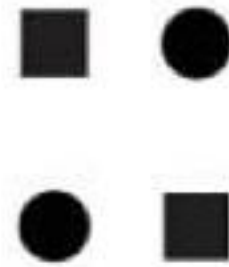
connectedness



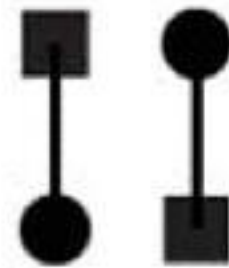
Ware, "Information Visualization"

grouping

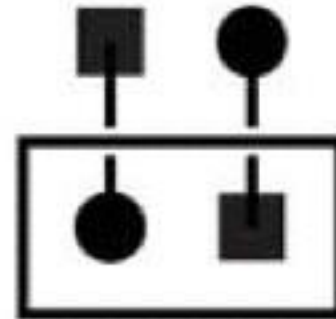
Similarity



Connection

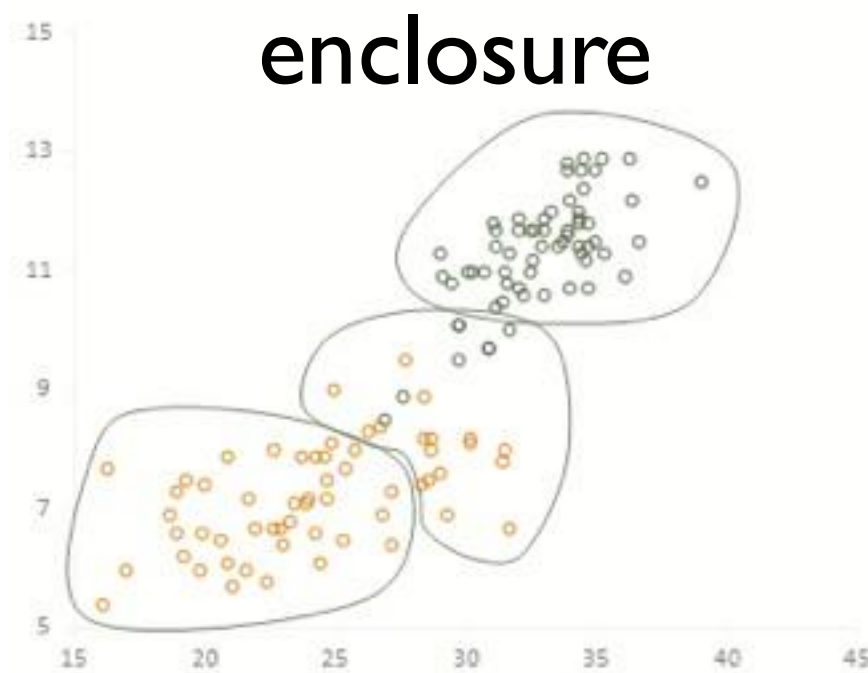
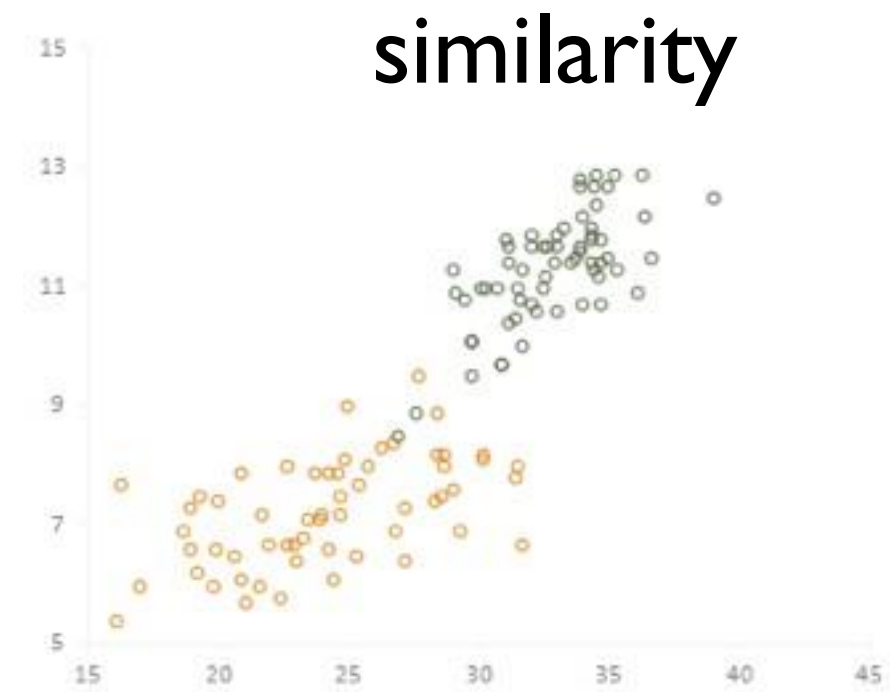
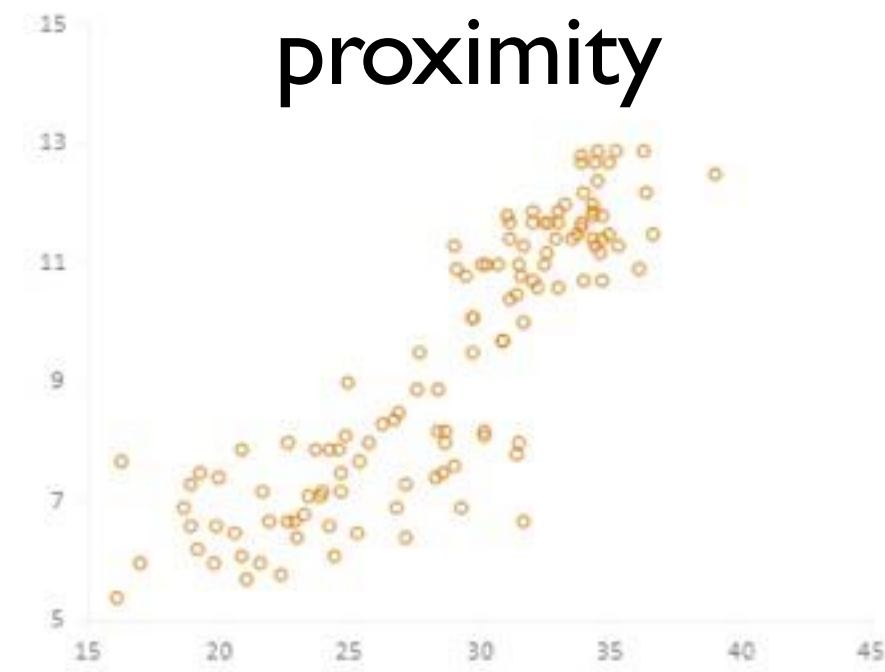


Enclosure

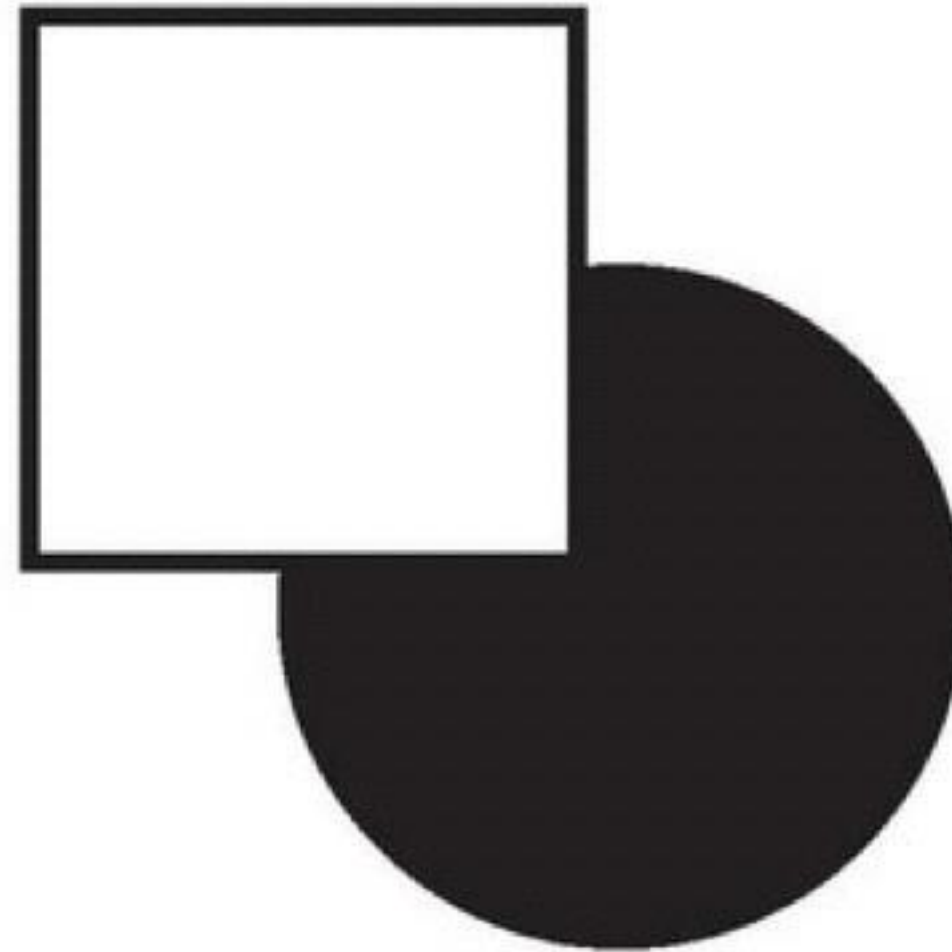


Bang Wong, "Gestalt Principles, I", Nature Methods

grouping

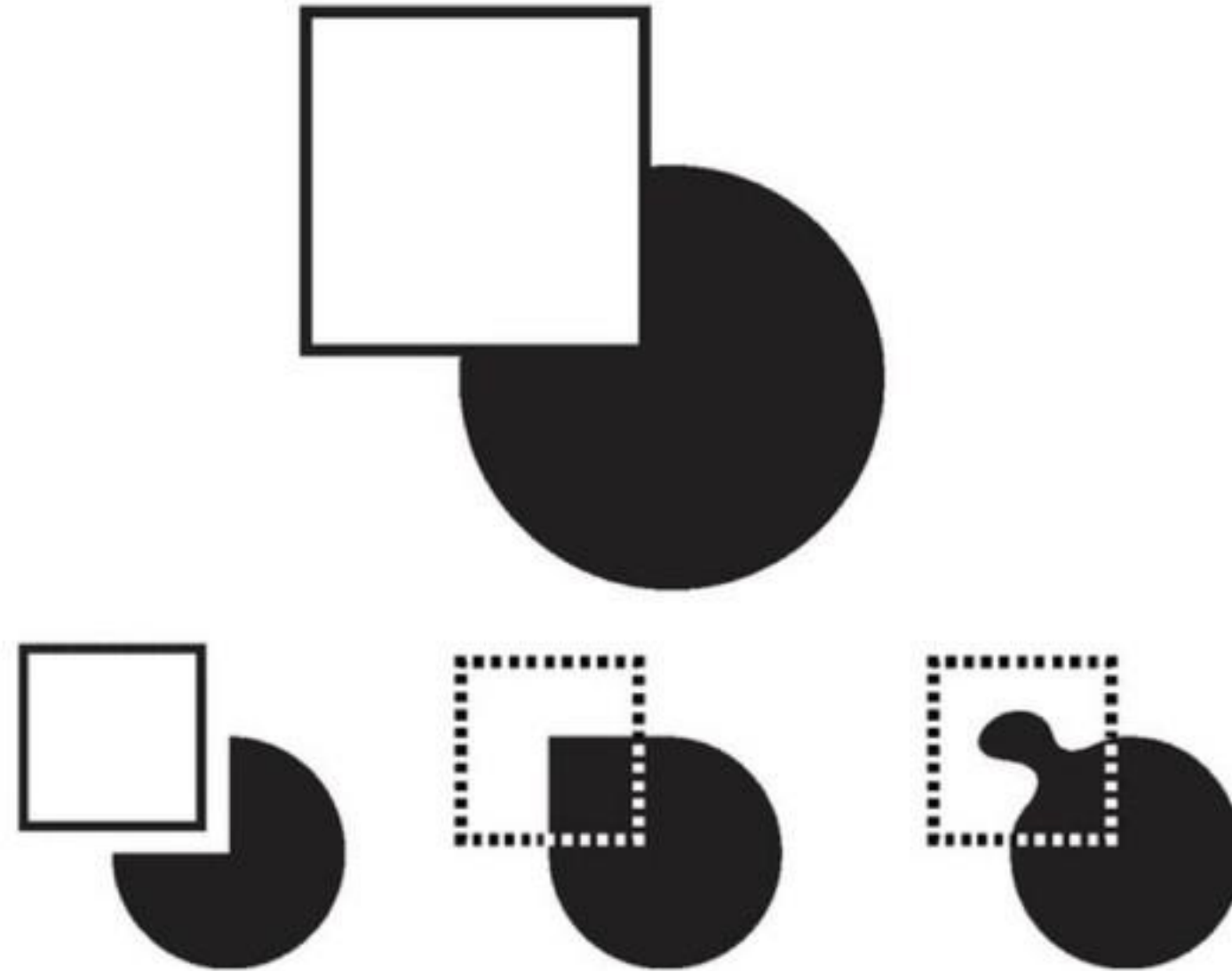


continuity



Bang Wong, “Gestalt Principles, I”, Nature Methods

continuity



Bang Wong, "Gestalt Principles, I", Nature Methods

closure

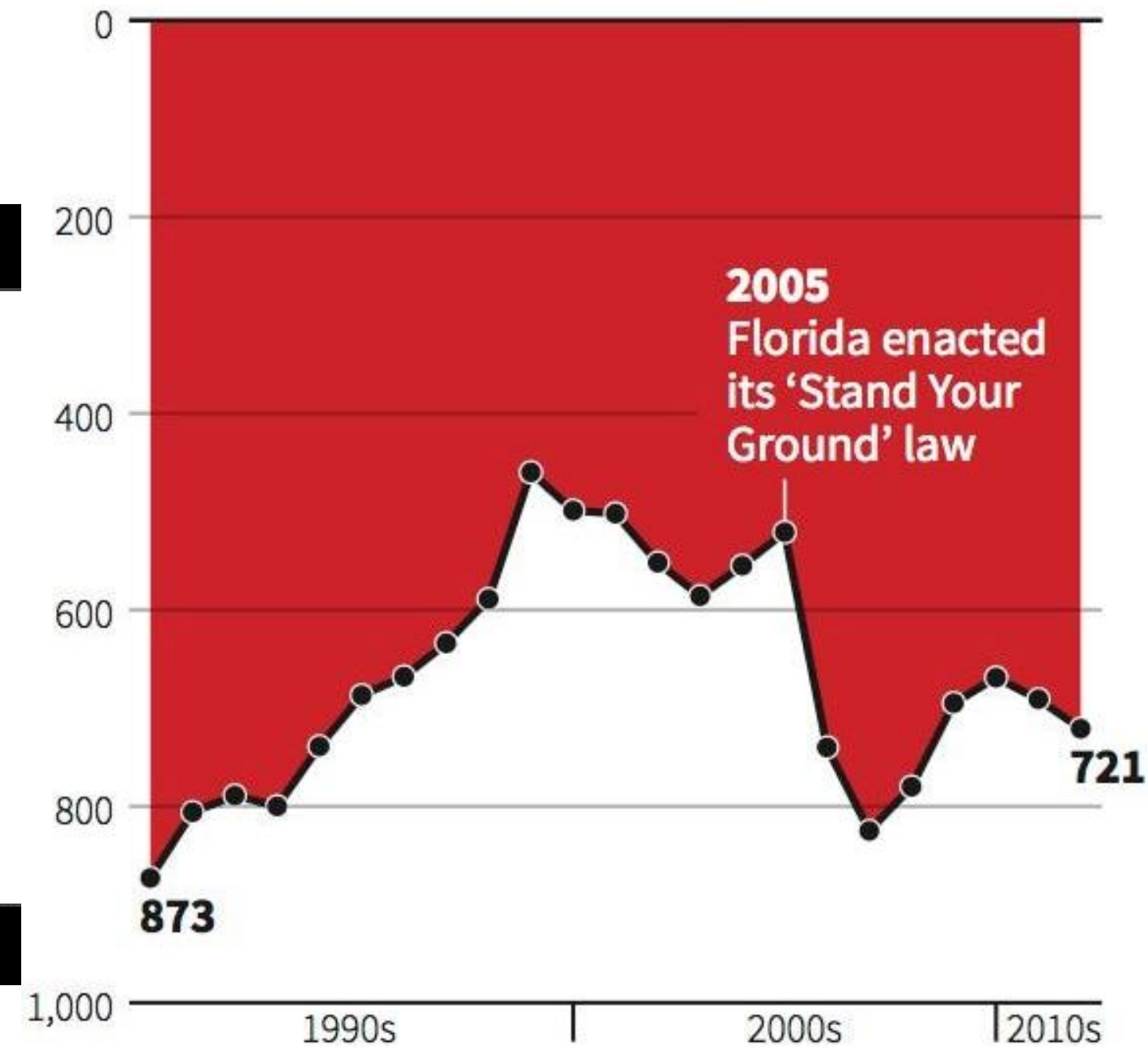


Bang Wong, "Gestalt Principles, I", Nature Methods

figure / group

Gun deaths in Florida

Number of murders committed using firearms



Source: Florida Department of Law Enforcement

C. Chan 16/02/2014

REUTERS

Gestalt Principles, I", Nature Methods

common fate

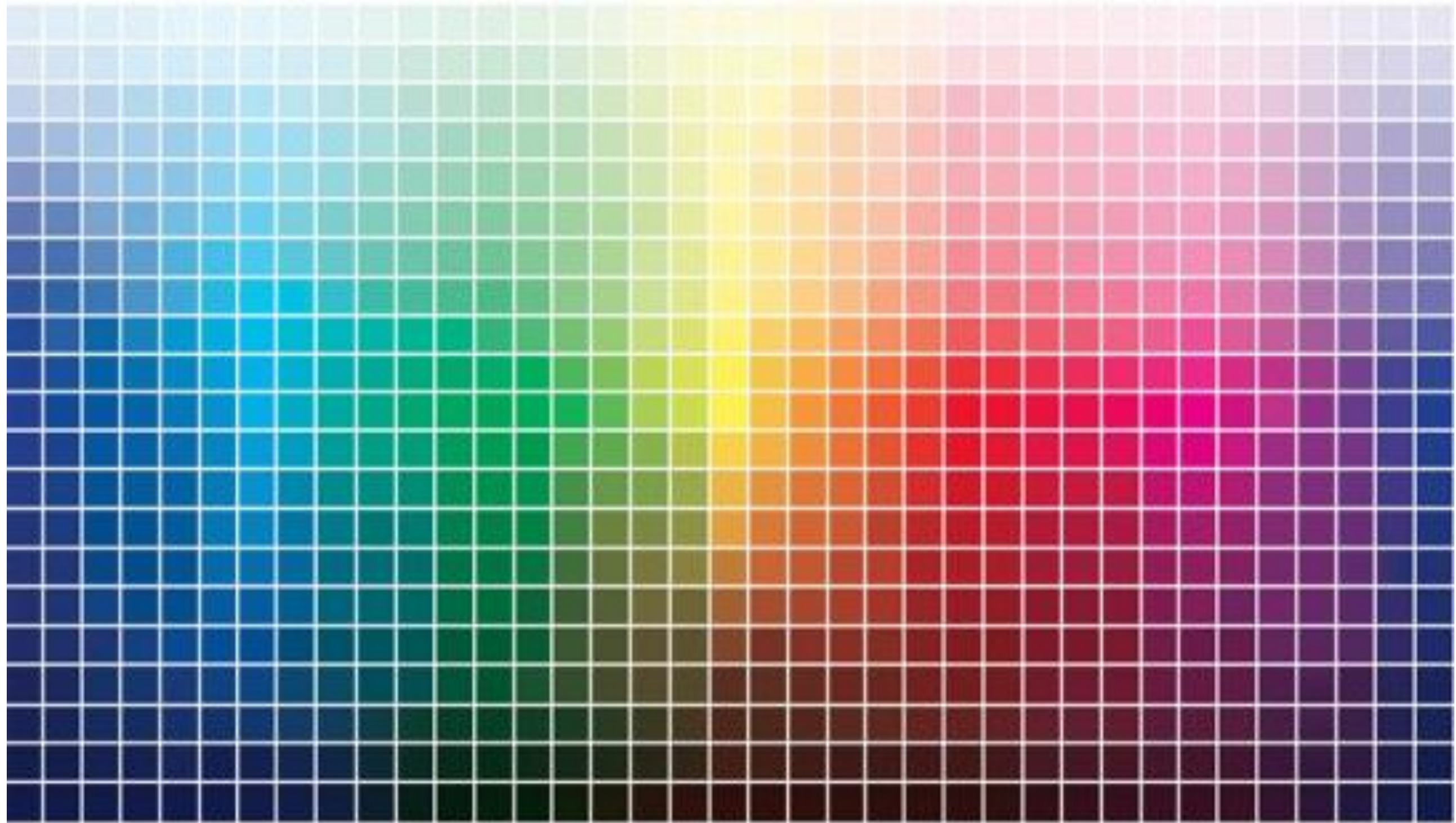


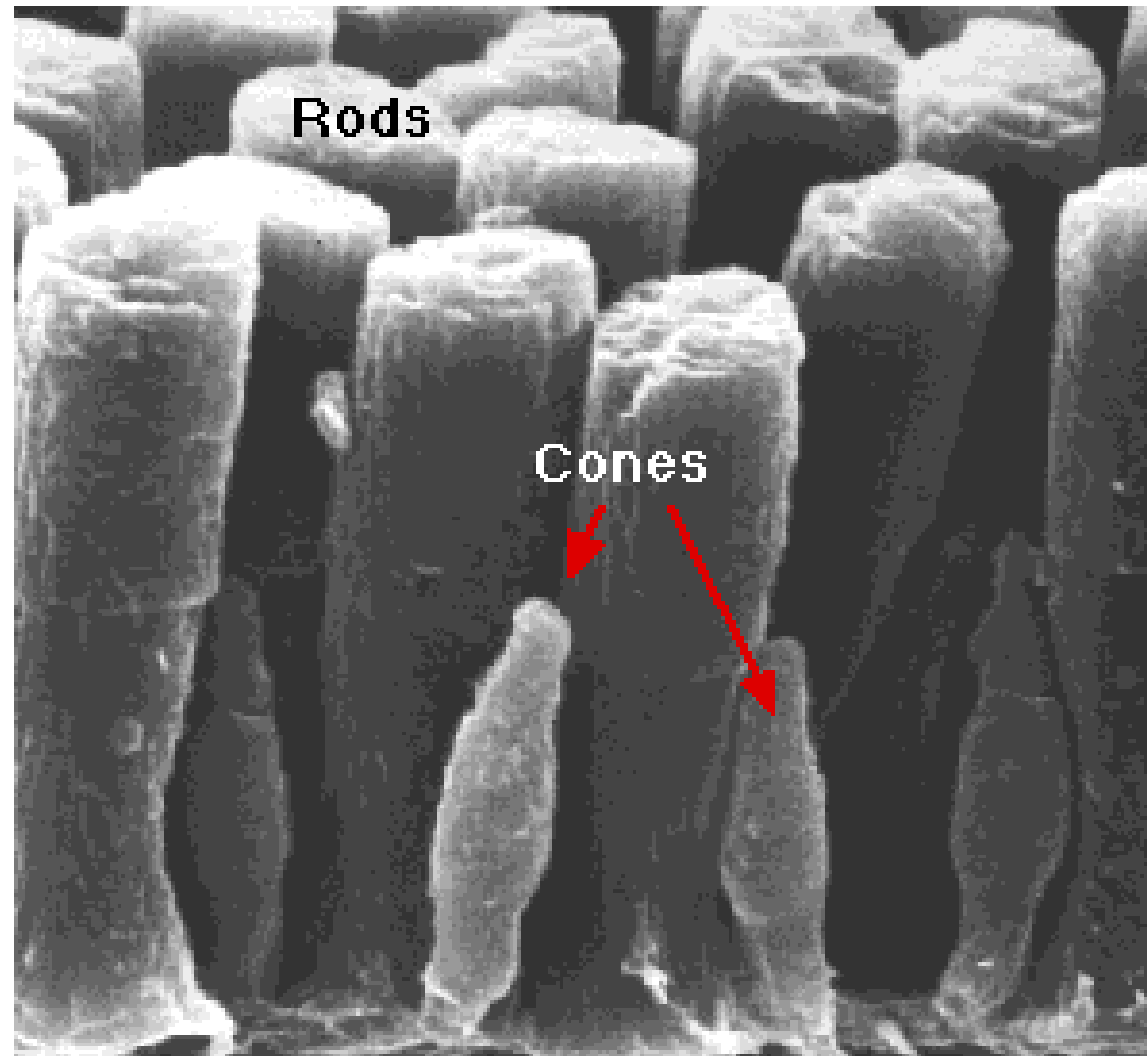
Andy Rutledge, “Gestalt Principles of Perception”

Takeaways

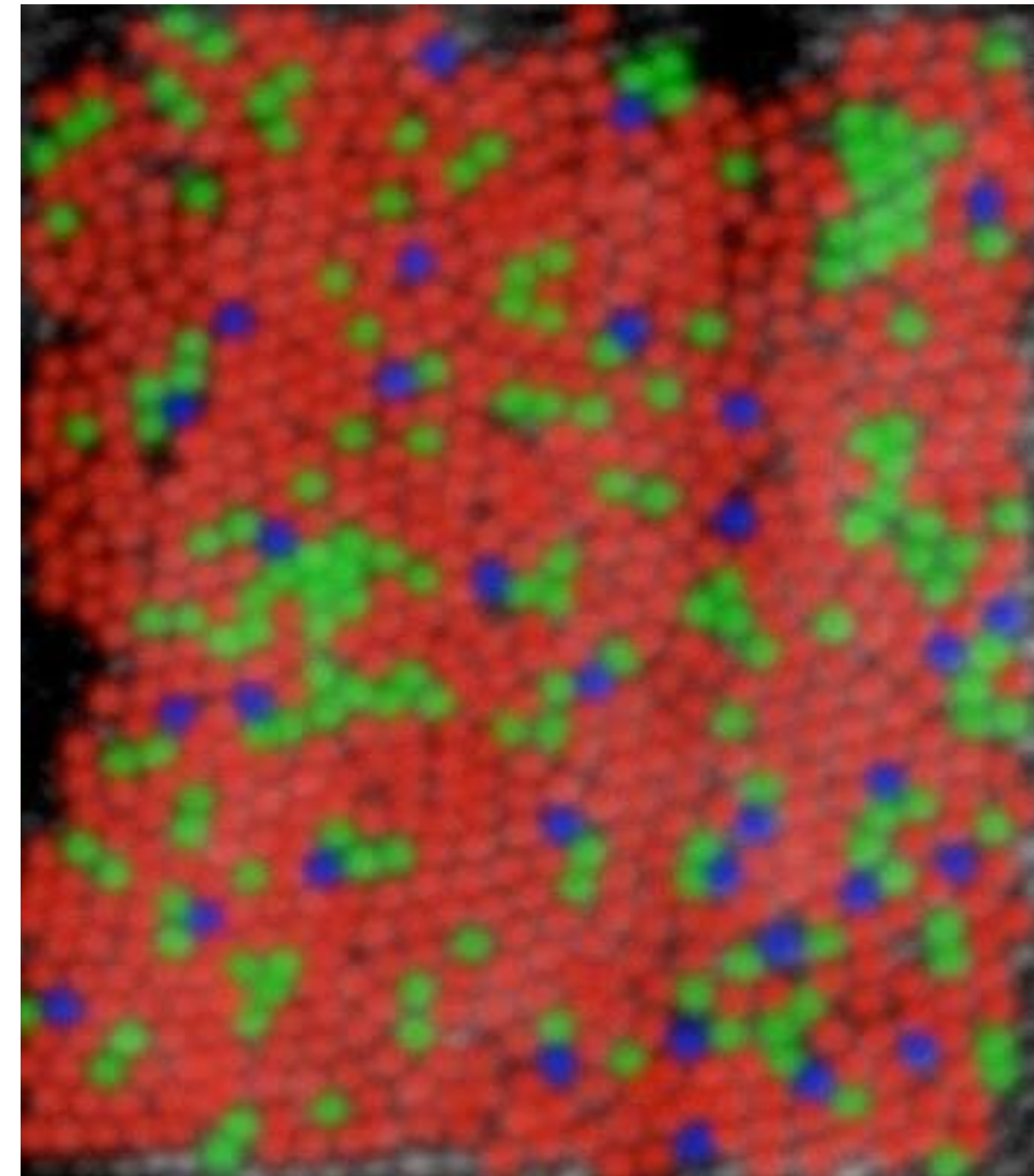
- Gestalt principles give us a conceptual understanding of the way our mind converts shapes into structured thought.
- **Using the Gestalt principles wisely will lead improve performance in interpretation of visualizations. Poor use may cause users to see things that aren't there...**

Color





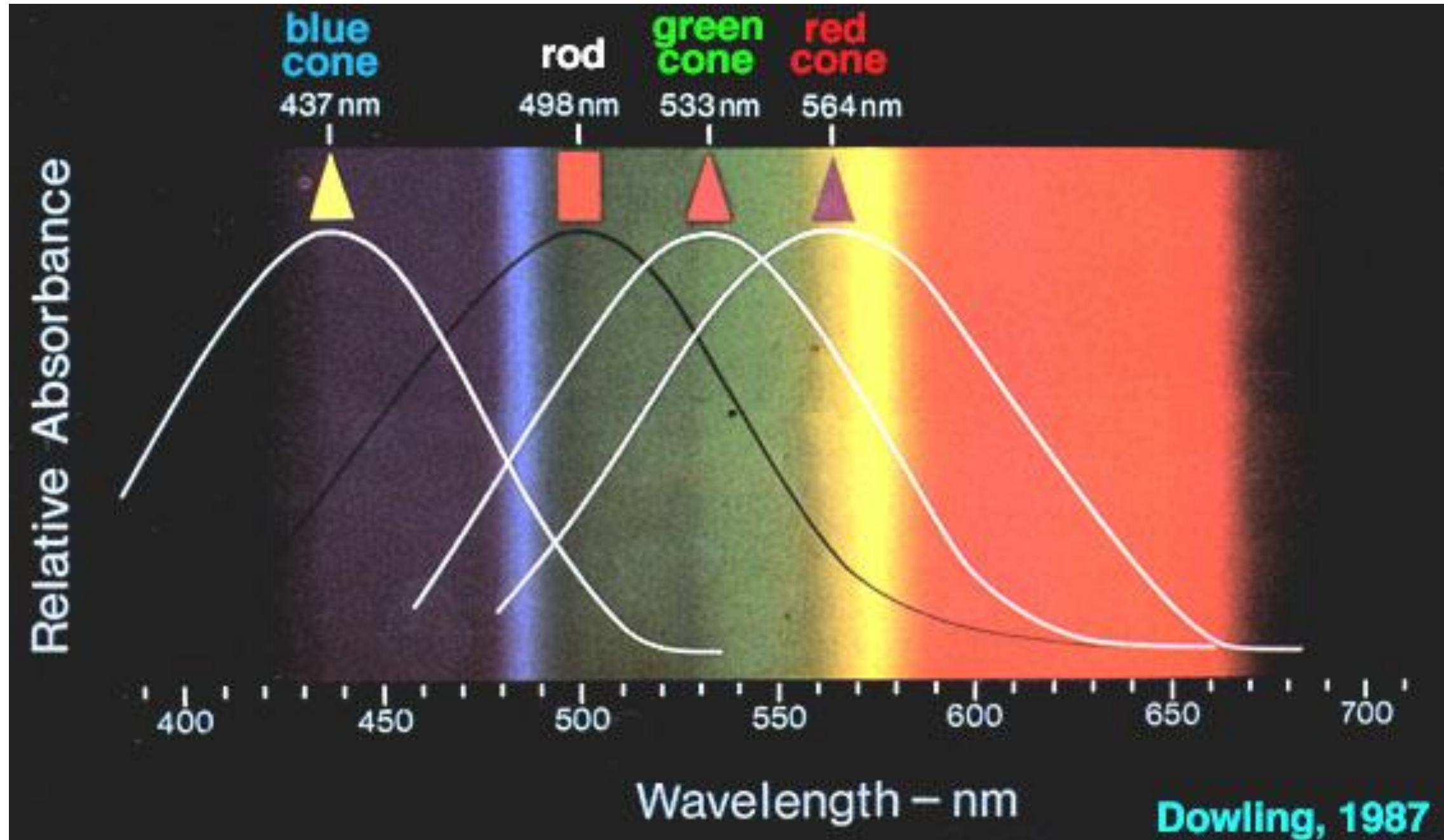
120 million rods



5-6 million cones

Wandell, "Foundations of Vision" (left) David R. Williams, Univ. of Rochester (right)

Cone Response



ADDITIVE COLOR

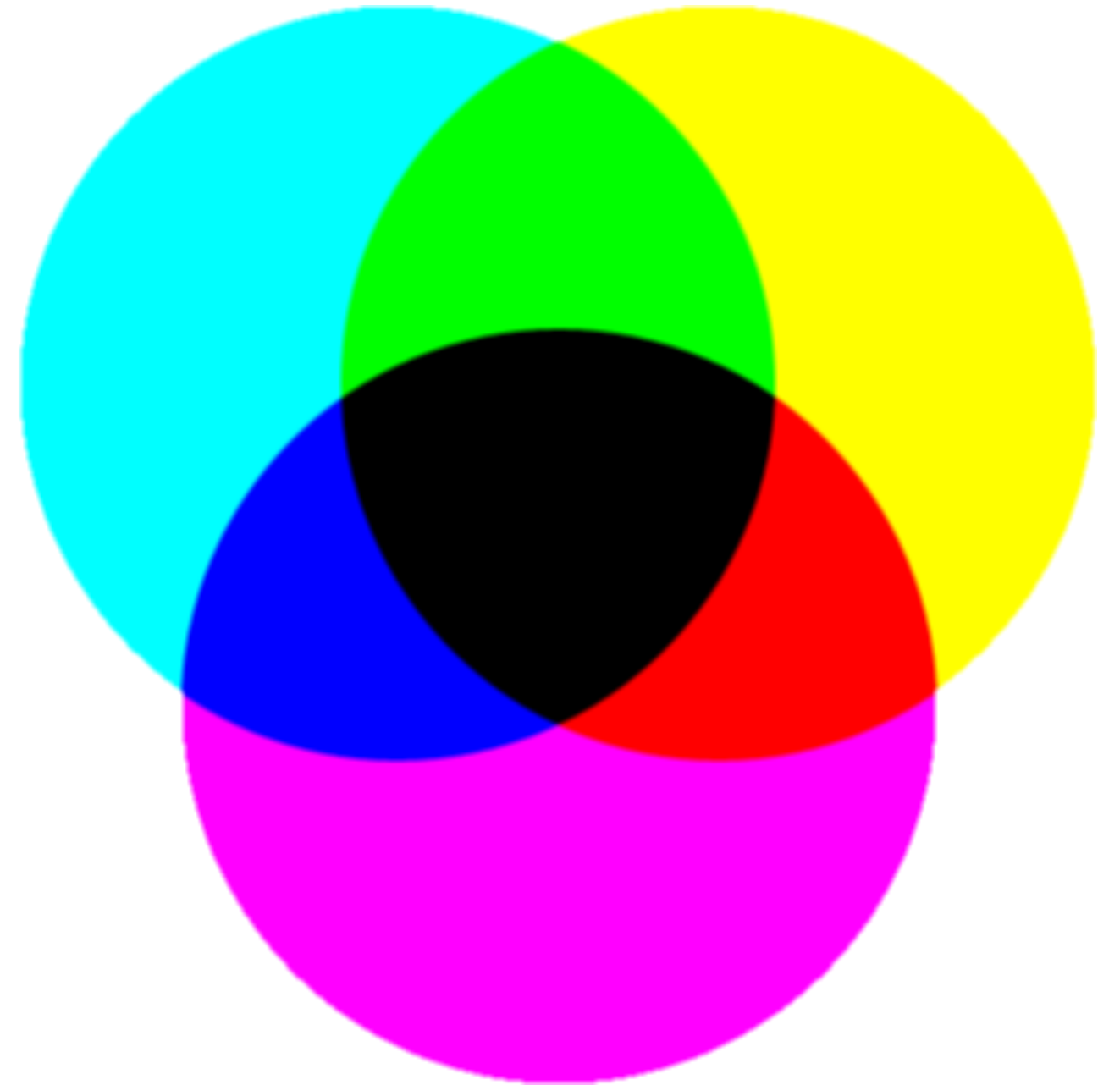
- (like we see in light)
- primary: RGB
- secondary: CMY

- very common color space
- not perceptually uniform
- actual color is device-dependent



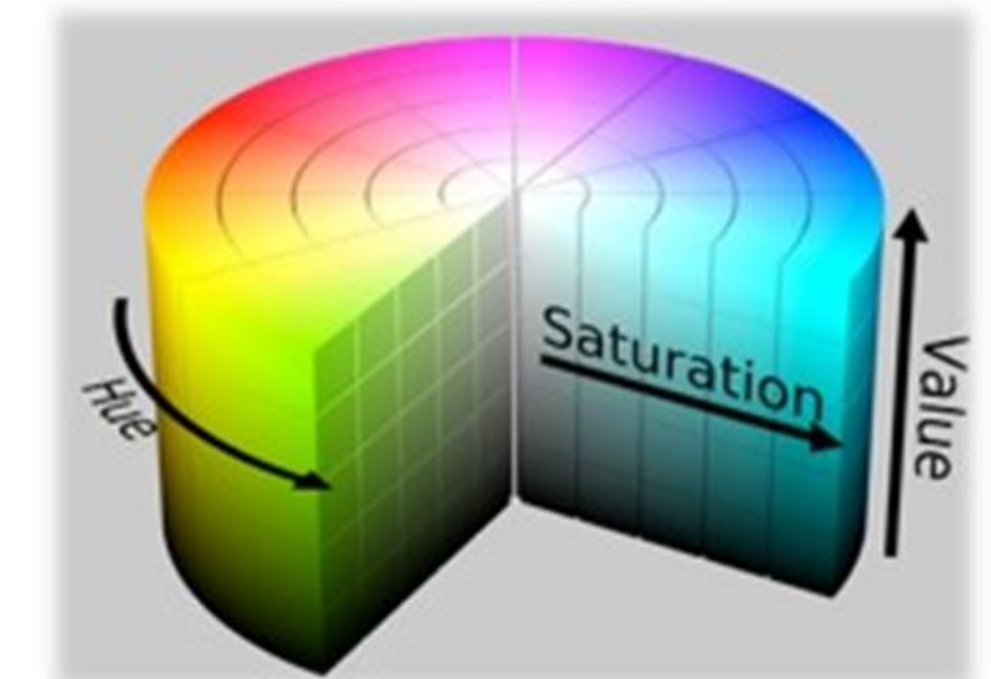
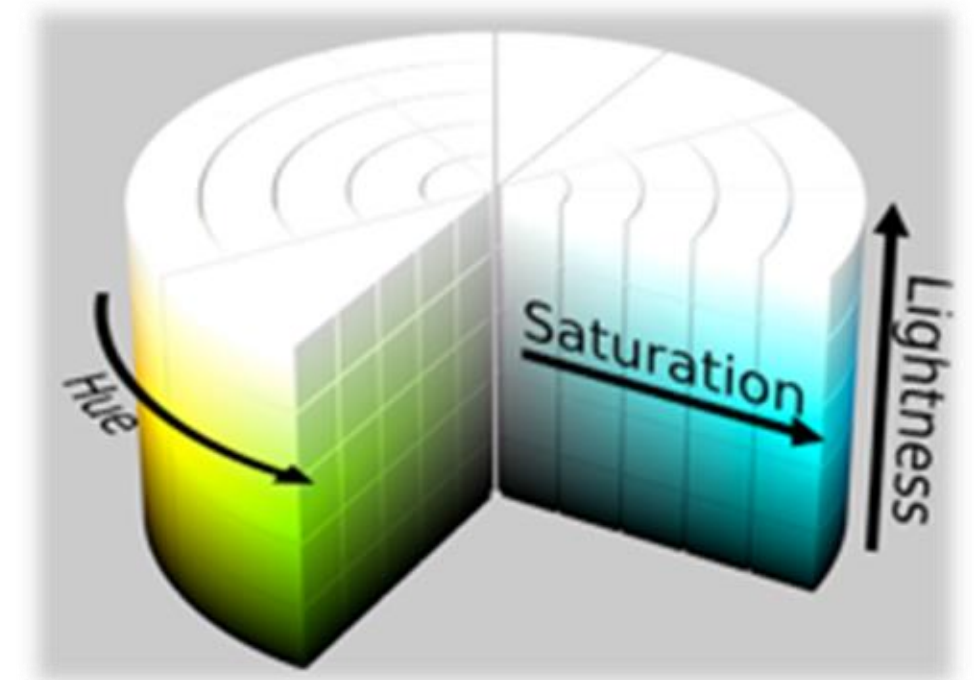
SUBTRACTIVE COLOR

- (used in print ink)
- primary: CMY
- secondary: RGB
- approx black = $C+M+Y$
- true black = $C+M+Y+K$
- actual color is device-dependent



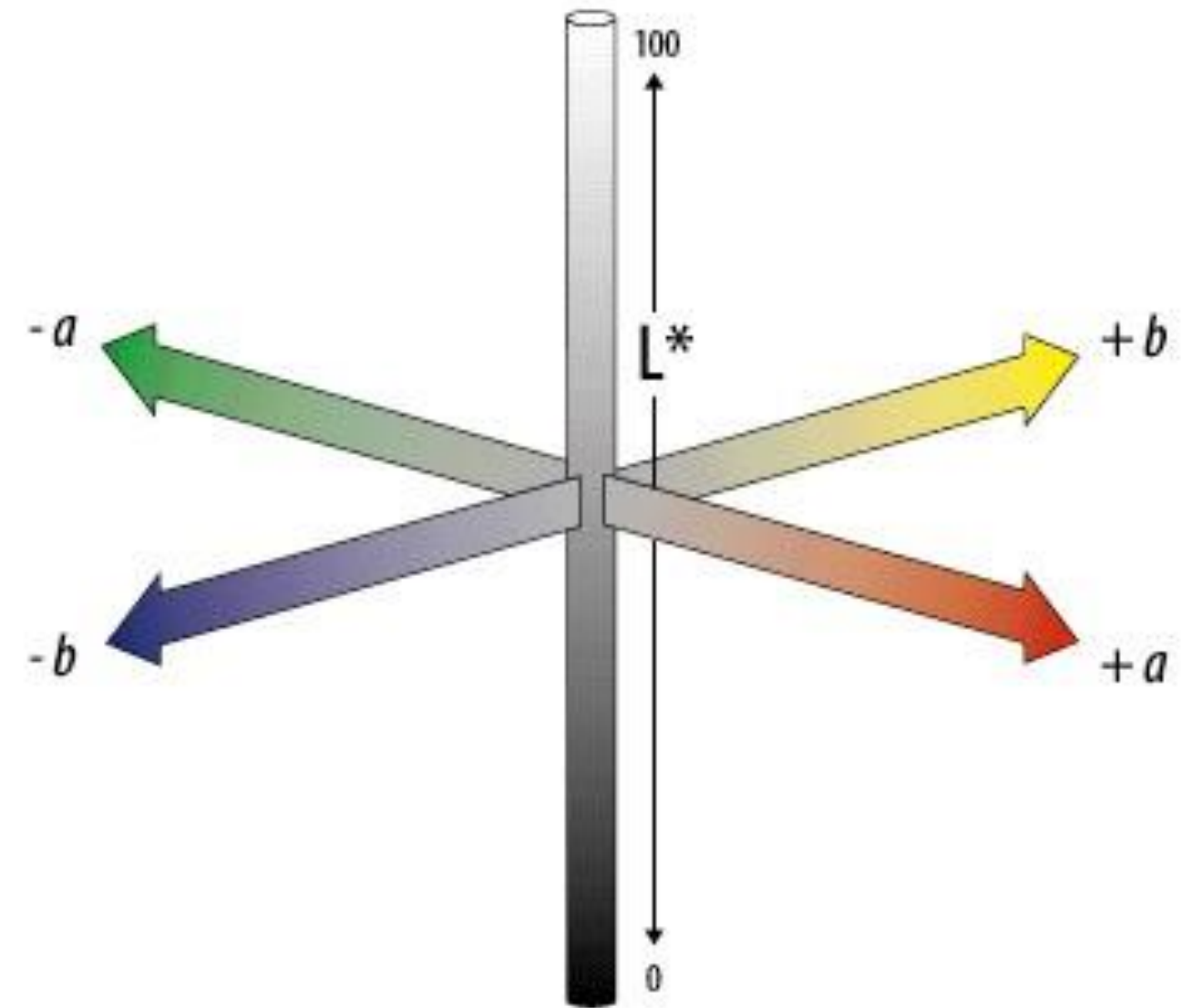
HSV [B, L, I] (additive)

- Hue, Saturation, [Value, Brightness, Lightness, Intensity]
 - polar coordinate representations of RGB space
 - conical or cylindrical shaped space
 - more intuitive than RGB for color tuning
-
- hue: what people think of as color
 - saturation: amount of white mixed in
 - luminance: amount of black mixed in
 - lightness vs value (or brightness)
 - intensity, in computer vision applications



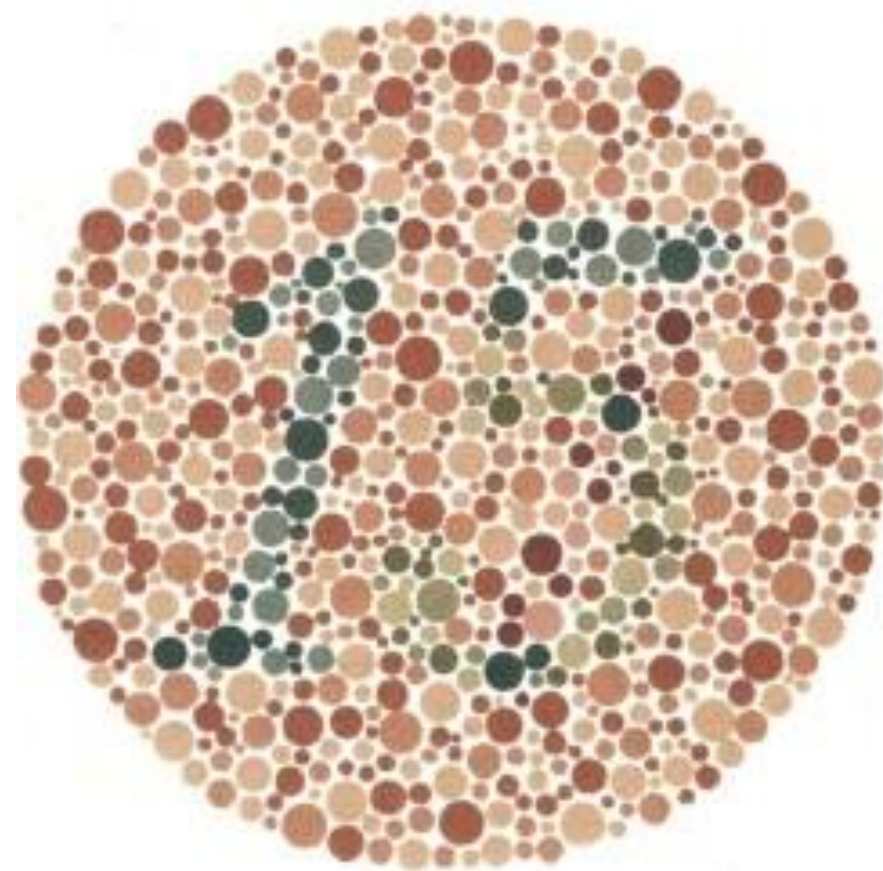
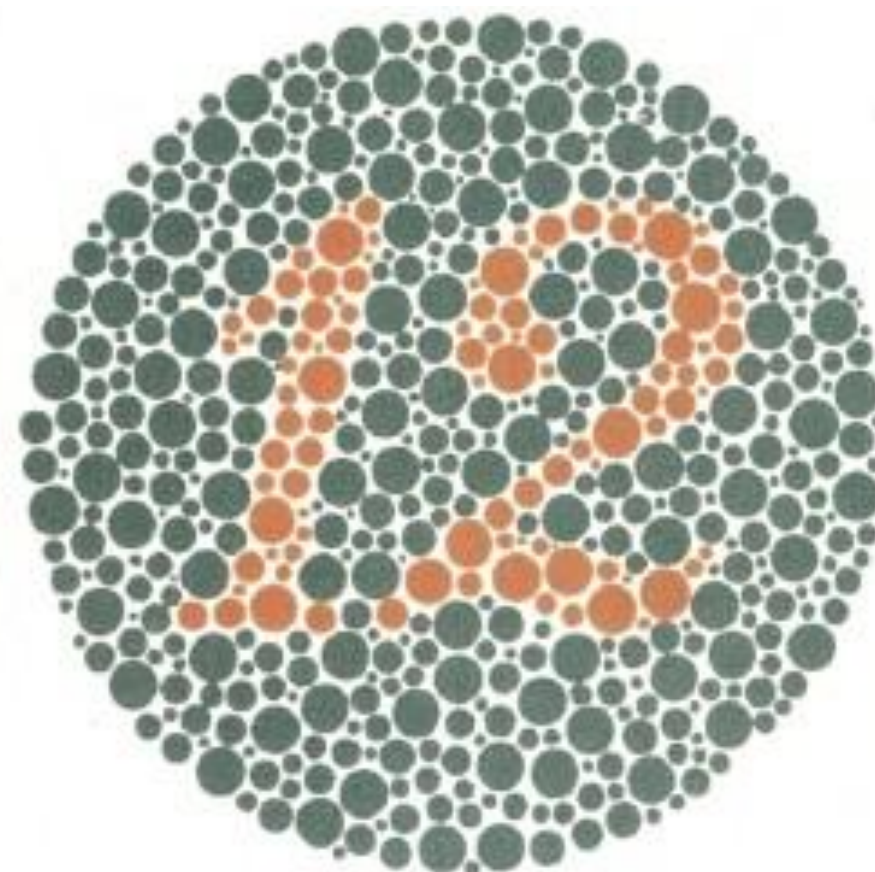
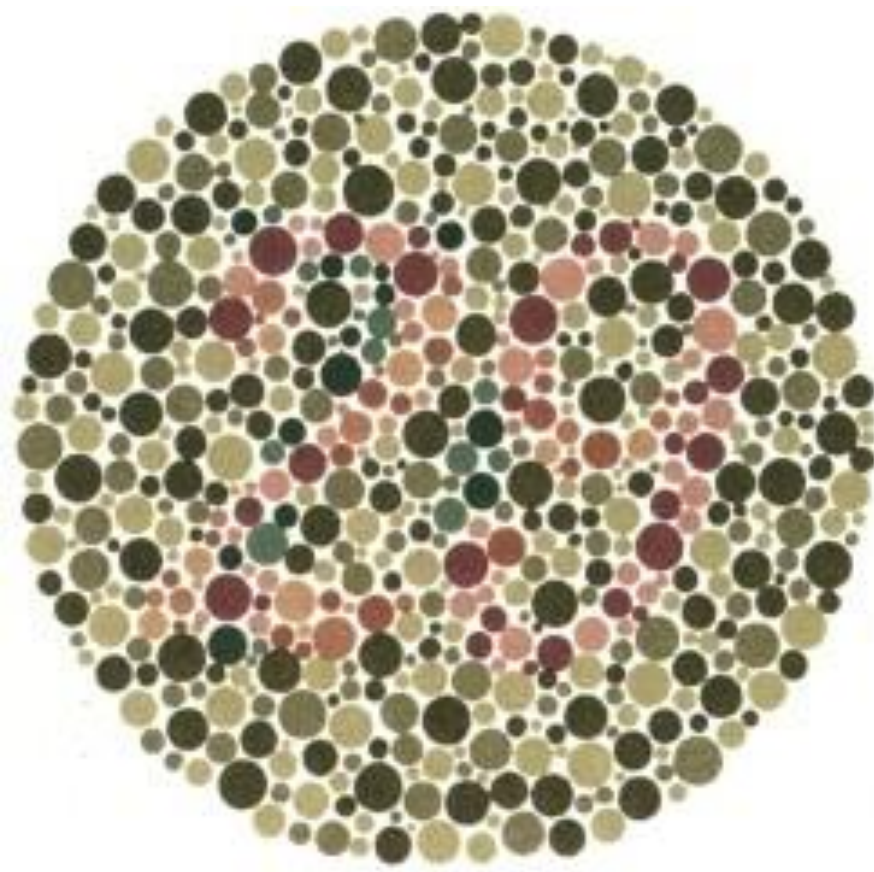
CIE LAB/LUV

- mathematically defined & perceptually based to include all perceivable colors
- a : red to green
- b : yellow to blue
- L^* : lightness (black to white)



COLOR BLINDNESS

- deficiency in color vision
- typically caused by faulty cone development
- found more in men than women
- photopigment genes carried in x-chromosome
- 5-8% of (European origin) men and 0.5% of women



The X-Rite Color Challenge and Hue Test

Are you among the 1 in 255 women and 1 in 12 men who have some form of color vision deficiency? If you work in a field where color is important, or you're just curious about your color IQ, take our online challenge to find out. Based on the [Farnsworth Munsell 100 Hue Test](#), this online challenge is a fun, quick way to better understand your color vision acuity.

Just remember, this is not a replacement for the full test!

Directions:

1. The first and last color chips are fixed.
2. Drag and drop the colors in each row to arrange them by hue color.
3. Complete all four color tests.
4. Click 'Score My Test' to review results.

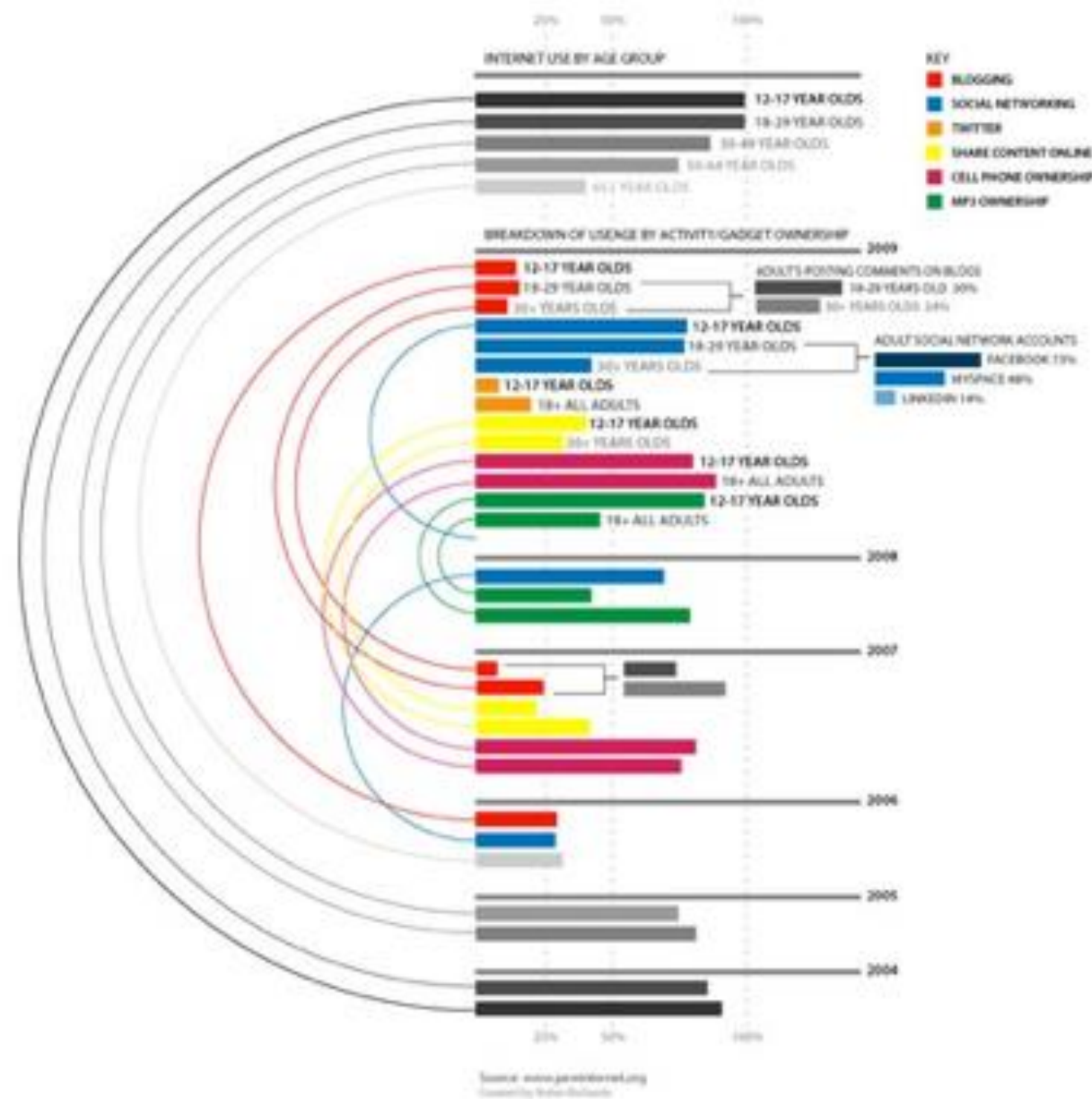
What's My Color IQ?

Score My Test

How different age groups are using the internet

With the growth of social media networks such as Facebook and Twitter, traditional blogging has been usurped by micro-blogging quick and short 140 character updates instead of lengthy, in-depth (and sometimes still equally pointless) articles.

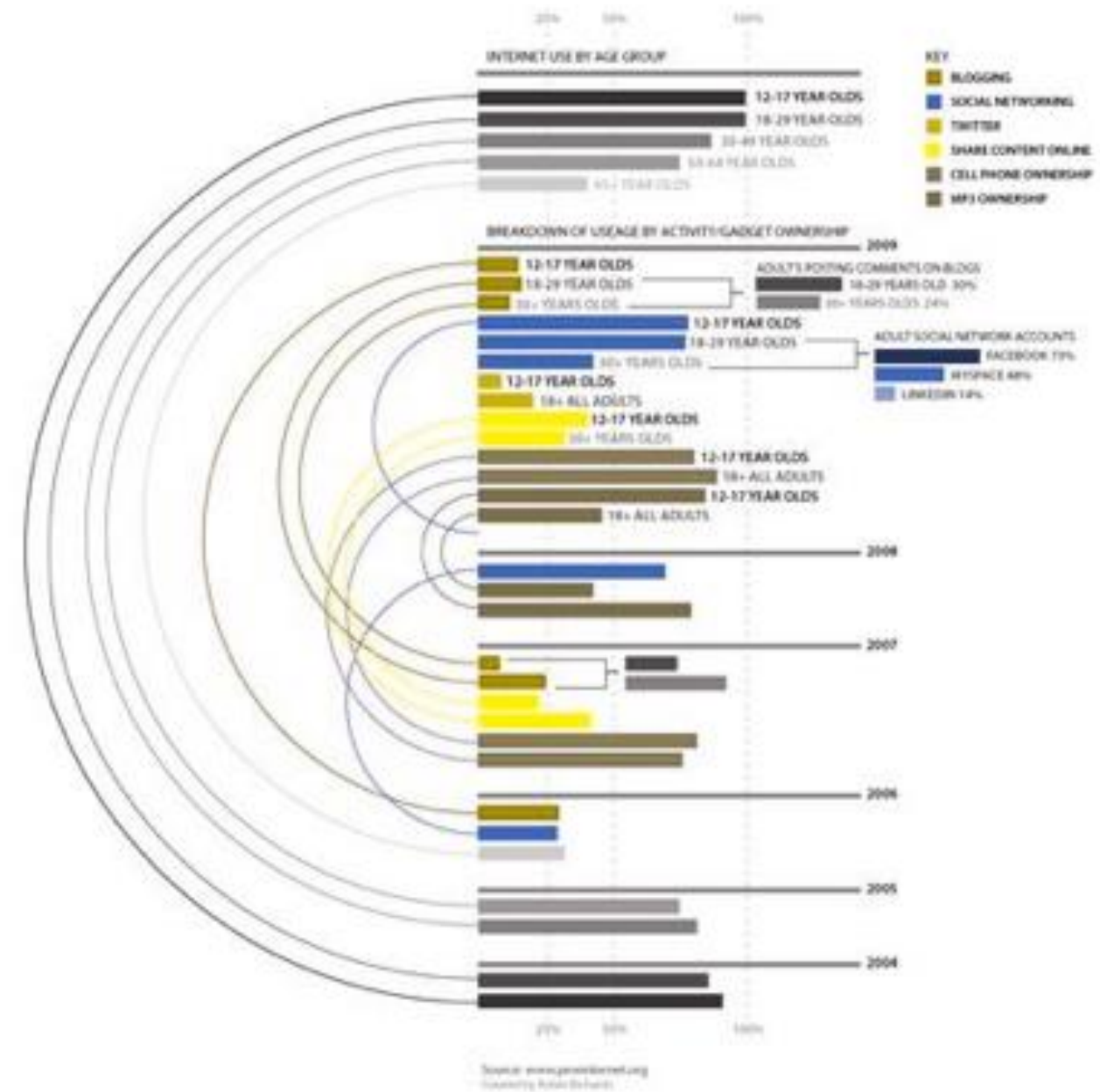
However, while teens and young adults seem to be shunning blogging, it is still strong among the over 30s...



How different age groups are using the internet

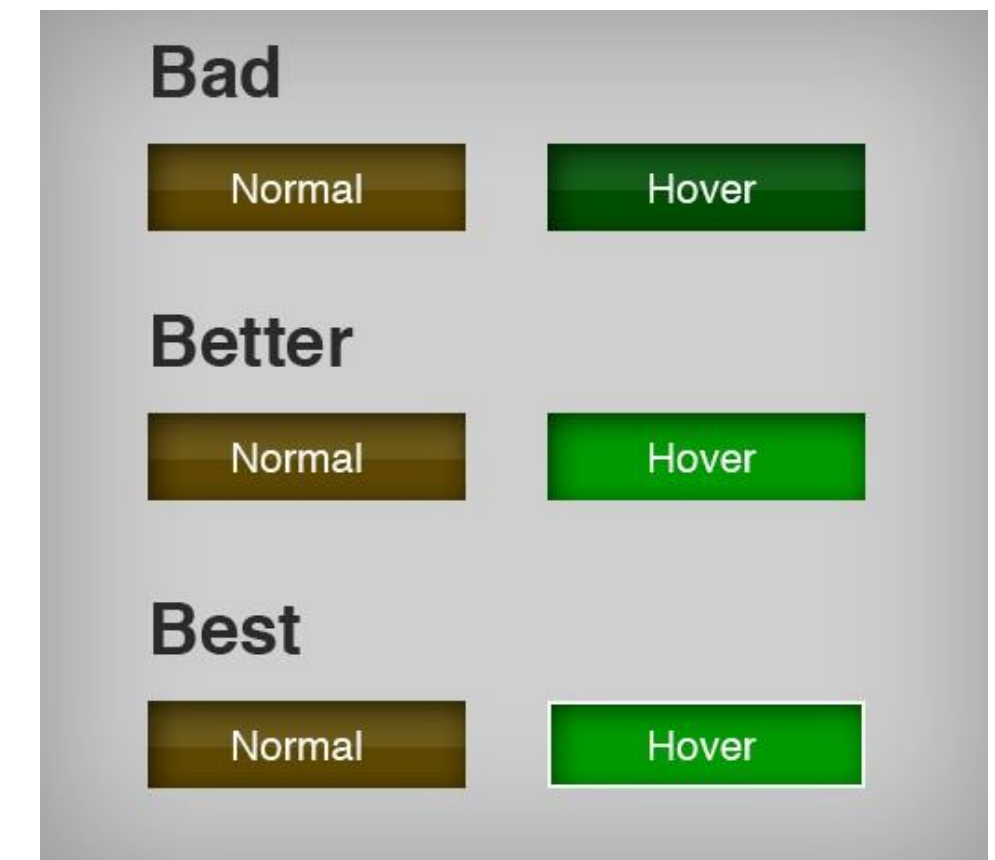
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However, while teens and young adults seem to be shunning blogging, it is still strong among the over 30s...



Takeaway

- Even if you aren't colorblind, someone you're working with could be
 - Be sure to design with colorblindness in mind by:
 - varying hue, saturation, brightness
 - using monochrome color schemes
 - using cues besides/in addition to color
 - software solution, vischeck (<http://www.vischeck.com>)



A note on accessibility in visualization design



Frank ↗ @FrankElavsky · Jan 18



Data visualization cares disproportionately far too much about designing for colorblindness relative to other disabilities that are more common (visual impairments included).

(A thread on disability, race, and patriarchy in data visualization.)



44



1.2K



3.3K



Frank ↗ @FrankElavsky · Jan 18



~4.5% of people with northern European ancestry are colorblind. But less than half of a percent of women are.

This means that nearly 8% of men from a northern European background have some form of colorblindness.

Colorblindness affects WHITE MEN the most.



7



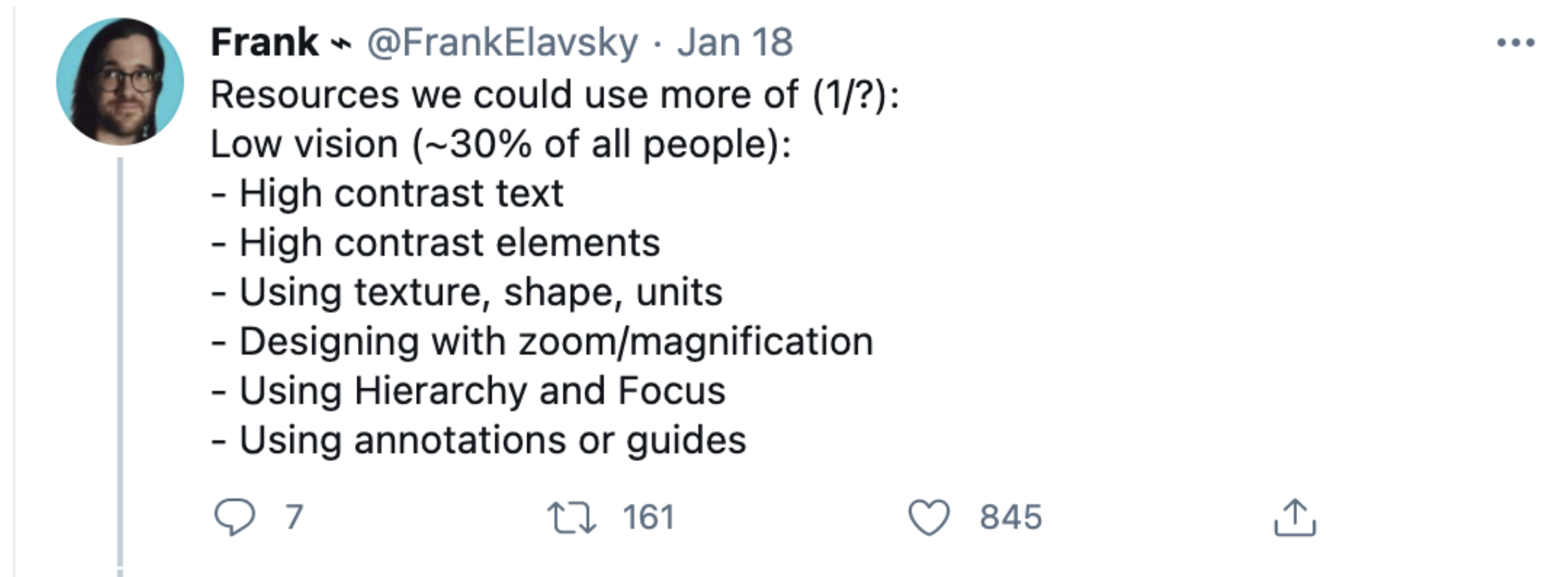
14



214



A note on accessibility in visualization design



A note on accessibility in visualization design



Frank ↗ @FrankElavsky · Jan 18



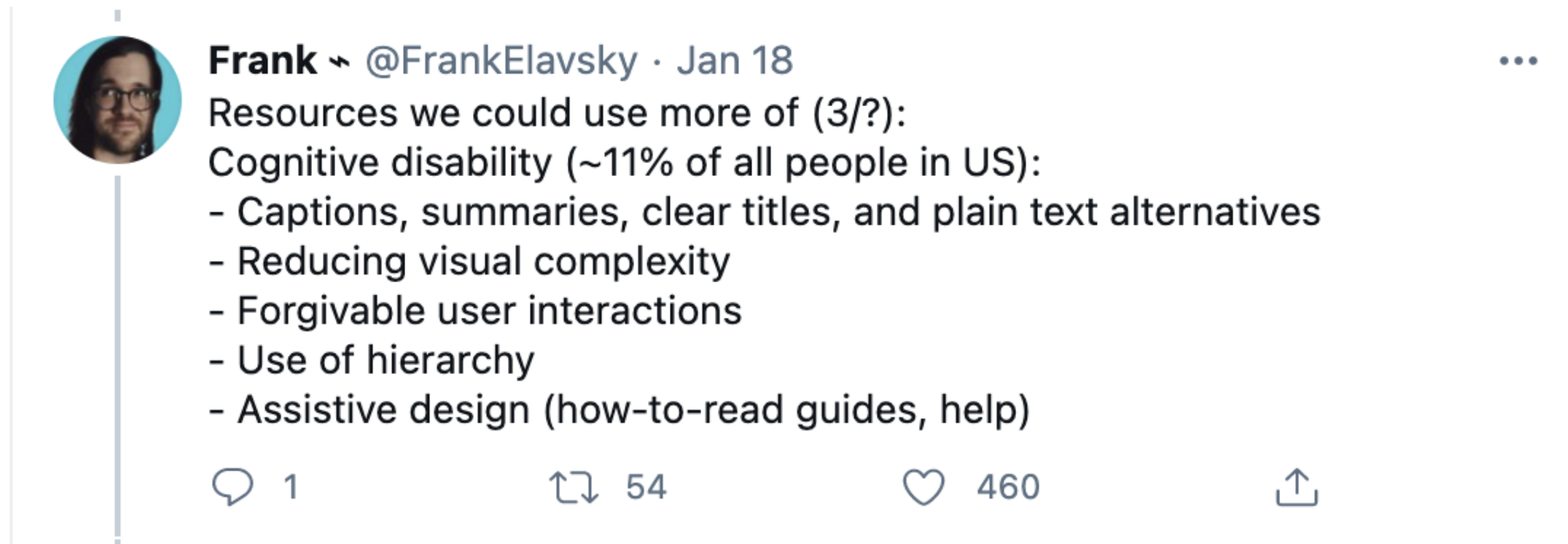
Resources we could use more of (2/?):

Functional/motor impairment (~13% of all people in US):

- Keyboard interactivity/navigation
- UI alternatives to in-chart controls (brushing, subselecting, etc)
- Alternative data navigation schemes
- Scrollytelling alternates



A note on accessibility in visualization design



A note on accessibility in visualization design



Frank ↗ @FrankElavsky · Jan 18



Resources we could use more of (4/?):

Attention deficit/hyperactive disorder (~9% of all people in US):

- Clear, short text summaries
- Object constancy
- Motion design and animation
- Use of breadcrumbs
- Interaction history (with undo/redo functions)



6



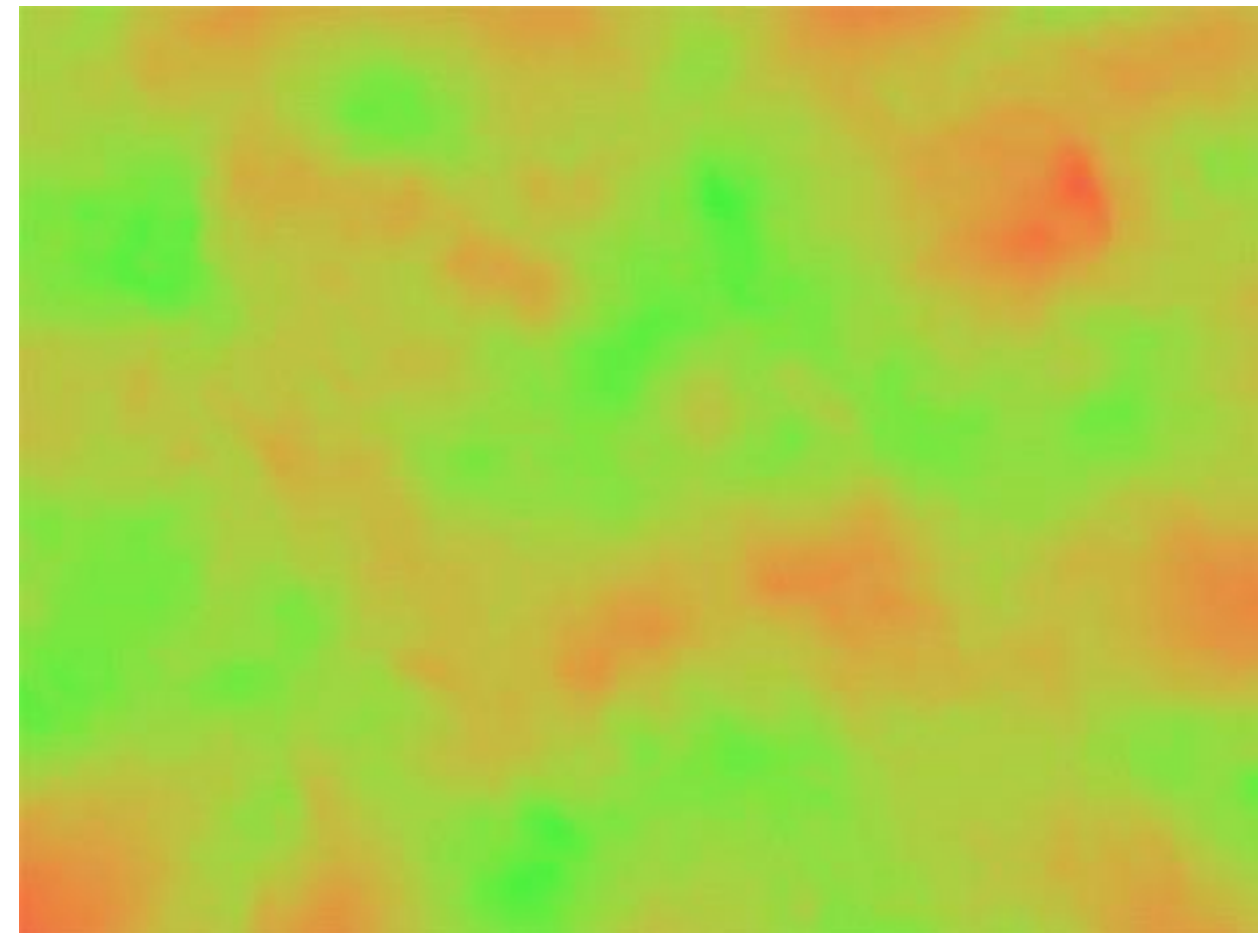
52



423



Contrast Sensitivity

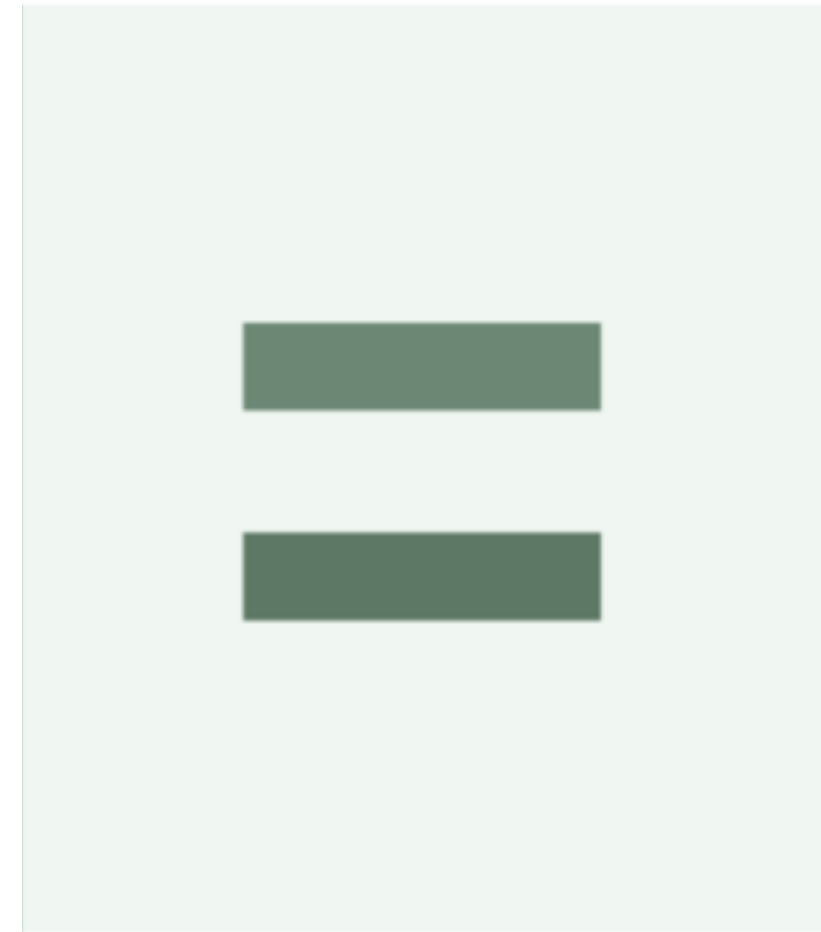
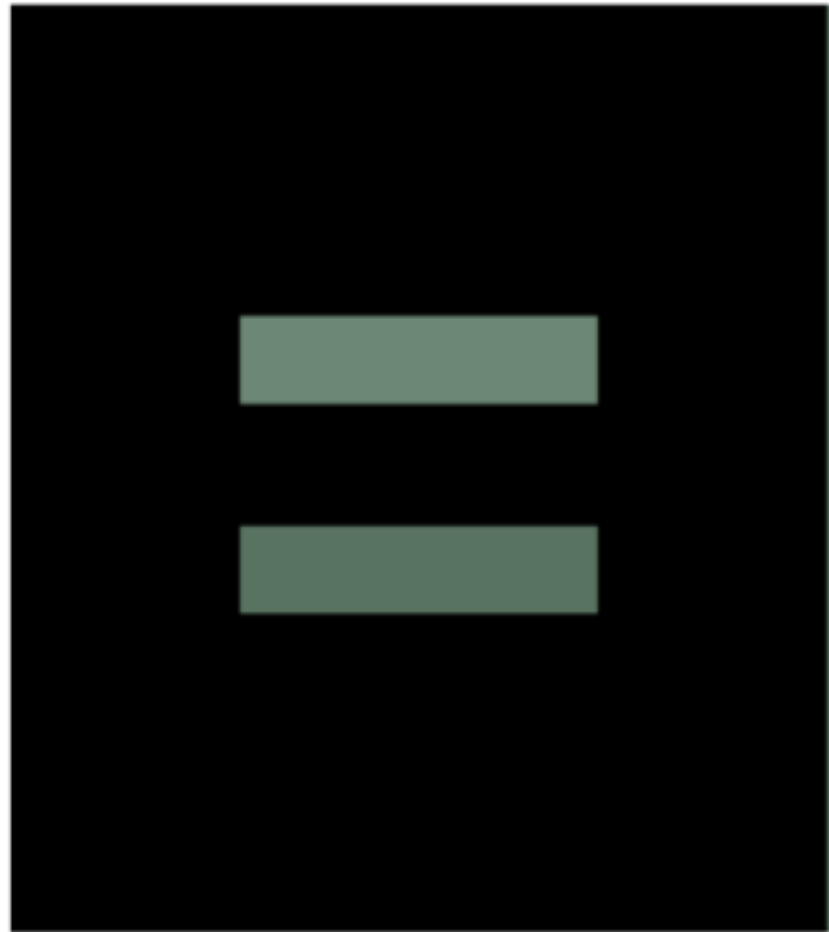


C.Ware, "Visual Thinking for Design"

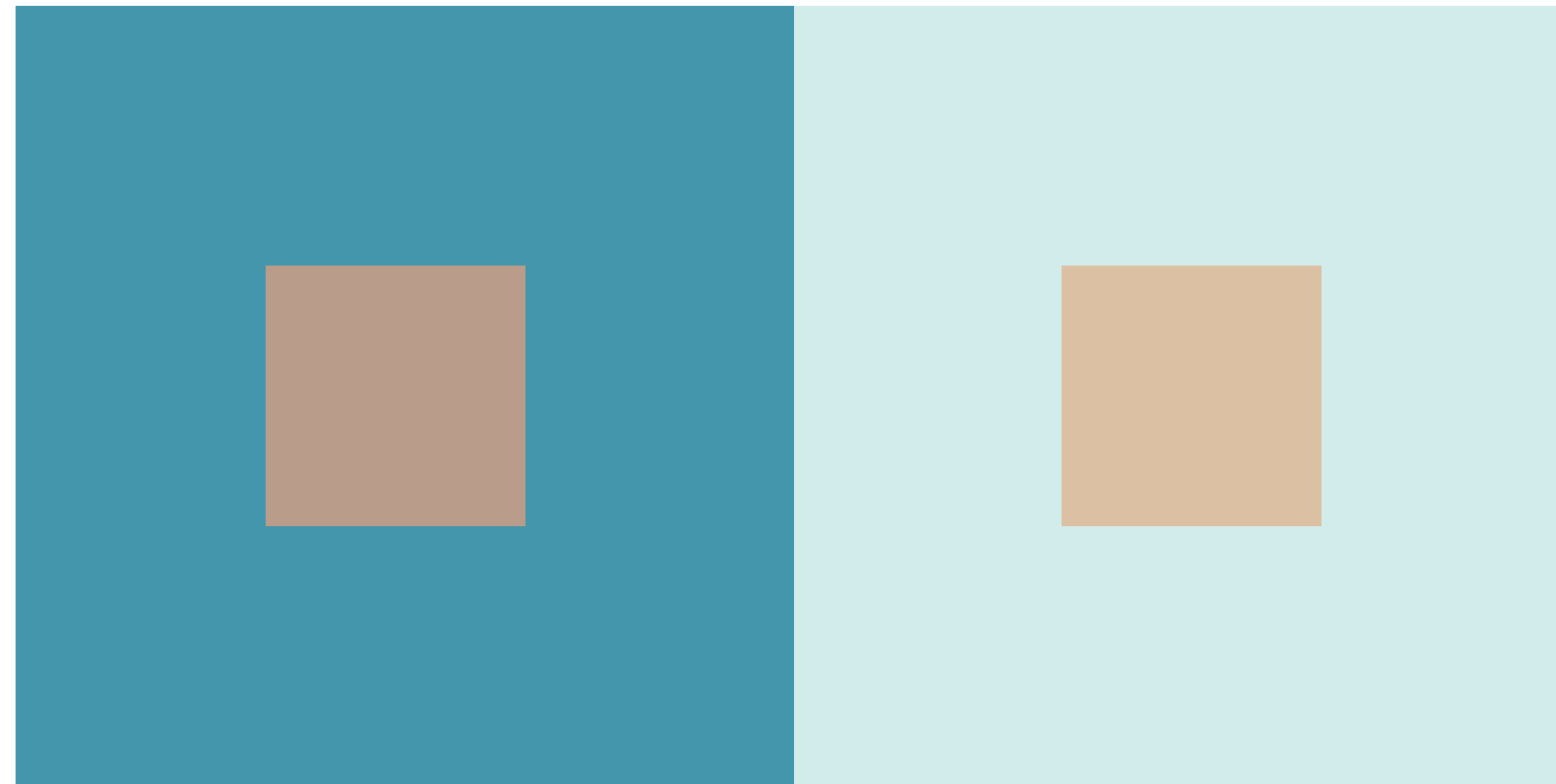
Takeaway

- We have higher contrast sensitivity in the luminance than in the chrominance channel. Show preference to luminance for encoding detail.

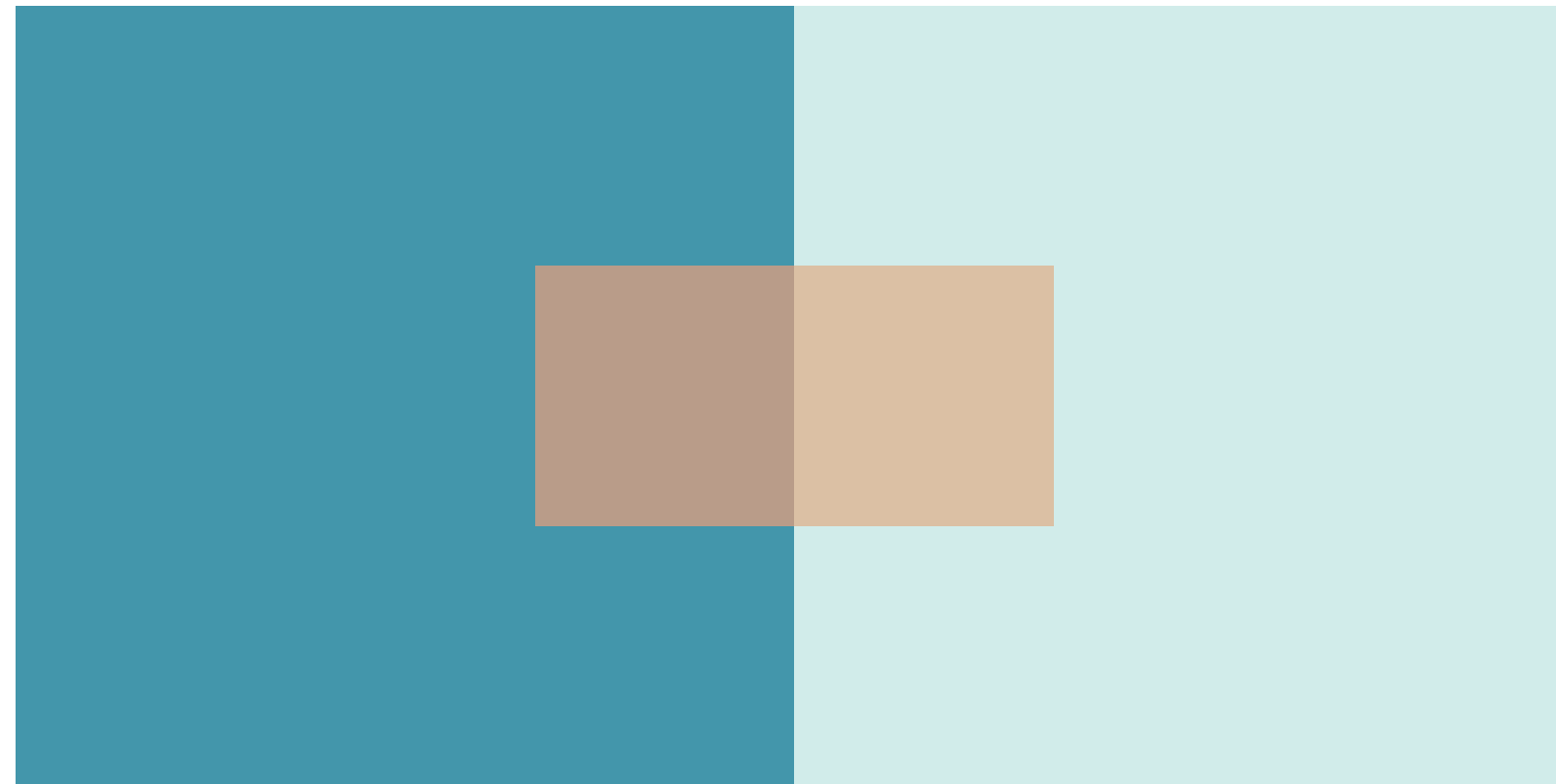
Color Relativity



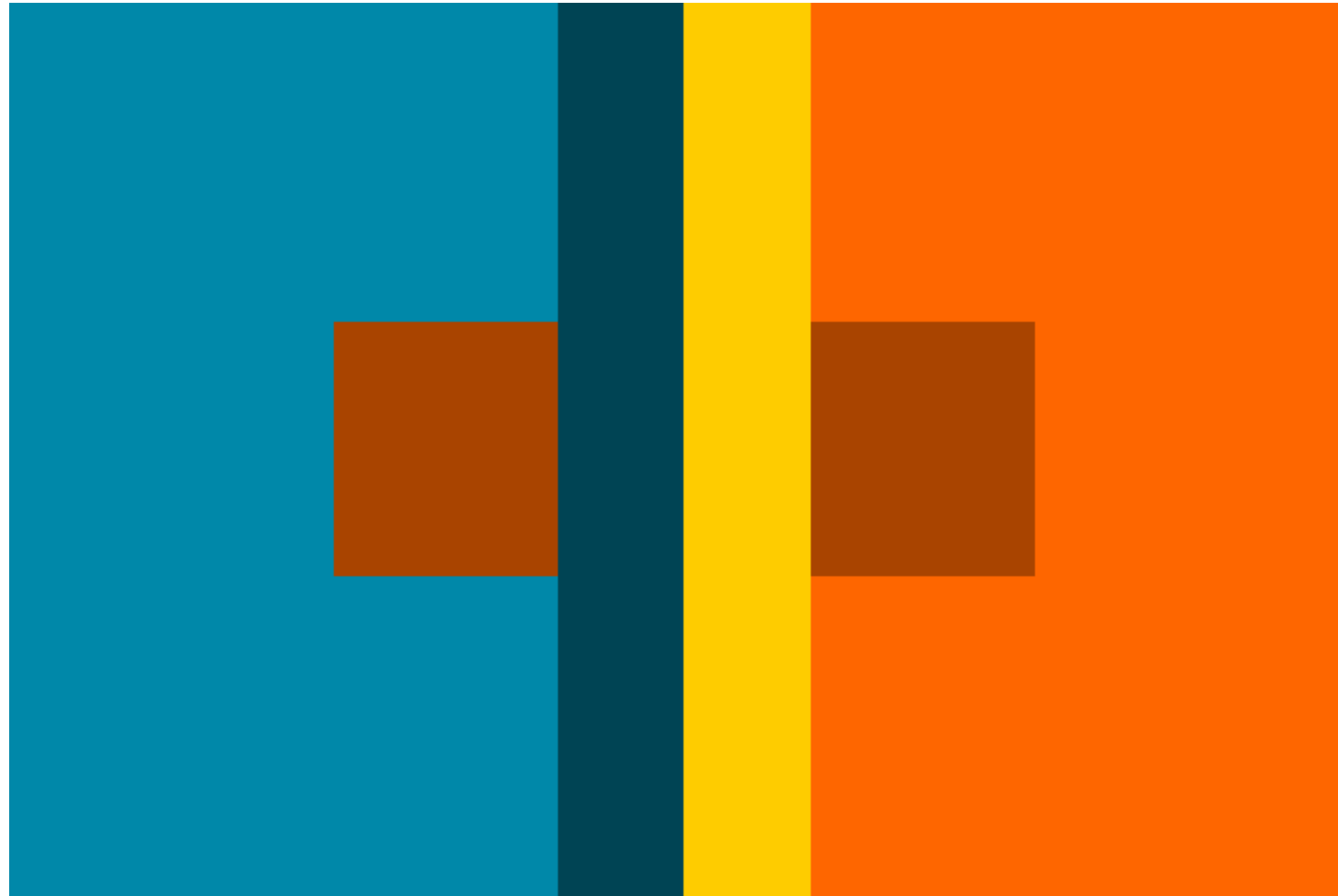
Color Relativity



Color Relativity

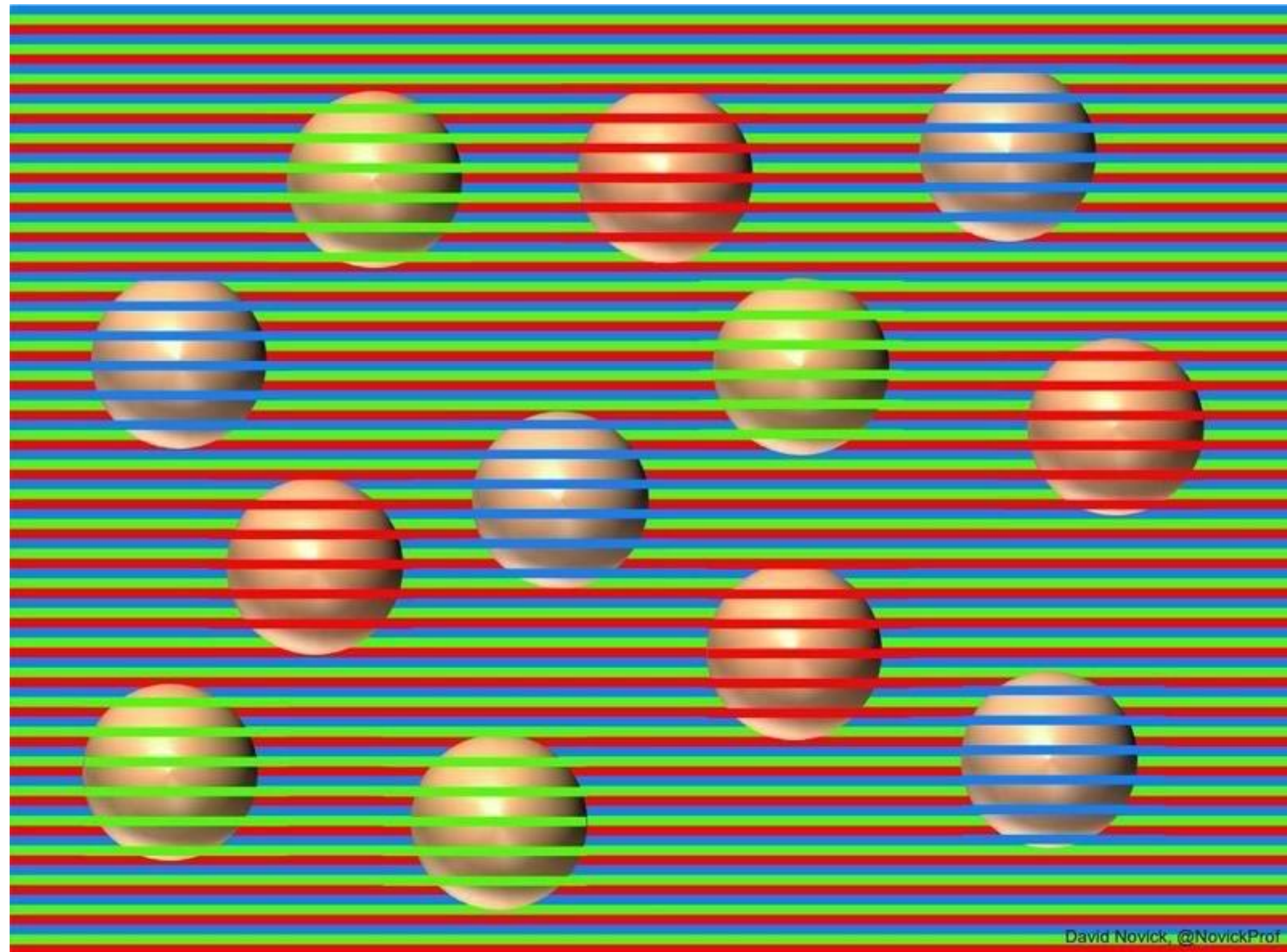


Color Relativity



Color Relativity





David Novick, @NovickProf

Color Relativity



Takeaway

- We have a strong propensity to assume our judgments of color are absolute, when in fact they are extremely relativistic.
- **Do your best to not place data in difficult contexts. Use color sparingly.**

luminance contrast

Showing small blue text on a black background is a bad idea.
There is insufficient luminance contrast.

Showing small blue text on a black background is a bad idea.
There is insufficient luminance contrast.

Showing small yellow text on a white background is a bad idea.
There is insufficient luminance contrast.

Showing small yellow text on a white background is a bad idea.
There is insufficient luminance contrast.

size & color



“the smaller the mark, the less distinguishable are the colors”

-Jacques Bertin

guidelines

- color is a relative medium—if encoding ordinal data with color, place marks on solid, neutral background
- because of contrast effects, it is difficult to perceive absolute luminance of noncontiguous regions
- in small regions use bright, highly saturated colors
- use low saturation pastel colors for large regions and backgrounds

what is a colormap?

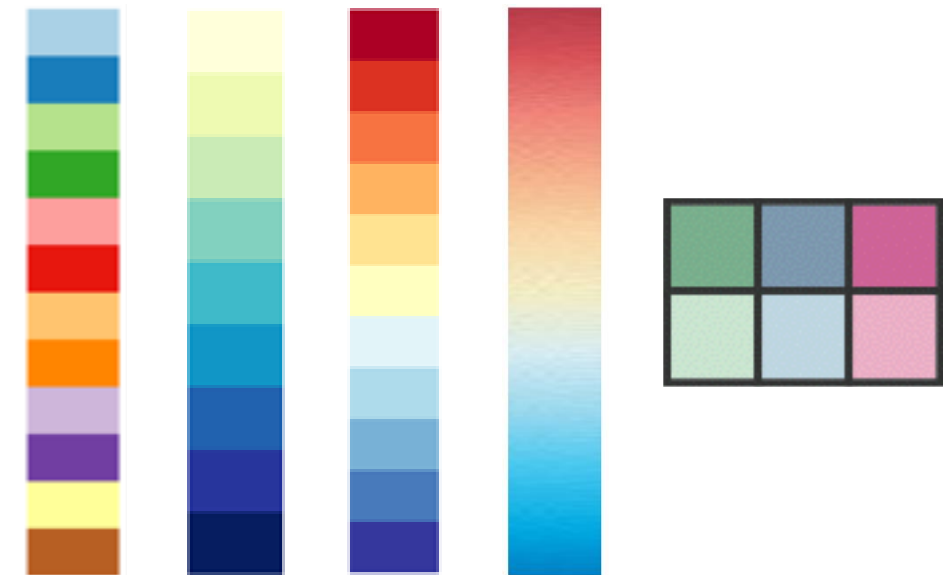
- specifies a mapping between color and values

- also called a transfer function

$[0, 8]$ →



- categorical vs ordered
 - sequential vs diverging
 - segmented vs continuous
 - univariate vs bivariate



guidelines

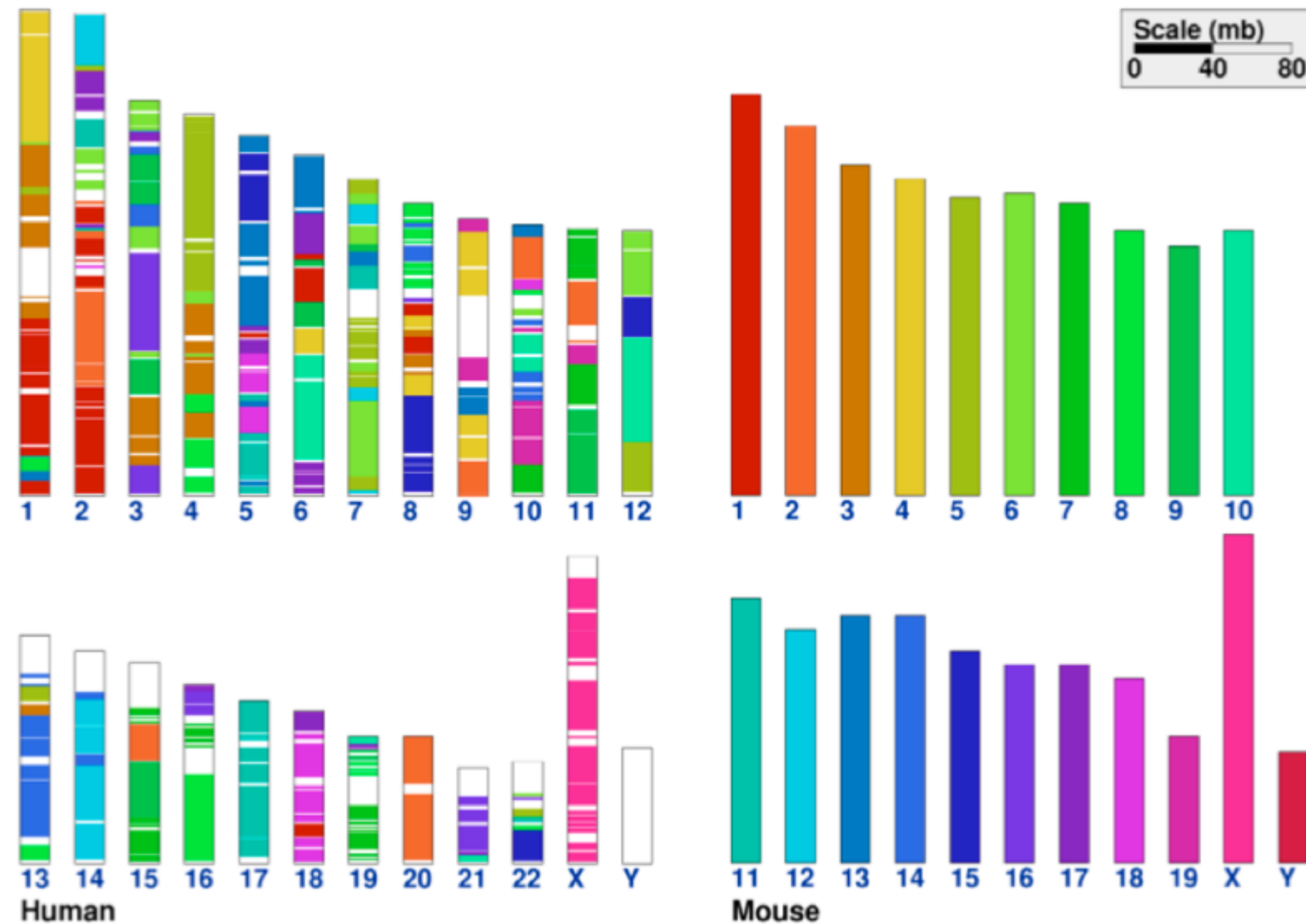
- categorical colors are easier to remember if they are nameable
- ordered colormaps should vary along saturation or luminance
- bivariate colormaps are difficult to interpret if at least one variable is not binary

hues for categories

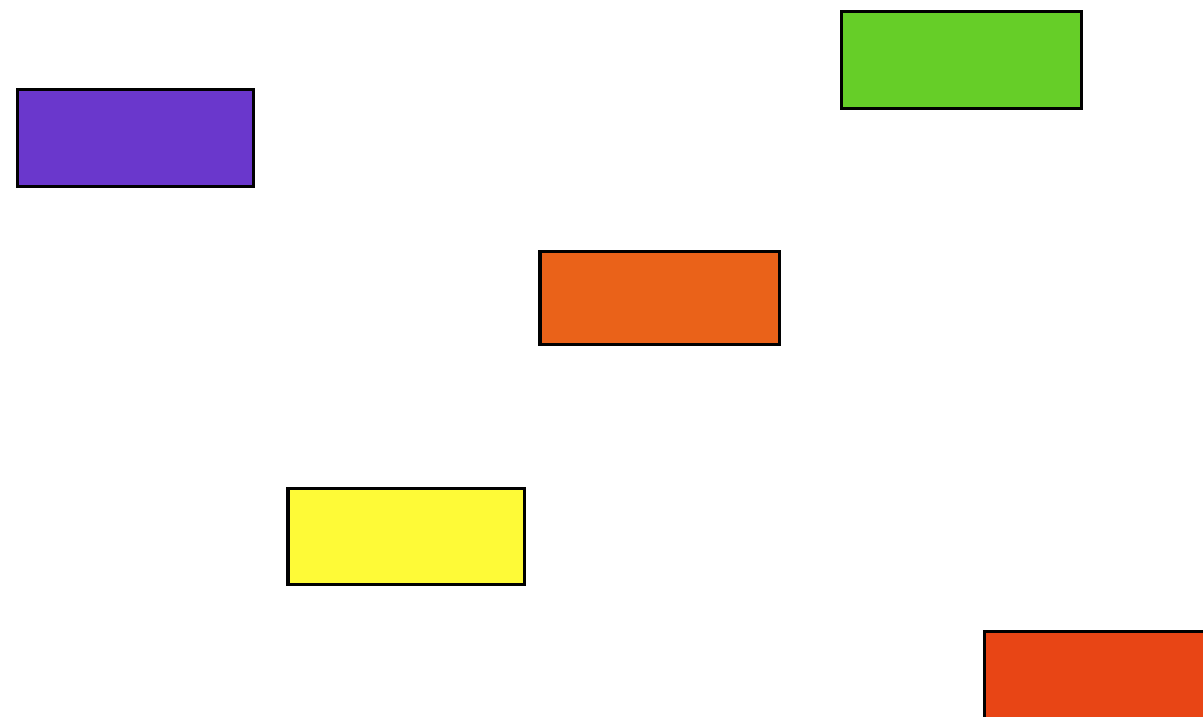


distinguishability

- only good at distinguishing 6-12 simultaneous colors



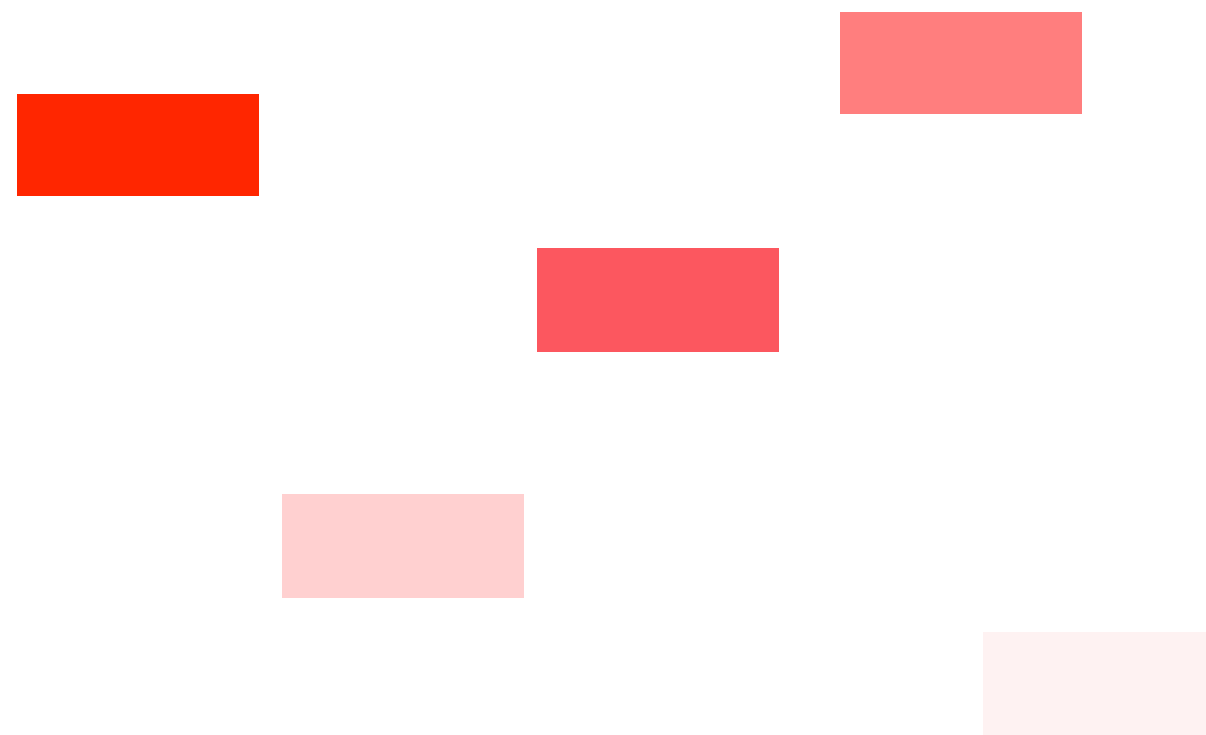
order these colors...



order these colors...



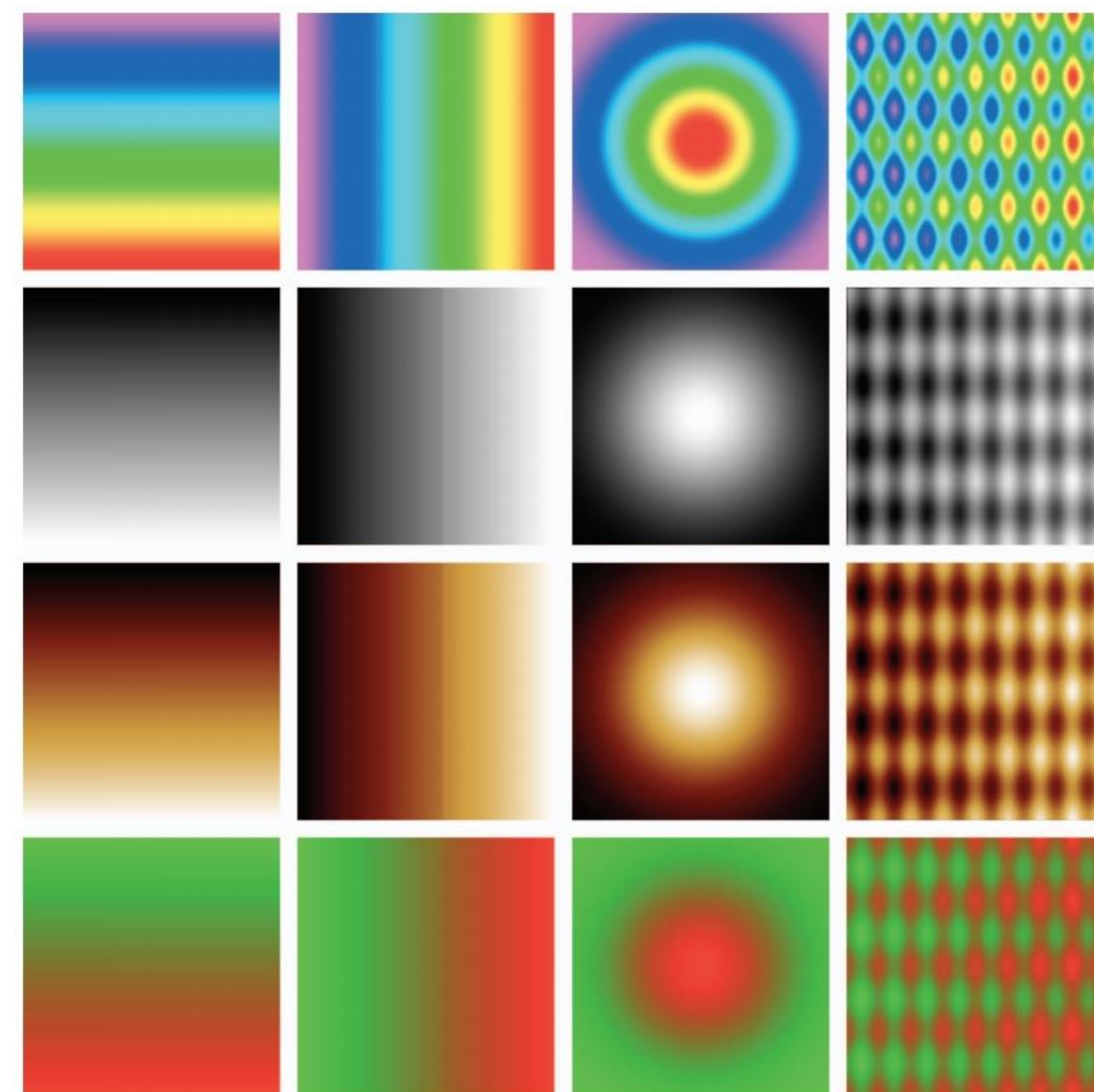
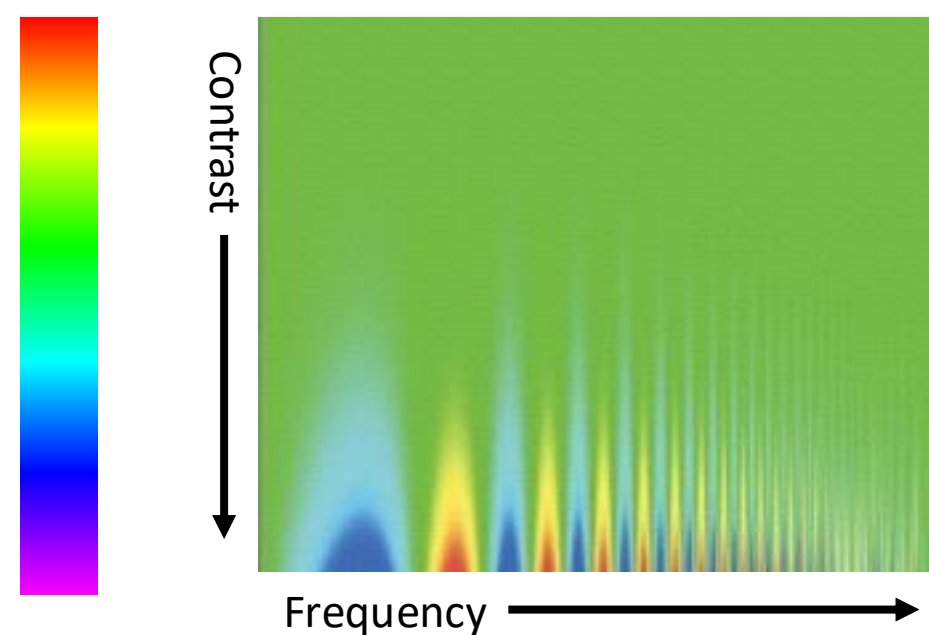
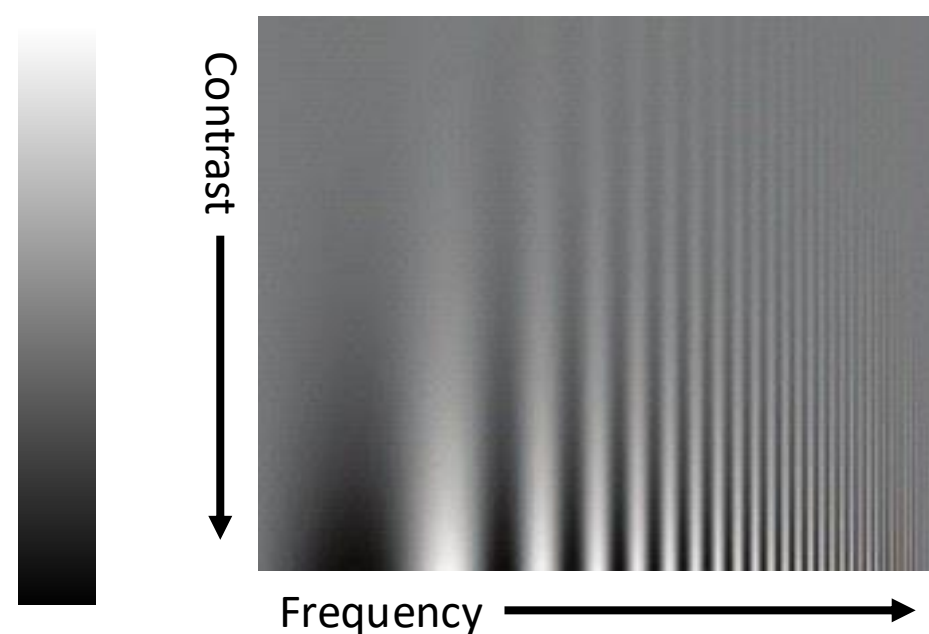
order these colors...



guidelines

- luminance and saturation are most effective for ordinal data because they have an inherent ordering
- hue is great for categorical data because there is no inherent ordering
 - but limit number of hues to 6-12 for distinguishability
- number of hues and distribution on the colormap should be related to which and how many structures in the data to emphasize

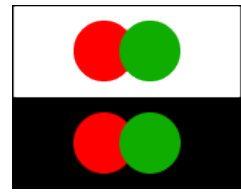
rainbow colormaps: challenges



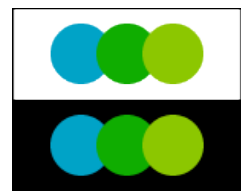
Borland 2007

Color section Guidelines

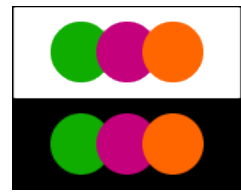
Color Harmonies



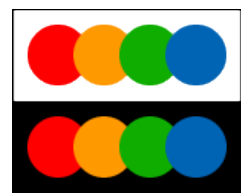
Complementary—high contrast creates a vibrant look



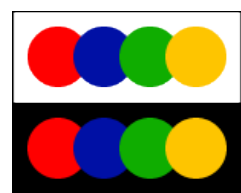
Analogous—often found in nature and are harmonious and pleasing to the eye



Triad—vibrant, even if you use pale or unsaturated versions of your hues

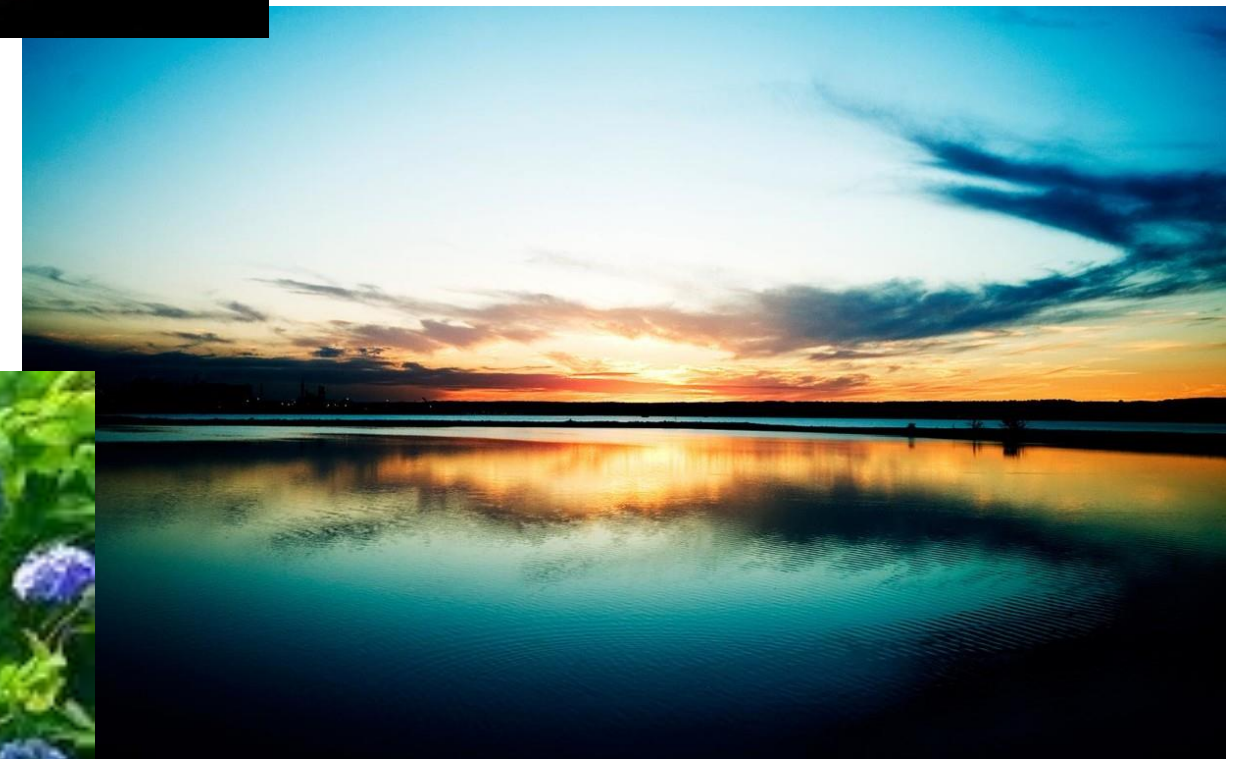


Split-complementary—same strong contrast as the complementary but less tension



Rectangle—rich color scheme offers plenty of possibilities for variation

LOOK TO NATURE



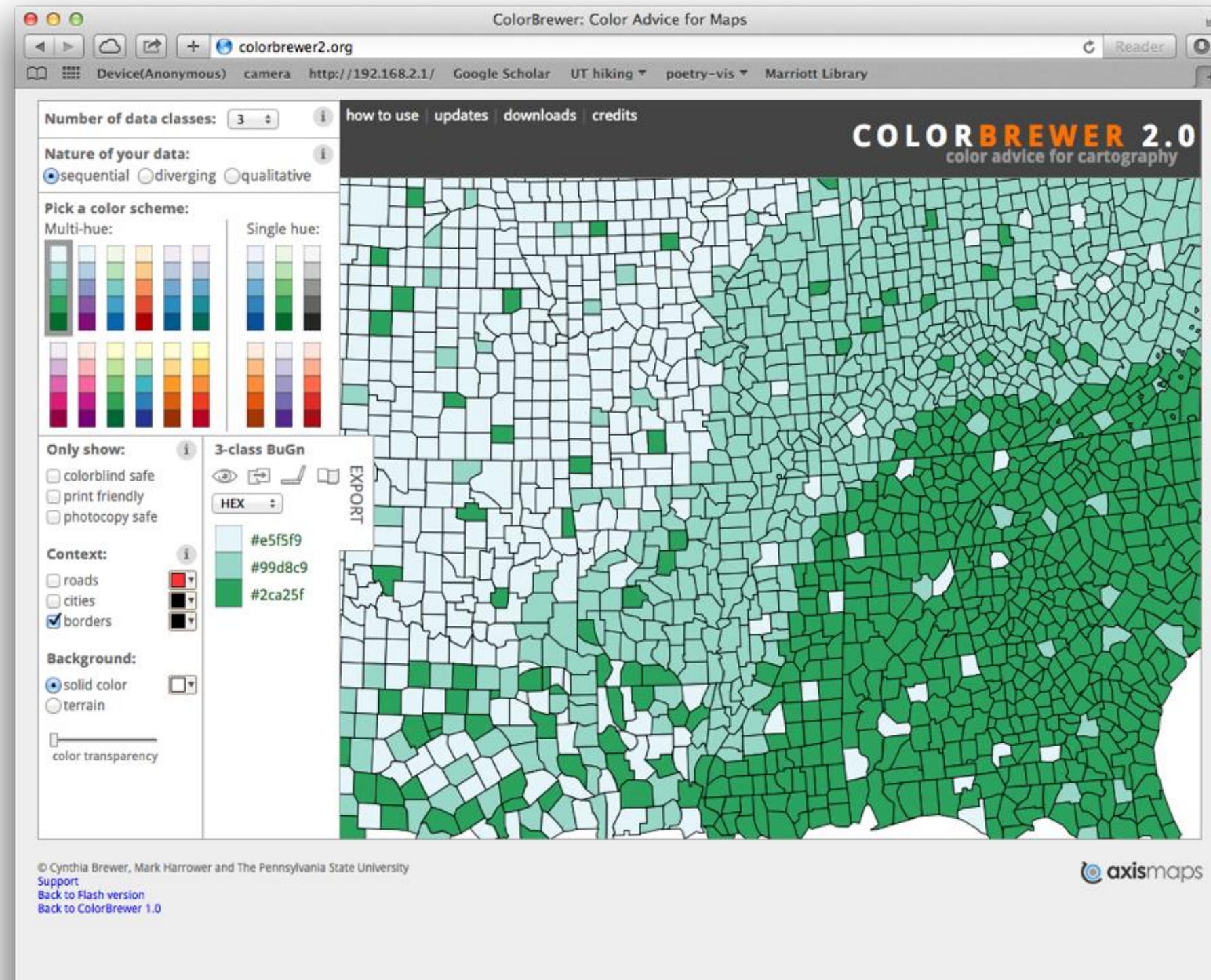
AVOIDANCE OF COLOR

- use neutrals (work with any scheme)
 - black, white, grey
- use diagrammatic marks (may be better encoding channels)
 - size, shape, texture, length, width, orientation, curvature and intensity

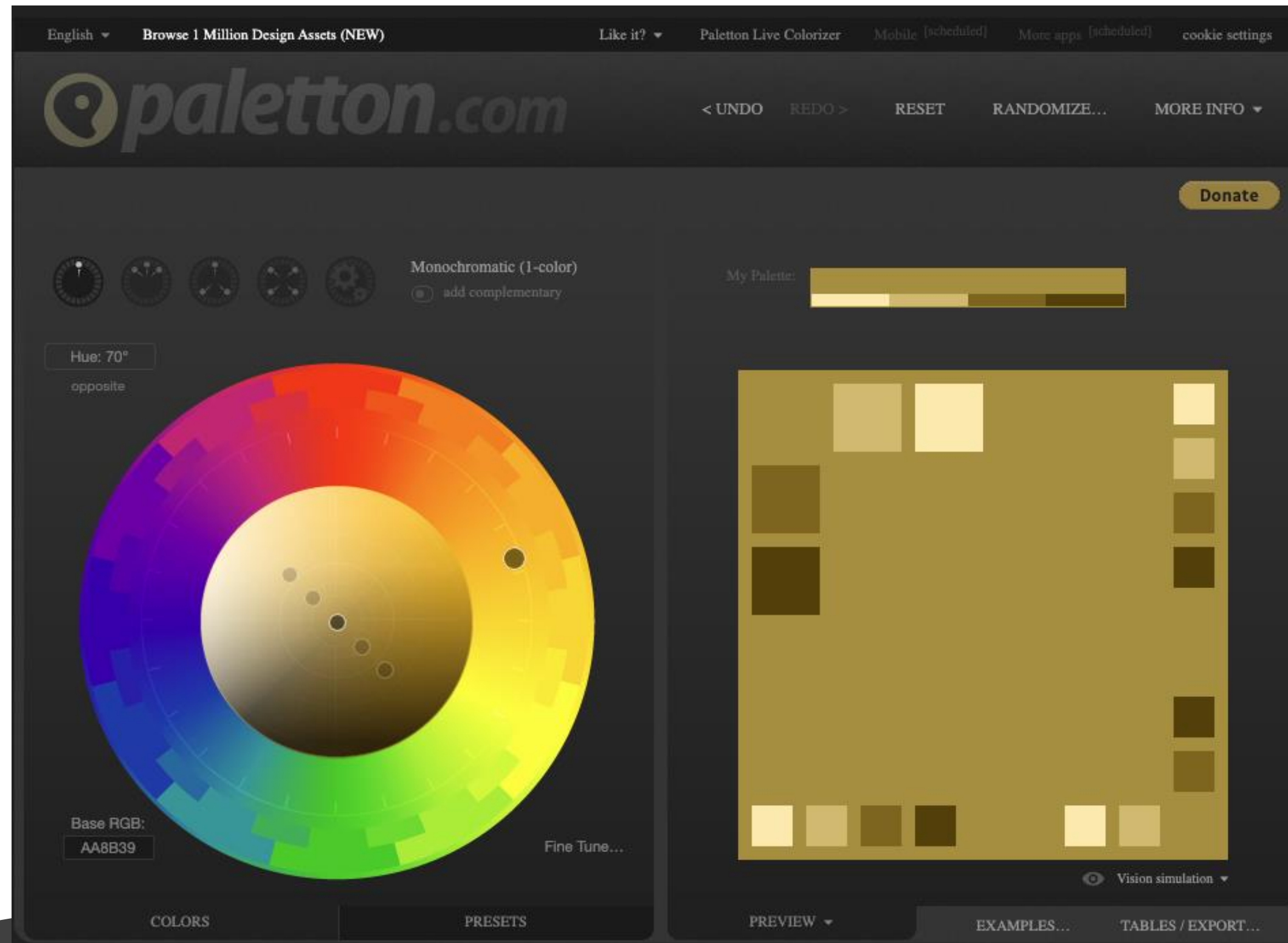
Get it right in black and white
- Maureen Stone

Tools for Color

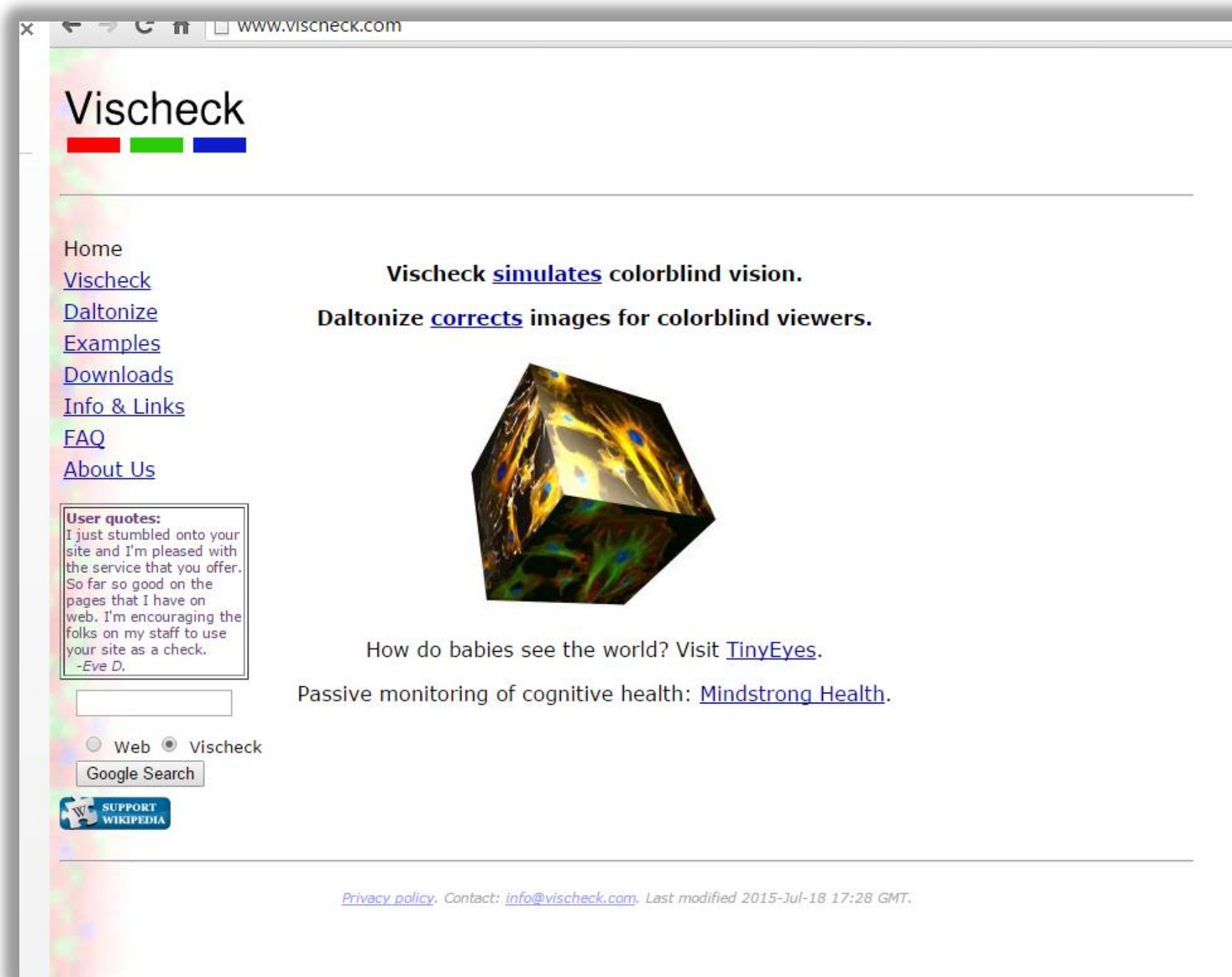
Colorbrewer2.org



paletton.com



Vischeck.com



Color Buddy (color-buddy.netlify.app)

